

THE RANKER'S MASTERBOOK

LIMITS.

The Edge of Calculus

Complete Solutions

100

PROBLEMS

10

CHAPTERS

3–5

DIFFICULTY

210

WORKED SOLUTIONS



One hundred original limits where the obvious move is the wrong move – for the very top of JEE Advanced and the Mathematical Olympiad. Every problem solved more than one way.

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Ten chapters, ten problems each.

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CHAPTER I

The Greatest Integer Function



Where the graph leaps, the limit hides.

№1 The Sinc Just Misses

●●●●● L4

$$\lim_{x \rightarrow 0^+} \left\lfloor \frac{100 \sin x}{x} \right\rfloor$$

floor

sinc

one-sided

$x \rightarrow 0^+$

squeeze

LIMIT

99

THE TRAP

Since $\frac{\sin x}{x} \rightarrow 1$, the bracket tends to 100, so "obviously" the floor tends to 100. This conflates the limit of $g(x)$ with the limit of $\lfloor g(x) \rfloor$. The floor is a step function: it cares only about which integer cell g lives in, not its limiting value. Because $\sin x < x$ strictly for $0 < x < \pi$, we have $\frac{100 \sin x}{x} < 100$ for every $x > 0$ near 0 — the value approaches 100 but never reaches it, sitting in the cell $(99, 100)$. So $\lfloor g(x) \rfloor = 99$ identically near 0, and the limit is 99, not 100.

Method I - Sign of the approach (one-sided analysis)

1. The floor of $g(x) = \frac{100 \sin x}{x}$ is determined entirely by which integer interval g falls into, so the whole game is whether g sits just below or just above 100.
2. For $0 < x < \pi$ the strict inequality $\sin x < x$ holds (the chord is shorter than the arc), hence $\frac{\sin x}{x} < 1$ and therefore

$$\frac{100 \sin x}{x} < 100.$$

The value is pinned strictly below 100 for all small $x > 0$.

3. We also need a lower bound to keep the value above 99. From the expansion $\sin x = x - \frac{x^3}{6} + \dots$,

$$\frac{100 \sin x}{x} = 100 \left(1 - \frac{x^2}{6} + \dots \right),$$

which exceeds 99 as soon as $\frac{100x^2}{6} < 1$, i.e. for $0 < x < \frac{\sqrt{6}}{10}$.

4. So for all sufficiently small $x > 0$,

$$99 < \frac{100 \sin x}{x} < 100,$$

meaning g never leaves the cell $(99, 100)$. Thus $\lfloor g(x) \rfloor = 99$ on that whole neighbourhood, and the limit is $\boxed{99}$.

Method II - Squeeze on the floor via explicit bounds

1. Write $g(x) = \frac{100 \sin x}{x}$. We trap g between two constants in a punctured right-neighbourhood of 0, then read off the floor.

2. Upper bound: $\sin x < x$ for $0 < x < \pi$ gives $g(x) < 100$.

3. Lower bound: the standard inequality $\sin x > x - \frac{x^3}{6}$ (valid for $x > 0$) gives

$$g(x) > 100 - \frac{100x^2}{6}.$$

Choosing $\delta = \frac{\sqrt{6}}{10}$ makes $\frac{100\delta^2}{6} = 1$, so for $0 < x < \delta$ we get $g(x) > 99$.

4. On $(0, \delta)$ we therefore have $g(x) \in (99, 100)$ — a single integer cell — so $\lfloor g(x) \rfloor = 99$ throughout. Since the floor is constant on a right-neighbourhood of 0, the limit equals 99.

INSIGHT

Multiplying a ratio that approaches 1 strictly from below by an integer N pins the floor at $N - 1$; the same statement with $\tan$ in place of $\sin$ (since $\frac{\tan x}{x} \rightarrow 1$ from above) would instead give N . The lesson: $\lim \lfloor g \rfloor \neq \lfloor \lim g \rfloor$ whenever the limit lands exactly on an integer.

№2 Sine From Below, Tangent From Above

●●●●● L4

$$\lim_{x \rightarrow 0^+} \left(\left\lfloor \frac{50 \sin x}{x} \right\rfloor + \left\lfloor \frac{50 \tan x}{x} \right\rfloor \right)$$

floor

sinc

tanc

one-sided

$x \rightarrow 0^+$

LIMIT

99

THE TRAP

Both ratios $\rightarrow 1$, so $50 \sin x/x$ and $50 \tan x/x$ both $\rightarrow 50$; "therefore" each floor is 50 and the sum is 100. The seduction is the apparent symmetry of the two terms. It fails because the floor reads the *direction* of approach, not the limiting value: $\sin x < x$ pushes the first ratio to 50 from *below* (floor 49), while $\tan x > x$ pushes the second to 50 from *above* (floor 50). They never both sit at 50, so the answer is $49 + 50 = 99$, not 100.

Method I - Opposite-side comparison

1. Handle the two floors separately using the strict monotone comparisons of $\sin$ and $\tan$ against their argument near 0.

2. For $0 < x < \pi$ we have $\sin x < x$, so $\frac{50 \sin x}{x} < 50$. Also $\sin x > x - \frac{x^3}{6}$ gives $\frac{50 \sin x}{x} > 50 - \frac{50x^2}{6} > 49$ once $x^2 < \frac{6}{50}$ (i.e. $x < 0.346$). Hence $\left\lfloor \frac{50 \sin x}{x} \right\rfloor = 49$.
3. For $0 < x < \frac{\pi}{2}$ we have $\tan x > x$, so $\frac{50 \tan x}{x} > 50$. Also $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots < x + x^3$ for small x , giving $\frac{50 \tan x}{x} < 50 + 50x^2 < 51$ once $x^2 < \frac{1}{50}$ (i.e. $x < 0.141$). Hence $\left\lfloor \frac{50 \tan x}{x} \right\rfloor = 50$.
4. On the common right-neighbourhood $(0, 0.141)$ both floors are constant, so the sum is identically $49 + 50 = 99$ and the limit is $\boxed{99}$.

Method II - Series bracket

1. Expand each quotient about $x = 0$:

$$\frac{50 \sin x}{x} = 50 - \frac{50x^2}{6} + O(x^4), \quad \frac{50 \tan x}{x} = 50 + \frac{50x^2}{3} + O(x^4).$$

2. For all sufficiently small $x > 0$ the first expression lies strictly in $(49, 50)$ (it is just below 50), and the second lies strictly in $(50, 51)$ (it is just above 50).
3. Therefore $\left\lfloor \frac{50 \sin x}{x} \right\rfloor = 49$ and $\left\lfloor \frac{50 \tan x}{x} \right\rfloor = 50$, independent of x . The sum is the constant 99, which is the limit.

INSIGHT

The asymmetry $\sin x < x < \tan x$ is exactly why halving the naive guess leaks one unit: the floor sees the direction of approach, not just the value. Note both ratios are even in x , so the same one-sided floors hold at 0^- and the two-sided limit also equals 99; the 0^+ phrasing is simply the cleanest way to state it.

№3 It Wanted To Be One

● ● ● ● ● L4

$$\lim_{x \rightarrow +\infty} \left\lfloor x \sin \frac{1}{x} \right\rfloor$$

floor sinc $x \rightarrow +\text{infinity}$ one-sided-from-below

LIMIT

0

THE TRAP

Since $x \sin \frac{1}{x} = \frac{\sin(1/x)}{1/x} \rightarrow 1$, it is tempting to pass the floor through the limit and answer 1. But $x \sin \frac{1}{x} = 1 - \frac{1}{6x^2} + O(x^{-4})$ approaches 1 strictly from below, so for every finite x it lies in $(0, 1)$ and its floor is 0. Floor and limit do not commute when the limit is an integer reached from below.

Method I - Strict inequality

1. Substitute $t = \frac{1}{x}$, so $t \rightarrow 0^+$ as $x \rightarrow +\infty$, and $x \sin \frac{1}{x} = \frac{\sin t}{t}$.
2. For $0 < t < \pi$ the sine is positive and lies strictly below its argument: $0 < \sin t < t$. Dividing by $t > 0$ gives $0 < \frac{\sin t}{t} < 1$.
3. A real number strictly inside $(0, 1)$ has floor 0. Hence $\left\lfloor x \sin \frac{1}{x} \right\rfloor = 0$ for all large x , and the limit is $\boxed{0}$.

Method II - Quantitative squeeze

- Use the sharp small-angle bounds $t - \frac{t^3}{6} < \sin t < t$ for $0 < t < 1$. With $t = 1/x$ and $x \geq 1$:

$$1 - \frac{1}{6x^2} < x \sin \frac{1}{x} < 1.$$
- For $x \geq 1$ the lower bound satisfies $1 - \frac{1}{6x^2} \geq 1 - \frac{1}{6} = \frac{5}{6} > 0$, so the expression is trapped in $(\frac{5}{6}, 1) \subset (0, 1)$.
- Therefore the floor equals 0 identically for all $x \geq 1$, and the limit is 0 — sharply different from the value 1 of the un-floored expression $\frac{\sin(1/x)}{1/x}$.

INSIGHT

Here $\lim f = 1$ yet $\lim \lfloor f \rfloor = 0$: a clean case where floor and limit fail to commute, because the limiting value is an integer approached strictly from below (the deficit is $\frac{1}{6x^2} > 0$).

№4 Half From The Triangle

● ● ● ● ● L3

$$\lim_{x \rightarrow +\infty} \frac{1}{x^2} \sum_{k=1}^{\lfloor x \rfloor} k$$

floor

summation

$x \rightarrow +\infty$

squeeze

LIMIT

$$\frac{1}{2}$$

THE TRAP

The careless route replaces $\lfloor x \rfloor$ by x and the sum by $\int_0^x t \, dt = \frac{x^2}{2}$, reading off $\frac{1}{2}$. The answer is right but the reasoning is not: you have silently assumed both that the discrete sum equals the integral and that the upper index $\lfloor x \rfloor$ may be swapped for x with no cost. Either swap changes the numerator by $\Theta(x)$; you must show that defect is lower order than the x^2 scale, not wave it away. A student who 'cancels' the floor without that estimate has guessed, not solved.

Method I - Exact sum then squeeze

- Let $m = \lfloor x \rfloor$, so $x - 1 < m \leq x$. The inner sum is given exactly by the triangular-number formula:

$$\sum_{k=1}^m k = \frac{m(m+1)}{2}.$$

- Hence

$$\frac{1}{x^2} \sum_{k=1}^m k = \frac{m(m+1)}{2x^2} = \frac{1}{2} \cdot \frac{m}{x} \cdot \frac{m+1}{x}.$$

- From $x - 1 < m \leq x$ we get $\frac{x-1}{x} < \frac{m}{x} \leq 1$ and $\frac{x}{x} < \frac{m+1}{x} \leq \frac{x+1}{x}$. As $x \rightarrow \infty$ both outer bounds tend to 1, so by the squeeze theorem $\frac{m}{x} \rightarrow 1$ and $\frac{m+1}{x} \rightarrow 1$.

- Therefore the product tends to $\frac{1}{2} \cdot 1 \cdot 1 = \boxed{\frac{1}{2}}$.

Method II - Bracketing the upper index

1. With $m = \lfloor x \rfloor$ and $x \geq 1$ we have $x - 1 < m \leq x$. Since $t \mapsto \frac{t(t+1)}{2}$ is increasing for $t \geq 0$, applying the triangular formula at $t = x - 1$, $t = m$, $t = x$ gives

$$\frac{(x-1)x}{2} \leq \frac{m(m+1)}{2} \leq \frac{x(x+1)}{2}.$$

2. Both flanking expressions equal $\frac{x^2}{2} + O(x)$, namely $\frac{x^2}{2} - \frac{x}{2}$ and $\frac{x^2}{2} + \frac{x}{2}$, so by squeezing

$$\sum_{k=1}^m k = \frac{m(m+1)}{2} = \frac{x^2}{2} + O(x).$$

3. Dividing by x^2 gives $\frac{1}{x^2} \sum_{k=1}^m k = \frac{1}{2} + O\left(\frac{1}{x}\right) \xrightarrow{x \rightarrow \infty} \frac{1}{2}$.

INSIGHT

The floor on the upper index perturbs the numerator only by an $O(x)$ amount against an x^2 main term, so it is invisible in the limit. Indeed at integer x one has the exact identity $\frac{1}{x^2} \sum_{k=1}^x k = \frac{1}{2} + \frac{1}{2x}$, showing the defect is precisely $\Theta(1/x)$ – small, but it must be estimated, not assumed away.

№5 Curvature Catches The Floor

●●●●● L4

$$\lim_{x \rightarrow 0^+} (1 - \cos x) \left\lfloor \frac{1}{x^2} \right\rfloor$$

floor

reciprocal-square

$x \rightarrow 0^+$

squeeze

cosine

LIMIT

$$\frac{1}{2}$$

THE TRAP

Replace $\lfloor 1/x^2 \rfloor$ by $1/x^2$ to collapse the product into $\frac{1 - \cos x}{x^2} \rightarrow \frac{1}{2}$. The number is right, but the move is illegal: $\lfloor 1/x^2 \rfloor \neq 1/x^2$, and the discarded piece $1/x^2 - \lfloor 1/x^2 \rfloor$ can sit anywhere in $[0, 1)$ — its gap is not $o(1)$. A correct solution must prove $x^2 \lfloor 1/x^2 \rfloor \rightarrow 1$ (the gap, scaled by x^2 , vanishes), not assume it away.

Method I - Factor and squeeze the floor

1. Split the product into a curvature factor and a floor factor:

$$(1 - \cos x) \left\lfloor \frac{1}{x^2} \right\rfloor = \frac{1 - \cos x}{x^2} \cdot \left(x^2 \left\lfloor \frac{1}{x^2} \right\rfloor \right).$$

2. First factor (curvature): $\frac{1 - \cos x}{x^2} \rightarrow \frac{1}{2}$ as $x \rightarrow 0$ (standard, since $1 - \cos x \sim x^2/2$).

3. Second factor (floor): from the universal bound $u - 1 < \lfloor u \rfloor \leq u$ with $u = 1/x^2$, multiply through by $x^2 > 0$ to get

$$1 - x^2 < x^2 \left\lfloor \frac{1}{x^2} \right\rfloor \leq 1.$$

As $x \rightarrow 0^+$ both ends $\rightarrow 1$, so by the squeeze theorem $x^2 \lfloor 1/x^2 \rfloor \rightarrow 1$.

4. Both factors converge, so the limit of the product is the product of the limits: $\frac{1}{2} \cdot 1 = \boxed{\frac{1}{2}}$.

Method II - Global squeeze (no splitting)

1. Apply the floor bound directly: $\frac{1}{x^2} - 1 < \left\lfloor \frac{1}{x^2} \right\rfloor \leq \frac{1}{x^2}$, valid for all $x \neq 0$. Near 0 we also have $1 - \cos x > 0$.

2. Multiply the chain by the positive quantity $(1 - \cos x)$ (inequalities are preserved):

$$(1 - \cos x) \left(\frac{1}{x^2} - 1 \right) < (1 - \cos x) \left\lfloor \frac{1}{x^2} \right\rfloor \leq \frac{1 - \cos x}{x^2}.$$

3. Right end: $\frac{1 - \cos x}{x^2} \rightarrow \frac{1}{2}$. Left end: $\frac{1 - \cos x}{x^2} - (1 - \cos x) \rightarrow \frac{1}{2} - 0 = \frac{1}{2}$ (the second term vanishes).

4. Both bounding sequences tend to $\frac{1}{2}$, so by the squeeze theorem the middle expression $\rightarrow \frac{1}{2}$.

INSIGHT

The $\frac{1}{2}$ is entirely a curvature effect ($1 - \cos x \sim x^2/2$); the floor contributes only the benign factor $x^2 \lfloor 1/x^2 \rfloor \rightarrow 1$.

The same mechanism gives $\lim_{x \rightarrow 0^+} \frac{x^2}{2} \lfloor 1/x^2 \rfloor = \frac{1}{2}$ and, more generally, $g(x) \lfloor 1/x^2 \rfloor \rightarrow L$ whenever $x^2 g(x) \rightarrow L$.

№6 The Inner Half

●●●●● L4

$$\lim_{n \rightarrow +\infty} \left\{ \sqrt{n^2 + n} \right\} \quad (n \in \mathbb{Z}^+)$$

floor

fractional-part

conjugate

$x \rightarrow +\infty$

integers

LIMIT

$$\frac{1}{2}$$

THE TRAP

Since $\sqrt{n^2 + n} \approx n$ for large n , the fractional part $\{\sqrt{n^2 + n}\}$ looks like it should shrink to 0. It does not. The error $\sqrt{n^2 + n} - n$ is not negligible: it stays in $(0, 1)$ and actually grows toward $\frac{1}{2}$. Equivalently, $\sqrt{n^2 + n}$ sits asymptotically halfway between the consecutive integers n and $n + 1$, so its fractional part tends to $\frac{1}{2}$, never 0.

Method I - Locate the integer part, then conjugate

1. First pin the integer part. For every $n \geq 1$,

$$n^2 < n^2 + n < n^2 + 2n + 1 = (n + 1)^2,$$

since $(n^2 + n) - n^2 = n > 0$ and $(n + 1)^2 - (n^2 + n) = n + 1 > 0$. Taking square roots, $n < \sqrt{n^2 + n} < n + 1$, so $\lfloor \sqrt{n^2 + n} \rfloor = n$.

2. Therefore the fractional part is

$$\{\sqrt{n^2 + n}\} = \sqrt{n^2 + n} - \lfloor \sqrt{n^2 + n} \rfloor = \sqrt{n^2 + n} - n.$$

3. Rationalise using the conjugate:

$$\sqrt{n^2 + n} - n = \frac{(n^2 + n) - n^2}{\sqrt{n^2 + n} + n} = \frac{n}{\sqrt{n^2 + n} + n}.$$

4. Divide numerator and denominator by n (valid since $n > 0$):

$$\frac{n}{\sqrt{n^2 + n} + n} = \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} \xrightarrow{n \rightarrow \infty} \frac{1}{\sqrt{1} + 1} = \frac{1}{2}.$$

5. Hence $\lim_{n \rightarrow \infty} \{\sqrt{n^2 + n}\} = \left\lfloor \frac{1}{2} \right\rfloor$.

Method II - Generalised binomial expansion

1. Write $\sqrt{n^2 + n} = n\sqrt{1 + \frac{1}{n}}$ and apply the generalised binomial series $\sqrt{1+u} = 1 + \frac{u}{2} - \frac{u^2}{8} + \dots$ with $u = \frac{1}{n} \rightarrow 0$ (so the series converges for large n):

$$\sqrt{n^2 + n} = n \left(1 + \frac{1}{2n} - \frac{1}{8n^2} + \dots \right) = n + \frac{1}{2} - \frac{1}{8n} + \dots$$

2. For $n \geq 1$ the tail $-\frac{1}{8n} + \dots$ lies in $(-\frac{1}{2}, 0)$, so $\sqrt{n^2 + n} \in (n, n+1)$ and again $\lfloor \sqrt{n^2 + n} \rfloor = n$.

3. Subtracting the integer part,

$$\{\sqrt{n^2 + n}\} = \sqrt{n^2 + n} - n = \frac{1}{2} - \frac{1}{8n} + \dots \xrightarrow{n \rightarrow \infty} \frac{1}{2}.$$

INSIGHT

The mechanism is that $\sqrt{n^2 + n}$ is squeezed strictly between the consecutive integers n and $n+1$ and drifts to their midpoint. More generally, for fixed a with $0 < a < 2$ one gets $\{\sqrt{n^2 + an}\} \rightarrow a/2$; the inner linear term n ($a = 1$) places the point exactly 'half a step' above n asymptotically.

№7 Two Floors Meet At An Integer

●●●●● L4

$$\lim_{x \rightarrow 2^-} (\lfloor x^2 \rfloor - \lfloor x \rfloor^2)$$

floor

integer-point

$x \rightarrow 2^-$

jump

composition

LIMIT

2

THE TRAP

At $x = 2$ both terms equal 4, so the difference is 0; a hasty reader plugs in the endpoint and answers 0. But the left-hand limit never reaches $x = 2$: for x just below 2, $\lfloor x \rfloor = 1$ so $\lfloor x \rfloor^2 = 1$, while x^2 is just below 4 giving $\lfloor x^2 \rfloor = 3$. The two floors jump at different instants, so the difference is $3 - 1 = 2$, not 0.

Method I - Evaluate each floor on the left

1. We need the value of the expression for x slightly less than 2, so restrict to a left-neighborhood of 2.
2. $\lfloor x \rfloor$: for $1 \leq x < 2$ we have $\lfloor x \rfloor = 1$, hence $\lfloor x \rfloor^2 = 1$.
3. $\lfloor x^2 \rfloor$: as $x \rightarrow 2^-$, $x^2 \rightarrow 4^-$. Moreover for $\sqrt{3} < x < 2$ we have $3 < x^2 < 4$, so $\lfloor x^2 \rfloor = 3$ throughout that interval.
4. Thus on the whole interval $(\sqrt{3}, 2)$ the expression equals the constant $3 - 1 = 2$, so the left-hand limit is $\boxed{2}$.
5. (For contrast, the right-hand limit at 2 is 0, which is why the problem is posed one-sided.)

Method II - Fractional-part decomposition

1. Write $\lfloor x^2 \rfloor = x^2 - \{x^2\}$ and $\lfloor x \rfloor = x - \{x\}$, so $\lfloor x \rfloor^2 = (x - \{x\})^2$.
2. On $(1, 2)$ we have $\{x\} = x - 1$, hence $x - \{x\} = 1$ and $\lfloor x \rfloor^2 = 1$ (a constant, not just a limit).
3. On $(\sqrt{3}, 2)$ we have $3 < x^2 < 4$, so $\{x^2\} = x^2 - 3$ and therefore $\lfloor x^2 \rfloor = x^2 - (x^2 - 3) = 3$ identically.
4. Hence on $(\sqrt{3}, 2)$ the difference is the constant $3 - 1 = 2$, and the left-hand limit is 2.

INSIGHT

At an integer the compositions $\lfloor x^2 \rfloor$ and $\lfloor x \rfloor^2$ jump at different instants (x^2 already crosses 3 at $x = \sqrt{3}$, well before x reaches 2); the left-limit gap measures that desynchronisation.

№8 The Logarithm Forgives The Floor

● ● ● ● ● L3

$$\lim_{x \rightarrow +\infty} (\ln \lfloor e^x \rfloor - x)$$

floor

logarithm

exponential

$x \rightarrow +\infty$

squeeze

LIMIT

0

THE TRAP

The absolute gap $e^x - \lfloor e^x \rfloor$ wanders chaotically in $[0, 1)$ and never settles, so it looks like $\ln \lfloor e^x \rfloor - x$ should jitter toward some nonzero constant (a common guess is a number like $-\frac{1}{2}$, the 'average' of the fractional part). But the limit only sees the *relative* error $\lfloor e^x \rfloor / e^x = 1 - O(e^{-x}) \rightarrow 1$, whose logarithm is crushed to 0 at exponential speed. The chaotic numerator is irrelevant once divided by something exponentially large.

Method I - Logarithm of a ratio

1. Since $x \rightarrow +\infty$ we have $e^x > 1$, so $\lfloor e^x \rfloor \geq 1$ and the logarithm is defined. Rewrite the difference as a single logarithm:

$$\ln \lfloor e^x \rfloor - x = \ln \lfloor e^x \rfloor - \ln e^x = \ln \frac{\lfloor e^x \rfloor}{e^x}.$$

2. Bound the ratio using $e^x - 1 < \lfloor e^x \rfloor \leq e^x$. Dividing by $e^x > 0$:

$$1 - e^{-x} < \frac{\lfloor e^x \rfloor}{e^x} \leq 1.$$

3. As $x \rightarrow +\infty$, $e^{-x} \rightarrow 0$, so by the squeeze theorem $\frac{\lfloor e^x \rfloor}{e^x} \rightarrow 1$.

4. By continuity of $\ln$ at 1,

$$\ln \frac{\lfloor e^x \rfloor}{e^x} \rightarrow \ln 1 = 0.$$

Hence the limit is $\boxed{0}$.

Method II - Quantitative bound on the difference

1. Let $r = \frac{\lfloor e^x \rfloor}{e^x}$. From Method I, $r \in (1 - e^{-x}, 1]$, and $\ln \lfloor e^x \rfloor - x = \ln r$, so $\ln r \leq 0$.

2. Write $r = 1 - u$ with $u = 1 - r \in [0, e^{-x}]$. The elementary inequality $\ln(1 - u) \geq -\frac{u}{1 - u}$ holds for all $u \in [0, 1)$ (verified numerically and provable since $f(u) = \ln(1 - u) + \frac{u}{1 - u}$ satisfies $f(0) = 0$, $f'(u) = \frac{u}{(1 - u)^2} \geq 0$).
3. Since $u \mapsto -\frac{u}{1 - u}$ is decreasing and $u < e^{-x}$, we get

$$0 \geq \ln r \geq -\frac{u}{1 - u} > -\frac{e^{-x}}{1 - e^{-x}}.$$

4. The right-hand bound $\rightarrow 0$ as $x \rightarrow +\infty$, so by squeeze $\ln r \rightarrow 0$. The floor disappears under the logarithm at exponential speed: $\ln \lfloor e^x \rfloor - x = O(e^{-x})$. The limit is 0.

INSIGHT

Subtracting the integer floor of an exponentially large quantity costs only $O(e^{-x})$ after taking logarithms. The moral: under a logarithm, what matters is the relative error $\lfloor e^x \rfloor / e^x \rightarrow 1$, not the absolute (and forever-chaotic) gap $e^x - \lfloor e^x \rfloor$.

№9 The Removable Ceiling

●●●●● L3

$$\lim_{x \rightarrow 0} \lfloor \cos x \rfloor$$

floor cosine $x \rightarrow 0$ two-sided value-vs-limit

LIMIT

0

THE TRAP

Substitute $x = 0$: $\cos 0 = 1$ and $\lfloor 1 \rfloor = 1$, so the limit 'must' be 1. This is exactly the value-vs-limit trap. A limit never reads the value at the point. For every $x \neq 0$ in a neighbourhood of 0 we have $\cos x < 1$ strictly (cosine peaks at 1 only at $x = 0$), and since $\cos x > 0$ there, $\cos x \in (0, 1)$, giving $\lfloor \cos x \rfloor = 0$. The function is the constant 0 on a punctured neighbourhood, so the limit is 0, not 1.

Method I - Strict maximum of cosine

- On the punctured interval $0 < |x| < \frac{\pi}{2}$, cosine is below its peak: $\cos x < 1$ strictly, since $\cos x = 1$ holds only at $x = 0$.
- Also $\cos x > 0$ on that interval, so together $0 < \cos x < 1$.
- Any value in $(0, 1)$ has floor 0, on both sides of 0. Hence $\lfloor \cos x \rfloor = 0$ for all small $x \neq 0$, so the two-sided limit is $\boxed{0}$.
- Contrast with the value: $f(0) = \lfloor \cos 0 \rfloor = \lfloor 1 \rfloor = 1 \neq 0$. This is a removable discontinuity, the limit exists ($= 0$) but differs from $f(0) = 1$, and the limit is blind to the single point.

Method II - Squeeze

- The standard bound $1 - \frac{x^2}{2} \leq \cos x$ holds for all real x , and $\cos x < 1$ for $x \neq 0$, so $\cos x \in [1 - \frac{x^2}{2}, 1)$ for $0 < |x| < \frac{\pi}{2}$.
- For $|x|$ small enough this interval sits inside $(0, 1)$, forcing $\lfloor \cos x \rfloor = 0$.

3. Thus $\lfloor \cos x \rfloor$ is the constant 0 on a punctured neighbourhood of 0, and a constant function has that constant as its limit: $\lim_{x \rightarrow 0} \lfloor \cos x \rfloor = 0$.

INSIGHT

Classic 'value $\neq$ limit' under the floor: an even function pinned strictly below its integer peak makes the floor flat at the integer just below, with a removable discontinuity exactly at the peak.

№10 Mind The Sign Of Zero

● ● ● ● ● L3

$$\lim_{x \rightarrow 0^-} \frac{\lfloor x \rfloor}{x}$$

floor one-sided $x \rightarrow 0$ non-existence comparison

LIMIT

$+\infty$

THE TRAP

It looks like a $\frac{0}{0}$ form: $\lfloor x \rfloor \rightarrow 0$ and $x \rightarrow 0$, tempting a rush to L'Hopital or a guess of 1 (and some recall the right side and guess that value too). Both reflexes fail. The fatal move is substituting $\lfloor 0 \rfloor = 0$ without resolving the floor by side. On the left, $-1 \leq x < 0$ forces $\lfloor x \rfloor = -1$ identically, so the ratio is $\frac{-1}{x} \rightarrow +\infty$ — there is no indeterminacy at all. (And even the 'analogy' guess of 1 is wrong: the right-side limit is 0, not 1.)

Method I - Resolve the floor on the left

- For $-1 \leq x < 0$ we have $\lfloor x \rfloor = -1$ (a constant, not 0).
- So on this interval

$$\frac{\lfloor x \rfloor}{x} = \frac{-1}{x}.$$

- As $x \rightarrow 0^-$, x is a small negative number, so $\frac{-1}{x} = \frac{-1}{\text{small negative}} \rightarrow +\infty$.
- Therefore $\lim_{x \rightarrow 0^-} \frac{\lfloor x \rfloor}{x} = \boxed{+\infty}$ (a definite extended-real value).

Method II - Two-sided contrast (why the limit must be posed one-sided)

- Right side: for $0 < x < 1$, $\lfloor x \rfloor = 0$, so $\frac{\lfloor x \rfloor}{x} = 0 \rightarrow 0$.
- Left side: as computed, $\frac{\lfloor x \rfloor}{x} = \frac{-1}{x} \rightarrow +\infty$.
- The two one-sided behaviours differ (0 vs $+\infty$), so the two-sided limit does not exist; the well-posed question is the left-hand limit, which is $+\infty$.
- This is the danger of the $\frac{0}{0}$ disguise: $\lfloor x \rfloor$ is 0 from the right but -1 from the left, so the apparent indeterminacy is fake on one side and not even an indeterminacy on the other.

INSIGHT

The floor's jump at 0 makes $\frac{\lfloor x \rfloor}{x}$ wildly asymmetric — a textbook reminder that $\frac{0}{0}$ from blind substitution is meaningless until you resolve the floor by side.

CHAPTER II

The Fractional Part



The sawtooth that never rests.

№11 The Wrong Branch at Two

●●●●● L3

$$\lim_{x \rightarrow 2^-} \frac{\{x\} - 1}{x - 2}$$

floor

one-sided

$x \rightarrow 2^-$

jump

removable-looking

LIMIT

1

THE TRAP

The tempting move is to evaluate $\{x\}$ at the limit point: $\{2\} = 0$, so the expression "looks like" $\frac{0-1}{2-2} = \frac{-1}{0^-}$, and a student declares the limit is $+\infty$ (or, mis-signing, $-\infty$, or simply DNE). This is wrong because $\{x\}$ is a discontinuous sawtooth: for $x \rightarrow 2^-$ we live on the branch $1 < x < 2$ where $\lfloor x \rfloor = 1$, so $\{x\} = x - 1$ (which approaches 1, not 0). The numerator $\{x\} - 1 = x - 2$ then cancels the denominator exactly, giving the finite value 1 — there is no blow-up at all.

Method I — Resolve the correct floor branch

1. For $x \rightarrow 2^-$ we have $1 < x < 2$, hence $\lfloor x \rfloor = 1$ and $\{x\} = x - \lfloor x \rfloor = x - 1$.
2. Substitute:

$$\frac{\{x\} - 1}{x - 2} = \frac{(x - 1) - 1}{x - 2} = \frac{x - 2}{x - 2} = 1$$

for every x in $(1, 2)$.

3. The expression is identically 1 on the entire left neighbourhood, so

$$\lim_{x \rightarrow 2^-} \frac{\{x\} - 1}{x - 2} = 1.$$

Method II — Substitution $x=2-h$

1. Let $x = 2 - h$ with $h \rightarrow 0^+$. Then $x \in (1, 2)$, so $\lfloor x \rfloor = 1$ and $\{x\} = x - 1 = 1 - h$.
2. Then

$$\frac{\{x\} - 1}{x - 2} = \frac{(1 - h) - 1}{(2 - h) - 2} = \frac{-h}{-h} = 1.$$

3. The ratio is independent of h , so the limit is 1. Writing $x = 2 - h$ exposes the live branch $\{x\} = 1 - h \rightarrow 1$ and removes the temptation to substitute the endpoint value $\{2\} = 0$.

INSIGHT

Never evaluate $\{x\}$ at an integer limit point itself: there the sawtooth drops from 1 back to 0. Only the one-sided branch — $\{x\} = x - \lfloor x \rfloor$ with the floor taken from the *interior* of the approach side — is a legal substitution, and here it turns a fake $\frac{1}{0}$ into a clean removable identity equal to 1.

№12 Approaching Zero From Below

●●●●● L4

$$\lim_{x \rightarrow 0^-} (1 + \{x\})^{1/x}$$

floor

one-sided

$x \rightarrow 0^-$

exponential

1^∞ -trap

LIMIT

0

THE TRAP

The form $(1 + \{x\})^{1/x}$ looks identical to the textbook $(1 + x)^{1/x} \rightarrow e$, and a student instinctively writes $\{x\} \approx x$ near 0, declares it a 1^∞ indeterminate form, and answers e . This is wrong on the side $x \rightarrow 0^-$: here $-1 < x < 0$ so $\lfloor x \rfloor = -1$ and $\{x\} = x + 1 \rightarrow 1$, which makes the base $1 + \{x\} = 2 + x \rightarrow 2$, NOT 1. There is no indeterminacy at all — the base sits at 2 while the exponent dives to $-\infty$, so the answer is 0, not e .

Method I — Identify base and exponent separately

- For $x \rightarrow 0^-$ we have $-1 < x < 0$, hence $\lfloor x \rfloor = -1$ and $\{x\} = x - \lfloor x \rfloor = x + 1$.
- Base: $1 + \{x\} = 2 + x \rightarrow 2$. Exponent: $1/x \rightarrow -\infty$.
- Since the base tends to $2 > 1$ with no indeterminacy,

$$(2 + x)^{1/x} \rightarrow 2^{-\infty} = 0.$$

Method II — Logarithmic transform

- Let $L = \lim_{x \rightarrow 0^-} (1 + \{x\})^{1/x}$. Taking logarithms of the expression (valid since the base $2 + x > 0$) gives

$$\ln(2 + x)^{1/x} = \frac{\ln(2 + x)}{x}.$$

- As $x \rightarrow 0^-$, the numerator $\ln(2 + x) \rightarrow \ln 2 > 0$ while $\frac{1}{x} \rightarrow -\infty$, so the product $\frac{\ln(2 + x)}{x} \rightarrow -\infty$.
- By continuity of the exponential, $L = \exp\left(\lim_{x \rightarrow 0^-} \frac{\ln(2 + x)}{x}\right) = e^{-\infty} = 0$.

INSIGHT

The 1^∞ reflex assumes the base $\rightarrow 1$; the jump discontinuity of $\{x\}$ at integers secretly pushes the base to 2 from the left, collapsing the indeterminate form entirely. (The two-sided limit does not exist: from 0^+ the base $\rightarrow 1$ and the value $\rightarrow e$.)

№13 Sawtooth Meets Taylor

●●●●● L4

$$\lim_{x \rightarrow 0^+} \frac{\{\tan x\} - \{\sin x\}}{x^3}$$

fractional-part

$x \rightarrow 0^+$

taylor

tan

sin

cubic

LIMIT

$$\frac{1}{2}$$

THE TRAP

Seeing a fractional part, a student writes $\{\tan x\} - \{\sin x\} \approx \tan x - \sin x \approx x - x = 0$ and concludes the limit is 0. The error is stopping at first order: both numerator pieces agree to order x (and even order x^2), so the leading terms cancel exactly. The cubic denominator is calibrated precisely to the first surviving term, $\frac{x^3}{2}$ — you must keep the x^3 Taylor data.

Method I — Strip the fractional parts, then expand

1. For $x \rightarrow 0^+$ with $x < \frac{\pi}{4}$, both $\sin x$ and $\tan x$ lie in $(0, 1)$, so $\lfloor \sin x \rfloor = \lfloor \tan x \rfloor = 0$, giving $\{\sin x\} = \sin x$ and $\{\tan x\} = \tan x$.

2. Thus the numerator is exactly $\tan x - \sin x$. Factor:

$$\tan x - \sin x = \sin x \left(\frac{1}{\cos x} - 1 \right) = \sin x \cdot \frac{1 - \cos x}{\cos x}.$$

3. Use the standard equivalents $\sin x \sim x$, $1 - \cos x \sim \frac{x^2}{2}$, and $\cos x \rightarrow 1$. Since each factor has a nonzero leading term, the equivalences multiply:

$$\tan x - \sin x \sim x \cdot \frac{x^2/2}{1} = \frac{x^3}{2}.$$

4. Therefore $\frac{\tan x - \sin x}{x^3} \rightarrow \frac{1}{2}$.

Method II — Maclaurin series

1. On a punctured right-neighbourhood $(0, \varepsilon)$ the floors vanish, so the ratio equals $\frac{\tan x - \sin x}{x^3}$.

2. Expand: $\tan x = x + \frac{x^3}{3} + O(x^5)$ and $\sin x = x - \frac{x^3}{6} + O(x^5)$.

3. Subtract: $\tan x - \sin x = \left(\frac{1}{3} + \frac{1}{6}\right)x^3 + O(x^5) = \frac{1}{2}x^3 + O(x^5)$.

4. Divide by x^3 and let $x \rightarrow 0^+$: the limit is $\frac{1}{2}$.

INSIGHT

The fractional part is a red herring on $(0, 1)$ — its real job is to bait you into treating $\{f(x)\} \approx x$ and stopping at first order; the genuine content is the order-three Taylor gap between $\tan$ and $\sin$. Note the next correction is $+\frac{1}{8}x^2$, so the convergence is from above.

№14 Bounded Sawtooth at Infinity

●●●●● L4

$$\lim_{n \rightarrow \infty} \left\{ \sqrt{n^2 + n} \right\}$$

fractional-part

$x \rightarrow \infty$

conjugate

sqrt

bounded-frac

LIMIT

$$\frac{1}{2}$$

THE TRAP

A student notes $\{\cdot\}$ is bounded in $[0, 1)$ and that $\sqrt{n^2 + n}$ runs through irrationals, then guesses the fractional part 'equidistributes' or oscillates with no limit, declaring DNE. In fact $\sqrt{n^2 + n}$ is squeezed strictly between the consecutive integers n and $n + 1$, pinning its floor to n and forcing convergence.

Method I — Locate the integer floor, then conjugate

1. Compare with consecutive squares: $n^2 < n^2 + n < n^2 + 2n + 1 = (n + 1)^2$ for every $n \geq 1$, so $n < \sqrt{n^2 + n} < n + 1$.
2. Hence $\lfloor \sqrt{n^2 + n} \rfloor = n$ and $\{\sqrt{n^2 + n}\} = \sqrt{n^2 + n} - n$.
3. Rationalise:

$$\sqrt{n^2 + n} - n = \frac{(n^2 + n) - n^2}{\sqrt{n^2 + n} + n} = \frac{n}{\sqrt{n^2 + n} + n} = \frac{1}{\sqrt{1 + \frac{1}{n}} + 1}.$$

4. As $n \rightarrow \infty$ the denominator $\rightarrow 2$, so the limit is $\frac{1}{2}$.

Method II — Binomial expansion

1. From the bound $n < \sqrt{n^2 + n} < n + 1$ we again get $\{\sqrt{n^2 + n}\} = \sqrt{n^2 + n} - n$.
2. Write $\sqrt{n^2 + n} = n\sqrt{1 + \frac{1}{n}} = n\left(1 + \frac{1}{2n} - \frac{1}{8n^2} + \dots\right)$.
3. So $\sqrt{n^2 + n} - n = \frac{1}{2} - \frac{1}{8n} + O\left(\frac{1}{n^2}\right)$, and every term past $\frac{1}{2}$ vanishes as $n \rightarrow \infty$, giving the limit $\frac{1}{2}$.

INSIGHT

Whenever $f(n)$ is trapped between two consecutive integers the floor is locked, so $\{f(n)\} = f(n) - \lfloor f(n) \rfloor$ inherits the smooth limit of $f(n)$ minus that integer – equidistribution never gets a chance.

№15 The Tamed Oscillator

●●●●● L4

$$\lim_{x \rightarrow 0^+} x \left\{ \frac{1}{x} \right\}$$

fractional-part

$x \rightarrow 0^+$

squeeze

oscillation

naive-DNE

LIMIT

$$0$$

THE TRAP

A student notes that $\{1/x\}$ sweeps densely across $[0, 1)$ infinitely often as $x \rightarrow 0^+$ — never approaching any single value — and, reasoning by analogy with the classic $\lim_{x \rightarrow 0} \sin(1/x)$ (which truly does not exist), declares the product DNE. The flaw: $\sin(1/x)$ stands alone, but here the divergent factor $\{1/x\}$ is merely *bounded* in $[0, 1)$, and the vanishing x in front multiplies it down to 0. Oscillation never settling is irrelevant once it is trapped between 0 and x .

Method I — Squeeze theorem

1. For every real t , the fractional part satisfies $0 \leq \{t\} < 1$. Putting $t = 1/x$ with $x > 0$ and multiplying by the positive number x :

$$0 \leq x \left\{ \frac{1}{x} \right\} < x.$$

2. As $x \rightarrow 0^+$ both outer bounds, 0 and x , tend to 0.
3. By the squeeze theorem the trapped quantity is forced to the common limit:

$$\lim_{x \rightarrow 0^+} x \left\{ \frac{1}{x} \right\} = 0.$$

Method II — Decompose via floor and a subsequence sanity check

1. Use $\{t\} = t - \lfloor t \rfloor$ with $t = 1/x$:

$$x \left\{ \frac{1}{x} \right\} = x \left(\frac{1}{x} - \left\lfloor \frac{1}{x} \right\rfloor \right) = 1 - x \left\lfloor \frac{1}{x} \right\rfloor.$$

2. The floor satisfies $\frac{1}{x} - 1 < \left\lfloor \frac{1}{x} \right\rfloor \leq \frac{1}{x}$; multiplying by $x > 0$ gives

$$1 - x < x \left\lfloor \frac{1}{x} \right\rfloor \leq 1.$$

3. As $x \rightarrow 0^+$ the left bound $1 - x \rightarrow 1$, so by squeeze $x \left\lfloor \frac{1}{x} \right\rfloor \rightarrow 1$, hence

$$x \left\{ \frac{1}{x} \right\} = 1 - x \left\lfloor \frac{1}{x} \right\rfloor \rightarrow 1 - 1 = 0.$$

4. This independently confirms the result: although $\{1/x\}$ alone has no limit, the floor decomposition shows the product is 1 minus a quantity that converges to 1.

INSIGHT

A bounded-but-divergent factor times a vanishing factor still converges – the existence of a product's limit does not require the existence of each factor's limit. Contrast $\lim_{x \rightarrow 0} \sin(1/x)$ (no vanishing prefactor, genuinely DNE) with $\lim_{x \rightarrow 0} x \sin(1/x) = 0$.

№16 Euler Hidden in the Sawtooth

● ● ● ● ● L5

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left\{ \frac{n}{k} \right\}$$

fractional-part

$n \rightarrow \infty$

riemann-sum

harmonic

euler-mascheroni

LIMIT

$$1 - \gamma$$

THE TRAP

The average $\frac{1}{n} \sum_{k=1}^n \{n/k\}$ looks exactly like a Riemann sum $\frac{1}{n} \sum_{k=1}^n g(k/n)$, tempting you to set $g(t) = \{1/t\}$ and write the limit as $\int_0^1 \{1/t\} dt = 1 - \gamma$. The final number is even correct — but the argument is bogus: the summand $\{n/k\}$ is NOT a function of k/n alone (e.g. $k/n = \frac{1}{2}$ gives $\{n/k\} = \{2\} = 0$ for every even n , yet $\{n/k\}$ jumps wildly as n varies with k/n held near other rationals), so no fixed g represents it and the Riemann-sum theorem simply does not apply. The honest route is the floor identity $\sum \{n/k\} = nH_n - \sum \lfloor n/k \rfloor$ together with Dirichlet's divisor asymptotic.

Method I — Floor identity and the harmonic asymptotic

1. Use $\{n/k\} = \frac{n}{k} - \lfloor \frac{n}{k} \rfloor$, so

$$\sum_{k=1}^n \left\{ \frac{n}{k} \right\} = n \sum_{k=1}^n \frac{1}{k} - \sum_{k=1}^n \left\lfloor \frac{n}{k} \right\rfloor = nH_n - D(n),$$

where $D(n) = \sum_{k=1}^n \lfloor n/k \rfloor$ is the divisor-summatory function.

2. Dirichlet's theorem: $D(n) = n \ln n + (2\gamma - 1)n + O(\sqrt{n})$. Also $H_n = \ln n + \gamma + O(1/n)$, so $nH_n = n \ln n + \gamma n + O(1)$.

3. Subtract: $\sum_{k=1}^n \{n/k\} = (n \ln n + \gamma n) - (n \ln n + (2\gamma - 1)n) + O(\sqrt{n}) = (1 - \gamma)n + O(\sqrt{n})$.

4. Divide by n :

$$\frac{1}{n} \sum_{k=1}^n \left\{ \frac{n}{k} \right\} = (1 - \gamma) + O\left(\frac{1}{\sqrt{n}}\right) \rightarrow 1 - \gamma.$$

Method II — Integral comparison via the Euler–Mascheroni integral

1. Compare the sum to $\frac{1}{n} \int_1^n \left\{ \frac{n}{x} \right\} dx$. For $x \in [k, k+1]$ with $k \geq \sqrt{n}$ the map $x \mapsto \{n/x\}$ has total variation $O(n/k^2)$ on that interval, so replacing the sum by the integral costs $\sum_{k \geq \sqrt{n}} O(n/k^2) \cdot \frac{1}{n} = O(1/\sqrt{n})$; the block $k < \sqrt{n}$ contributes $O(\sqrt{n})$ to the sum, i.e. $O(1/\sqrt{n})$ to the average. Hence both have the same limit.

2. Substitute $u = n/x$, $dx = -\frac{n}{u^2} du$:

$$\frac{1}{n} \int_1^n \left\{ \frac{n}{x} \right\} dx = \int_1^n \{u\} \frac{du}{u^2}.$$

3. As $n \rightarrow \infty$ this tends to $\int_1^\infty \frac{\{u\}}{u^2} du$. Writing $\{u\} = u - \lfloor u \rfloor$ and summing the integer blocks gives the classical value $\int_1^\infty \{u\}/u^2 du = 1 - \gamma$.

4. Hence the average converges to $1 - \gamma$, matching Method I.

INSIGHT

$\int_1^\infty \frac{\{u\}}{u^2} du = 1 - \gamma$ is the integral form of the Euler–Mascheroni constant; the sawtooth $\{n/k\}$ is exactly the discrepancy between the harmonic sum nH_n and the divisor sum $D(n)$, which is why γ appears twice and cancels down to $1 - \gamma$.

№17 Cosine Slips Below One

●●●●● L3

$$\lim_{x \rightarrow 0^+} (\{\sin x\} + \{\cos x\})$$

fractional-part

$x \rightarrow 0^+$

one-sided

sin

cos

LIMIT

1

THE TRAP

A student plugs in the limit point: $\sin 0 = 0$, $\cos 0 = 1$, so $\{\sin 0\} + \{\cos 0\} = \{0\} + \{1\} = 0 + 0 = 0$, and answers 0. The fatal slip is evaluating the fractional part AT the boundary instead of taking the limit: for every $x > 0$

near 0, $\cos x$ sits strictly *below* 1, so $\lfloor \cos x \rfloor = 0$ and $\{\cos x\} = \cos x \rightarrow 1^-$ — it never reaches the integer 1 where the fractional part would reset to 0. Hence the cosine term contributes 1, not 0, and the limit is 1.

Method I — Track each fractional part's branch

1. For small $x > 0$: $\sin x \in (0, 1)$, so $\lfloor \sin x \rfloor = 0$ and $\{\sin x\} = \sin x \rightarrow 0^+$.
2. Also $\cos x \in (0, 1)$ (strictly below 1 for $x > 0$), so $\lfloor \cos x \rfloor = 0$ and $\{\cos x\} = \cos x \rightarrow 1^-$.
3. Therefore $\{\sin x\} + \{\cos x\} = \sin x + \cos x \rightarrow 0 + 1 = 1$.

Method II — Continuity after removing the floors

1. On the punctured right neighbourhood $(0, \delta)$ both $\sin x$ and $\cos x$ stay in $(0, 1)$, hence both floors are identically 0.
2. Thus the function equals the continuous $\sin x + \cos x$ there, with no discontinuity to navigate.
3. $\lim_{x \rightarrow 0^+} (\sin x + \cos x) = 0 + 1 = 1$. (Contrast: from 0^- , $\sin x < 0$ forces $\{\sin x\} = \sin x + 1 \rightarrow 1$, giving the *different* one-sided value 2.)

INSIGHT

Whether $\{f(x)\} \rightarrow f(a)$ or jumps depends on which side of an integer f approaches from; here $\cos x \uparrow 1$ keeps $\{\cos x\} \rightarrow 1^-$ instead of dropping to 0.

№18 Two Sawtooths Cancel

● ● ● ● ● L4

$$\lim_{x \rightarrow 1} (\{x\} + \{2 - x\})$$

fractional-part

$x \rightarrow 1$

removable

two-sided

complementary

LIMIT

1

THE TRAP

Substituting $x = 1$ gives $\{1\} + \{2 - 1\} = \{1\} + \{1\} = 0 + 0 = 0$, tempting the answer 0. But the value at the point is irrelevant to a limit: both $\{x\}$ and $\{2 - x\}$ have a jump exactly at $x = 1$, and as $x \rightarrow 1$ one fractional part rises to 1^- while the other falls to 0^+ . The jumps are equal and opposite, so the sum approaches 1 — the function has a removable discontinuity (hole) at $x = 1$ where it dips to 0.

Method I — Two-sided branch analysis

1. Right limit ($x \rightarrow 1^+$, so $1 < x < 2$): $\{x\} = x - 1 \rightarrow 0^+$; and $2 - x \in (0, 1)$ so $\{2 - x\} = 2 - x \rightarrow 1^-$. Sum $\rightarrow 0 + 1 = 1$.
2. Left limit ($x \rightarrow 1^-$, so $0 < x < 1$): $\{x\} = x \rightarrow 1^-$; and $2 - x \in (1, 2)$ so $\{2 - x\} = (2 - x) - 1 = 1 - x \rightarrow 0^+$. Sum $\rightarrow 1 + 0 = 1$.
3. Both one-sided limits equal 1, so the two-sided limit is 1 (while the point value is 0 — a removable discontinuity).

Method II — Complementary identity $\{t\} + \{1 - t\} = 1$

1. For non-integer t , $\{t\} + \{-t\} = 1$. Put $t = x - 1$ (non-integer for $x \neq 1$): then $\{x - 1\} + \{1 - x\} = 1$.
2. Since adding an integer leaves the fractional part unchanged, $\{x\} = \{x - 1\}$ and $\{2 - x\} = \{1 + (1 - x)\} = \{1 - x\}$.
3. Hence $\{x\} + \{2 - x\} = \{x - 1\} + \{1 - x\} = 1$ for every $x \neq 1$, so the limit as $x \rightarrow 1$ is 1.

INSIGHT

The graphs of $\{x\}$ and $\{2 - x\}$ are reflections of each other about the line $x = 1$; their downward unit jumps are equal and opposite, so the sum is the constant 1 off the integers with a removable hole (value 0) at each integer.

№19 One Side Lives, One Side Explodes

● ● ● ● ● L4

$$\lim_{x \rightarrow 0} \frac{\{x\}}{x} \quad (\text{compare the two one-sided limits})$$

fractional-part

$x \rightarrow 0$

one-sided-comparison

non-existence

floor

LIMIT

does not exist: $\lim_{x \rightarrow 0^+} = 1$, $\lim_{x \rightarrow 0^-} = -\infty$

THE TRAP

A student writes $\{x\} = x$ 'near 0', cancels, and answers 1. That identity holds only for $0 \leq x < 1$, i.e. it silently assumes $x > 0$. For $x \rightarrow 0^-$ we have $-1 < x < 0$, so $\lfloor x \rfloor = -1$ and $\{x\} = x + 1 \rightarrow 1^+$; then $\frac{\{x\}}{x} \rightarrow \frac{1}{0^-} = -\infty$. The two sides disagree catastrophically, so the limit does not exist.

Method I — Compute each side from its floor branch

1. Right ($x \rightarrow 0^+$, $0 < x < 1$): $\lfloor x \rfloor = 0$, $\{x\} = x$, so $\frac{\{x\}}{x} = \frac{x}{x} = 1 \rightarrow 1$.
2. Left ($x \rightarrow 0^-$, $-1 < x < 0$): $\lfloor x \rfloor = -1$, $\{x\} = x + 1 \rightarrow 1^+$, while $x \rightarrow 0^-$, so $\frac{\{x\}}{x} \rightarrow \frac{1}{0^-} = -\infty$.
3. The one-sided limits are 1 and $-\infty$; since they differ, the two-sided limit does not exist.

Method II — Substitution exposes the asymmetry

1. Right: set $x = h$, $h \rightarrow 0^+$: $\frac{\{h\}}{h} = \frac{h}{h} = 1$.
2. Left: set $x = -h$, $h \rightarrow 0^+$. For $0 < h < 1$ we have $-1 < -h < 0$, so $\lfloor -h \rfloor = -1$ and $\{-h\} = -h - (-1) = 1 - h$. Hence $\frac{\{-h\}}{-h} = \frac{1 - h}{-h} = -\frac{1}{h} + 1 \rightarrow -\infty$.
3. The substitution shows the jump discontinuity of $\{x\}$ at 0 injects a $+1$ on the left that the vanishing denominator turns into a blow-up. Hence no two-sided limit.

INSIGHT

$\{x\}/x$ is the cleanest demonstration that the sawtooth's jump at an integer destroys two-sidedness: the same algebraic surface gives a finite limit on one side and an infinite one on the other.

№20 The Arctangent Plateau

● ● ● ● ● L5

$$\lim_{x \rightarrow 0^+} \frac{\left\{ \arctan \frac{1}{x} \right\} - \left(\frac{\pi}{2} - 1 \right)}{x}$$

fractional-part

$x \rightarrow 0^+$

$\arctan$

floor

derivative-form

LIMIT

$$-1$$

THE TRAP

A student sees $\arctan \frac{1}{x} \rightarrow \frac{\pi}{2} \approx 1.571$, takes $\left\{ \arctan \frac{1}{x} \right\} \rightarrow \frac{\pi}{2} - 1$, declares the numerator $\rightarrow 0$ and the answer 0 — or, mishandling the floor, thinks $\arctan \frac{1}{x} < 1$ so the fractional part is just $\arctan \frac{1}{x}$ and the floor never matters. Both miss that this is a $0/0$ derivative-form whose value is the slope -1 , and that the floor is pinned to 1 (not 0).

Method I — Pin the floor, then expand

1. For small $x > 0$, $\arctan \frac{1}{x}$ lies in $(1, \frac{\pi}{2}) \subset (1, 2)$, so $\lfloor \arctan \frac{1}{x} \rfloor = 1$ and $\left\{ \arctan \frac{1}{x} \right\} = \arctan \frac{1}{x} - 1$.
2. Use the identity $\arctan \frac{1}{x} = \frac{\pi}{2} - \arctan x$ (valid for $x > 0$). Then the numerator becomes $\left(\frac{\pi}{2} - \arctan x - 1 \right) - \left(\frac{\pi}{2} - 1 \right) = -\arctan x$.
3. So the expression is $\frac{-\arctan x}{x}$, and since $\arctan x \sim x$,

$$\lim_{x \rightarrow 0^+} \frac{-\arctan x}{x} = -1.$$

Method II — Series after removing the floor

1. As above, on $(0, \delta)$ the floor is 1, so the numerator equals $\arctan \frac{1}{x} - 1 - \left(\frac{\pi}{2} - 1 \right) = \arctan \frac{1}{x} - \frac{\pi}{2}$.
2. For $x \rightarrow 0^+$, expand $\arctan \frac{1}{x} = \frac{\pi}{2} - x + \frac{x^3}{3} - \dots$ (from $\arctan \frac{1}{x} = \frac{\pi}{2} - \arctan x$ and the $\arctan$ series).
3. Hence numerator $= -x + \frac{x^3}{3} - \dots$, and dividing by x gives $-1 + \frac{x^2}{3} - \dots \rightarrow -1$.

INSIGHT

Because $\arctan \frac{1}{x}$ approaches its plateau $\frac{\pi}{2}$ from below while sitting in $(1, 2)$, the floor is frozen at 1 and the fractional part inherits the smooth slope $\frac{d}{dx} \arctan \frac{1}{x} \Big|_{0^+} = -1$.

CHAPTER III

The Signum Function

sgn

A function with only three answers – and it changes its mind.

№21 Sign of a Vanishing Gap

●●●●● L4

$$\lim_{x \rightarrow 0^+} \operatorname{sgn}(x - \sin x) \cdot \frac{1 - \cos x}{x^2}$$

signum

one-sided

$x \rightarrow 0^+$

taylor

1-cos

LIMIT

$$\frac{1}{2}$$

THE TRAP

Reading $\operatorname{sgn}(x - \sin x)$ as $\operatorname{sgn}(0) = 0$ because both x and $\sin x$ vanish at $x = 0$, and concluding the whole limit is 0. This evaluates the sign at the endpoint instead of along the approach. The signum must be read on the punctured neighbourhood: for every $x > 0$ near 0 we have $x - \sin x = \frac{x^3}{6} - \dots > 0$, so the factor is a constant $+1$, never 0. (Equally seductive: noticing that from the left it would be -1 and declaring the limit nonexistent — but the problem is posed one-sided, so only $x \rightarrow 0^+$ matters.)

Method I - Series sign analysis + squeeze

1. Expand the argument of sgn : $x - \sin x = x - \left(x - \frac{x^3}{6} + \dots\right) = \frac{x^3}{6} - \frac{x^5}{120} + \dots$.
2. For all small $x > 0$ this is > 0 (leading term $x^3/6 > 0$), hence $\operatorname{sgn}(x - \sin x) = +1$ on a punctured right-neighbourhood of 0.
3. The sign factor is now the constant $+1$, so the limit reduces to $\lim_{x \rightarrow 0^+} \frac{1 - \cos x}{x^2}$.
4. Using $1 - \cos x = \frac{x^2}{2} - \frac{x^4}{24} + \dots$, we get $\frac{1 - \cos x}{x^2} = \frac{1}{2} - \frac{x^2}{24} + \dots \rightarrow \frac{1}{2}$.
5. Therefore the limit is $\boxed{\frac{1}{2}}$.

Method II - Half-angle identity

1. On small $x > 0$ the sign factor is $+1$ (since $\sin x < x$ there), so we only need $\frac{1 - \cos x}{x^2}$.
2. Use the identity $1 - \cos x = 2 \sin^2 \frac{x}{2}$.
3.
$$\frac{1 - \cos x}{x^2} = \frac{2 \sin^2(x/2)}{x^2} = \frac{1}{2} \left(\frac{\sin(x/2)}{x/2} \right)^2.$$

4. As $x \rightarrow 0^+$, $\frac{\sin(x/2)}{x/2} \rightarrow 1$, so the expression tends to $\frac{1}{2} \cdot 1^2 = \frac{1}{2}$.

INSIGHT

The whole difficulty is realising sgn must be evaluated along the path, not at the limit point: $x - \sin x$ is a one-signed $O(x^3)$ quantity, so its sign is locally constant (+1 on the right, -1 on the left) even though it 'looks' indeterminate. This is exactly why the one-sided posing is essential – the two-sided limit would not exist.

№22 Sign Guarding a Difference of Squares

●●●●● L3

$$\lim_{x \rightarrow 2^-} \text{sgn}(4 - x^2) \cdot \frac{\arctan(x - 2)}{x - 2}$$

signum

one-sided

$x \rightarrow 2^-$

arctan

factoring

LIMIT

1

THE TRAP

The careless student answers 1 from $\frac{\arctan(x - 2)}{x - 2} \rightarrow 1$ while ignoring the sign factor. The careful student over-corrects: $4 - x^2$ flips sign at $x = 2$, so they call the limit nonexistent. But it is one-sided: as $x \rightarrow 2^-$, $x < 2$ gives $x^2 < 4$, so $4 - x^2 > 0$ and $\text{sgn}(4 - x^2) = +1$, making the limit 1 (from the right it would be -1).

Method I - One-sided sign plus standard arctan limit

- For $x \rightarrow 2^-$ we have $x \in (0, 2)$ so $x^2 < 4$, giving $4 - x^2 > 0$ and $\text{sgn}(4 - x^2) = +1$; the sign factor is locally constant.
- Let $t = x - 2 \rightarrow 0^-$. The expression becomes $\frac{\arctan t}{t}$.
- Since $\lim_{t \rightarrow 0} \frac{\arctan t}{t} = 1$ and $\frac{\arctan t}{t}$ is even, the left value as $t \rightarrow 0^-$ equals the two-sided value, namely 1.
- Multiplying by +1 gives $\boxed{1}$.

Method II - Series of arctan

- Sign factor = +1 as established. Put $t = x - 2 \rightarrow 0^-$.
- Use $\arctan t = t - \frac{t^3}{3} + \frac{t^5}{5} - \dots$ for $|t| < 1$.
- Then $\frac{\arctan t}{t} = 1 - \frac{t^2}{3} + \frac{t^4}{5} - \dots \rightarrow 1$ as $t \rightarrow 0^-$.
- Limit = 1.

№23 The Hidden Signum

●●●●● L5

$$\lim_{x \rightarrow 0^+} \frac{2}{\pi} \arctan\left(\frac{1}{x - \tan x}\right)$$

signum

hidden

arctan

tan

one-sided

$x \rightarrow 0^+$

composition

LIMIT

−1

THE TRAP

The fast guess is +1: students see $x - \tan x \rightarrow 0$, recall the familiar $x - \sin x = \frac{x^3}{6} + \dots > 0$, and assume the denominator is positive, so the fraction $\rightarrow +\infty$ and $\frac{2}{\pi} \arctan(+\infty) = +1$. But tangent overshoots: $\tan x = x + \frac{x^3}{3} + \dots > x$ for small $x > 0$, so $x - \tan x = -\frac{x^3}{3} - \dots < 0$. The fraction $\rightarrow -\infty$ and the true value is $\frac{2}{\pi} \arctan(-\infty) = -1$. The sign flips precisely because $\tan$ bulges above the line $y = x$, unlike $\sin$.

Method I - $\arctan(\pm\infty)$ as a disguised signum

1. Recall $\frac{2}{\pi} \arctan(u) \rightarrow \text{sgn}(u)$ as $|u| \rightarrow \infty$; here $\arctan$ is secretly acting as a signum detector.
2. So the whole limit equals $\text{sgn}\left(\frac{1}{x - \tan x}\right) = \text{sgn}(x - \tan x)$ in the limit, provided $x - \tan x \rightarrow 0$.
3. Sign of the denominator: $\tan x = x + \frac{x^3}{3} + \dots$, so $x - \tan x = -\frac{x^3}{3} - \dots < 0$ for all small $x > 0$.
4. Thus $\frac{1}{x - \tan x} \rightarrow -\infty$, and $\frac{2}{\pi} \arctan(-\infty) = \frac{2}{\pi} \cdot \left(-\frac{\pi}{2}\right) = -1$.
5. Answer: −1.

Method II - Direct substitution chase

1. Let $D(x) = x - \tan x$. For $x \rightarrow 0^+$, $D(x) \rightarrow 0^-$ (negative, by the cubic expansion above).
2. Hence $u(x) = \frac{1}{D(x)} \rightarrow -\infty$.
3. $\arctan$ is continuous and increasing with horizontal asymptote $-\frac{\pi}{2}$ as $u \rightarrow -\infty$, so $\arctan(u) \rightarrow -\frac{\pi}{2}$.
4. Multiply by $\frac{2}{\pi}$: limit = −1.

INSIGHT

The signum is fully camouflaged inside $\frac{2}{\pi} \arctan$, whose two horizontal asymptotes ± 1 ARE the two signum values; the real test is the sign of the cubic $x - \tan x$.

№24 Sign of the Tangent Excess

●●●●● L4

$$\lim_{x \rightarrow 0^+} \text{sgn}(\tan x - \sin x) \cdot \frac{1 - \cos x}{x \tan x}$$

signum

$x \rightarrow 0^+$

one-sided

$\tan$

$\sin$

$1 - \cos$

LIMIT

$\frac{1}{2}$

THE TRAP

Seeing $\tan x - \sin x \rightarrow 0$ and concluding $\text{sgn}(\tan x - \sin x) \rightarrow \text{sgn}(0) = 0$, which would force the whole limit to 0. But sgn depends on the sign of the argument near 0, not on its limiting value: $\tan x - \sin x = \sin x(\sec x - 1) > 0$

for all small $x > 0$, so the factor is identically $+1$ on a punctured right-neighborhood and the limit is $\frac{1}{2}$, not 0.

Method I - Factor the sign argument

1. Factor: $\tan x - \sin x = \sin x \left(\frac{1}{\cos x} - 1 \right) = \frac{\sin x (1 - \cos x)}{\cos x}$.
2. For small $x > 0$ all three factors $\sin x$, $1 - \cos x$, $\cos x$ are positive, so $\text{sgn}(\tan x - \sin x) = +1$ on a punctured right-neighborhood of 0.
3. The limit therefore reduces to $\lim_{x \rightarrow 0^+} \frac{1 - \cos x}{x \tan x}$.
4. Write $\frac{1 - \cos x}{x \tan x} = \frac{1 - \cos x}{x^2} \cdot \frac{x}{\tan x}$.
5. As $x \rightarrow 0^+$: $\frac{1 - \cos x}{x^2} \rightarrow \frac{1}{2}$ and $\frac{x}{\tan x} \rightarrow 1$, so the product $\rightarrow \frac{1}{2}$.

Method II - Half-angle on numerator, small-angle on denominator

1. Sign factor = $+1$ on a right-neighborhood of 0 (as established in Method I).
2. Use $1 - \cos x = 2 \sin^2 \frac{x}{2}$ and $\tan x = \frac{\sin x}{\cos x}$, so $\frac{1 - \cos x}{x \tan x} = \frac{2 \sin^2(x/2) \cos x}{x \sin x}$.
3. Replace $\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$:
4.
$$\frac{2 \sin^2(x/2) \cos x}{x \cdot 2 \sin(x/2) \cos(x/2)} = \frac{\sin(x/2) \cos x}{x \cos(x/2)} = \frac{1}{2} \cdot \frac{\sin(x/2)}{x/2} \cdot \frac{\cos x}{\cos(x/2)}$$
5. Each factor tends to its limit: $\frac{1}{2} \cdot 1 \cdot 1 = \frac{1}{2}$.

INSIGHT

$\tan x - \sin x = \frac{x^3}{2} + O(x^5)$ is positive and cubic-small; the lesson is that a quantity vanishing in the limit still has a definite, locally constant sign – sgn reads that local sign, not the limit value.

№25 Signum at the Horizon

●●●●● L3

$$\lim_{x \rightarrow +\infty} \text{sgn} \left(\sin \frac{1}{x} \right) \cdot \arctan x$$

signum

$x \rightarrow +\infty$

sin

arctan

composition

LIMIT

$$\frac{\pi}{2}$$

THE TRAP

Pattern-matching on $\text{sgn}(\sin(\cdot))$: "sin oscillates, so its sign flips forever and the limit cannot exist." The flaw is that the oscillation lives in the argument, and here the argument is $1/x \rightarrow 0^+$, which stays trapped in $(0, \pi)$ where $\sin$ is strictly positive. The sign never gets to flip even once, so the factor is the constant $+1$ and the limit is simply $\lim \arctan x = \frac{\pi}{2}$.

Method I - Localise the argument

1. As $x \rightarrow +\infty$, $\frac{1}{x} \rightarrow 0^+$, so eventually $0 < \frac{1}{x} < \pi$.

2. On $(0, \pi)$, $\sin$ is strictly positive, hence $\sin \frac{1}{x} > 0$ and $\operatorname{sgn}(\sin \frac{1}{x}) = +1$ for all large x .

3. The sign factor is therefore eventually the constant $+1$.

4. Thus the limit equals $\lim_{x \rightarrow +\infty} \arctan x = \frac{\pi}{2}$.

5. Answer: $\boxed{\frac{\pi}{2}}$.

Method II - Explicit tail threshold

1. For $0 < u < \pi$ we have $\sin u > 0$. Take $u = \frac{1}{x}$: the condition $0 < \frac{1}{x} < \pi$ is exactly $x > \frac{1}{\pi}$.

2. Hence $\operatorname{sgn}(\sin \frac{1}{x}) = 1$ for every $x > \frac{1}{\pi}$ — a tail condition, which is all a limit at $+\infty$ depends on.

3. On that entire tail the expression equals $\arctan x$ identically (no zero crossing of $\sin \frac{1}{x}$ occurs there).

4. Therefore the limit is $\lim_{x \rightarrow +\infty} \arctan x = \frac{\pi}{2}$.

INSIGHT

Oscillation of $\operatorname{sgn} \circ \sin$ only happens when the argument sweeps across many half-periods; squeezing the argument to 0^+ pins the sign down to $+1$ on a whole tail.

№26 Sign of the Sine Across an Even Integer

●●●●● L4

$$\lim_{x \rightarrow 2^-} \operatorname{sgn}(\sin(\pi x)) \cdot \frac{2^x - 4}{x - 2}$$

signum

$x \rightarrow 2^-$

one-sided

$\sin(\pi x)$

exponential

$\ln$

LIMIT

$$-4 \ln 2$$

THE TRAP

The quotient $\frac{2^x - 4}{x - 2}$ is the difference quotient of 2^x at $x = 2$, so it tends to $2^2 \ln 2 = 4 \ln 2$. The seductive slip is to stop there and report $4 \ln 2$, treating $\operatorname{sgn}(\sin \pi x)$ as harmlessly $+1$ since $\sin \pi x \rightarrow 0$. But the side matters: as $x \rightarrow 2^-$, $\pi x \rightarrow 2\pi^-$ and $\sin(2\pi - \varepsilon) = -\sin \varepsilon < 0$, so $\operatorname{sgn}(\sin \pi x) = -1$ throughout the left neighbourhood, flipping the answer to $-4 \ln 2$.

Method I - Sign from the sine's left approach + derivative limit

1. Near $x = 2^-$, write $x = 2 - \varepsilon$ with $\varepsilon \rightarrow 0^+$. Then $\sin(\pi x) = \sin(2\pi - \pi\varepsilon) = -\sin(\pi\varepsilon) < 0$.

2. Hence $\operatorname{sgn}(\sin(\pi x)) = -1$ on the left of 2, and this is a constant factor as $x \rightarrow 2^-$.

3. The remaining factor is the difference quotient of $f(x) = 2^x$ at $x = 2$: $\lim_{x \rightarrow 2} \frac{2^x - 2^2}{x - 2} = f'(2) = 2^2 \ln 2 = 4 \ln 2$.

4. Multiply by the sign: $(-1) \cdot 4 \ln 2 = -4 \ln 2$.

5. Answer: $\boxed{-4 \ln 2}$.

Method II - Substitution and exponential expansion

1. Sign factor = -1 as established above. Put $t = x - 2 \rightarrow 0^-$, so $2^x = 4 \cdot 2^t = 4e^{t \ln 2}$.

$$2. \quad \frac{2^x - 4}{x - 2} = \frac{4(e^{t \ln 2} - 1)}{t} = 4 \ln 2 \cdot \frac{e^{t \ln 2} - 1}{t \ln 2}.$$

3. As $t \rightarrow 0^-$, $\frac{e^{t \ln 2} - 1}{t \ln 2} \rightarrow 1$, so the quotient $\rightarrow 4 \ln 2$.

4. Including the sign factor -1 : the limit is $-4 \ln 2$.

INSIGHT

At an even integer $\sin(\pi x)$ crosses zero downward from the left; the sgn silently flips the otherwise textbook derivative quotient. The one-sided posing is essential: from the right the limit is $+4 \ln 2$, so the two-sided limit does not exist.

№27 Inverse-Trig Tie-Breaker

● ● ● ● ● L5

$$\lim_{x \rightarrow 0^+} \text{sgn}(\arcsin x - \arctan x) \cdot \frac{\arcsin x - \arctan x}{x^3}$$

signum

$x \rightarrow 0^+$

one-sided

arcsin

arctan

series

composition

LIMIT

$$\frac{1}{2}$$

THE TRAP

Naive first-order expansions $\arcsin x \approx x$ and $\arctan x \approx x$ give $\arcsin x - \arctan x \approx 0$, so the numerator looks like 0 and the whole limit seems to be 0. The x^1 terms are precisely the part that cancels; the entire answer lives in the cubic terms ($+\frac{x^3}{6}$ vs $-\frac{x^3}{3}$), which combine to $+\frac{x^3}{2}$ and divide the x^3 exactly. The sgn is not decoration: it certifies the numerator is positive for small $x > 0$, but it does not change the magnitude.

Method I - Cubic Taylor terms

1. Expand to cubic order: $\arcsin x = x + \frac{x^3}{6} + O(x^5)$ and $\arctan x = x - \frac{x^3}{3} + O(x^5)$.
2. Subtract; the x terms cancel exactly: $\arcsin x - \arctan x = \left(\frac{x^3}{6} + \frac{x^3}{3}\right) + O(x^5) = \frac{x^3}{2} + O(x^5)$.
3. For small $x > 0$ the head $\frac{x^3}{2}$ is positive, so $\text{sgn}(\arcsin x - \arctan x) = +1$.
4. Then $\frac{\arcsin x - \arctan x}{x^3} = \frac{x^3/2 + O(x^5)}{x^3} = \frac{1}{2} + O(x^2) \rightarrow \frac{1}{2}$.
5. Multiplying by the sign factor $+1$, the limit is $\frac{1}{2}$.

Method II - Sign by monotonicity, value by L'Hopital

1. Let $g(x) = \arcsin x - \arctan x$. Then $g'(x) = \frac{1}{\sqrt{1-x^2}} - \frac{1}{1+x^2}$. For $x \in (0, 1)$ we have $\frac{1}{\sqrt{1-x^2}} > 1 > \frac{1}{1+x^2}$, so $g'(x) > 0$; with $g(0) = 0$ this gives $g(x) > 0$, hence $\text{sgn } g = +1$ on $(0, 1)$.
2. Evaluate $L = \lim_{x \rightarrow 0^+} \frac{g(x)}{x^3}$, a $\frac{0}{0}$ form. Differentiating numerator and denominator: $\frac{g'(x)}{3x^2} = \frac{1}{3x^2} \left(\frac{1}{\sqrt{1-x^2}} - \frac{1}{1+x^2} \right)$.

3. Use $\frac{1}{\sqrt{1-x^2}} = 1 + \frac{1}{2}x^2 + O(x^4)$ and $\frac{1}{1+x^2} = 1 - x^2 + O(x^4)$, so the bracket is $\frac{3}{2}x^2 + O(x^4)$, giving $\frac{g'(x)}{3x^2} = \frac{\frac{3}{2}x^2 + O(x^4)}{3x^2} \rightarrow \frac{1}{2}$. Hence $L = \frac{1}{2}$.
4. Therefore the original limit is $(+1) \cdot \frac{1}{2} = \frac{1}{2}$.

INSIGHT

The sgn certifies that the cubic head $+\frac{x^3}{2}$ controls the sign; the trap is the universal x -cancellation that makes naive first-order Taylor read 0. Note that on $x \rightarrow 0^-$ the same factor $\text{sgn } g$ flips with g , so the two-sided limit also equals $\frac{1}{2}$ -- the one-sided form is used only for a clean, well-posed statement.

№28 Signum at Negative Infinity

●●●●● L4

$$\lim_{x \rightarrow -\infty} \text{sgn}(x) \cdot (\sqrt{x^2 + x} + x)$$

signum

$x \rightarrow -\infty$

conjugate

surd

$\text{sgn}(x)$

LIMIT

$$\frac{1}{2}$$

THE TRAP

Treating it like $x \rightarrow +\infty$: rationalise $\sqrt{x^2 + x} + x = \frac{x}{\sqrt{x^2 + x} - x} \rightarrow \frac{1}{2}$ (using $\sqrt{x^2 + x} \approx +x$), then take $\text{sgn}(x) = +1$, declaring the limit $\frac{1}{2}$. Both moves are wrong: at $-\infty$ one needs $\sqrt{x^2 + x} = |x|\sqrt{1 + \frac{1}{x}} = -x\sqrt{1 + \frac{1}{x}}$, so the bracket actually tends to $-\frac{1}{2}$, and $\text{sgn}(x) = -1$. The seduction is that the two sign errors cancel and the careless student still writes $\frac{1}{2}$ — the right number for entirely wrong reasons, so the slip never gets caught.

Method I - Honest surd handling at $-\infty$

- For $x < 0$ we have $|x| = -x$, so $\sqrt{x^2 + x} = |x|\sqrt{1 + \frac{1}{x}} = -x\sqrt{1 + \frac{1}{x}}$ (the surd is real because $x^2 + x = x(x+1) > 0$ once $x < -1$).
- Hence the bracket is $-x\sqrt{1 + \frac{1}{x}} + x = x\left(1 - \sqrt{1 + \frac{1}{x}}\right)$.
- Expand $\sqrt{1 + \frac{1}{x}} = 1 + \frac{1}{2x} - \frac{1}{8x^2} + \dots$, so $1 - \sqrt{1 + \frac{1}{x}} = -\frac{1}{2x} + \frac{1}{8x^2} - \dots$.
- Multiply by x : bracket $= -\frac{1}{2} + \frac{1}{8x} - \dots \rightarrow -\frac{1}{2}$ as $x \rightarrow -\infty$.
- Since $\text{sgn}(x) = -1$ for $x \rightarrow -\infty$, the product is $(-1) \cdot \left(-\frac{1}{2}\right) = \frac{1}{2}$.
- Answer: $\boxed{\frac{1}{2}}$.

Method II - Conjugate with correct sign of the surd

- Multiply and divide by the conjugate: $\sqrt{x^2 + x} + x = \frac{(x^2 + x) - x^2}{\sqrt{x^2 + x} - x} = \frac{x}{\sqrt{x^2 + x} - x}$.
- For $x < 0$, $\sqrt{x^2 + x} = -x\sqrt{1 + \frac{1}{x}}$, so the denominator is $\sqrt{x^2 + x} - x = -x\left(\sqrt{1 + \frac{1}{x}} + 1\right)$.

3. Hence the bracket $= \frac{x}{-x(\sqrt{1+1/x}+1)} = \frac{-1}{\sqrt{1+1/x}+1} \rightarrow \frac{-1}{2}$ as $x \rightarrow -\infty$.

4. Multiply by $\text{sgn}(x) = -1$: the limit is $(-1) \cdot \left(-\frac{1}{2}\right) = \frac{1}{2}$.

INSIGHT

The $\text{sgn}(x)$ factor is honest bait: doing the surd correctly at $-\infty$ forces $|x| = -x$, and the explicit signum makes the two minus signs visible instead of letting them cancel silently. Note $\lim_{x \rightarrow +\infty} \text{sgn}(x)(\sqrt{x^2+x}+x) = +\infty$, so the one-sided framing at $-\infty$ is essential.

№29 Sign of a Fourth-Order Remnant

●●●●● L5

$$\lim_{x \rightarrow 0^+} \text{sgn}\left(\cos x - 1 + \frac{x^2}{2}\right) \cdot \frac{x^4}{\cos x - 1 + \frac{x^2}{2}}$$

signum

$x \rightarrow 0^+$

one-sided

cosine

fourth-order

series

LIMIT

24

THE TRAP

Approximating $\cos x \approx 1 - \frac{x^2}{2}$ makes the inner expression ≈ 0 , so the denominator looks like 0 and the ratio like $\pm\infty$. That second-order truncation is exactly the part designed to cancel; the surviving term is $+\frac{x^4}{24}$, which is finite and positive, so the signum is +1 and the limit is the finite value 24.

Method I - Fourth-order expansion

1. Expand to fourth order: $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \dots$.
2. So $\cos x - 1 + \frac{x^2}{2} = \frac{x^4}{24} - \frac{x^6}{720} + \dots$.
3. For small $x > 0$ this is positive (leading term $x^4/24 > 0$), so the sgn factor is +1.
4. Then $\frac{x^4}{\cos x - 1 + x^2/2} = \frac{x^4}{\frac{x^4}{24}\left(1 - \frac{x^2}{30} + \dots\right)} = \frac{24}{1 - \frac{x^2}{30} + \dots} \rightarrow 24$.
5. Answer: 24.

Method II - Repeated L'Hopital on the reciprocal

1. Sign factor is +1 (Method I). Compute $M = \lim_{x \rightarrow 0^+} \frac{\cos x - 1 + x^2/2}{x^4}$ — a 0/0 form.
2. Differentiate numerator and denominator four times (each stage stays 0/0): numerator derivatives are $-\sin x + x$, $-\cos x + 1$, $\sin x$, $\cos x$, all vanishing at 0 through the third; denominators $4x^3$, $12x^2$, $24x$, 24 .
3. At $x \rightarrow 0^+$ the fourth stage gives $M = \frac{\cos 0}{24} = \frac{1}{24}$.
4. The required limit is the reciprocal: $\frac{x^4}{\cos x - 1 + x^2/2} \rightarrow \frac{1}{M} = 24$.
5. With sign factor +1, the answer is 24.

INSIGHT

A textbook 'engineered cancellation': subtracting the $1 - \frac{x^2}{2}$ part of $\cos x$ exposes the $\frac{x^4}{24}$ remnant, and sgn verifies that remnant is positive.

№30 Floor Trap Beside the Signum

●●●●● L4

$$\lim_{x \rightarrow 0^+} \left(\operatorname{sgn}(x) + \lfloor \cos x \rfloor \right) \cdot \frac{e^x - 1}{x}$$

signum

floor

$x \rightarrow 0^+$

one-sided

exponential

composition

LIMIT

1

THE TRAP

Three seductions all give the wrong reading. (1) Evaluating the bracket literally at $x = 0$: $\operatorname{sgn}(0) + \lfloor \cos 0 \rfloor = 0 + \lfloor 1 \rfloor = 1$ — right number, wrong reason (the expression isn't even defined at 0). (2) Plugging $\cos x \rightarrow 1$ and reading $\lfloor 1 \rfloor = 1$ with $\operatorname{sgn}(x) = 1$, summing to 2 — this misuses continuity inside a floor. (3) A naive numerical check is treacherous: in double precision $\cos x$ rounds to exactly 1.0 for $|x| \lesssim 10^{-8}$, so a calculator reports $\lfloor \cos x \rfloor = 1$ and the limit 2. The truth: for $x > 0$ near 0, $\cos x \in (0, 1)$ strictly, so $\lfloor \cos x \rfloor = 0$ (never 1), $\operatorname{sgn}(x) = 1$, bracket = 1, and the limit is 1.

Method I - Evaluate floor and sgn along the path

1. For $x \rightarrow 0^+$ we have $x > 0$, so $\operatorname{sgn}(x) = +1$.
2. For $0 < x < \frac{\pi}{2}$ the cosine is strictly between 0 and 1: $0 < \cos x < 1$. In particular $\cos x < 1$ because the value 1 is attained only at $x = 0$. Hence $\lfloor \cos x \rfloor = 0$.
3. Therefore the bracket equals $1 + 0 = 1$ on a punctured right-neighbourhood of 0.
4. The remaining factor is the classic limit $\lim_{x \rightarrow 0^+} \frac{e^x - 1}{x} = 1$.
5. Therefore the limit is $1 \cdot 1 = \boxed{1}$.

Method II - Series and continuity bookkeeping

1. $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots$, so $1 - \cos x = \frac{x^2}{2} + O(x^4) > 0$ for every $x \neq 0$; thus $\cos x \in (0, 1)$ near 0^+ and $\lfloor \cos x \rfloor = 0$.
2. Hence the bracket is identically $\operatorname{sgn}(x) = 1$ for small $x > 0$.
3. Expand $e^x - 1 = x + \frac{x^2}{2} + \dots$, so $\frac{e^x - 1}{x} = 1 + \frac{x}{2} + \dots \rightarrow 1$.
4. Product $\rightarrow 1 \cdot 1 = 1$.

INSIGHT

Two endpoint traps stacked on a removable singularity: $\lfloor \cos x \rfloor$ equals 0 for every $x \neq 0$ but jumps to 1 exactly at $x = 0$, and sgn is discontinuous at 0. The one-sided limit never sees either value at 0. (From the left the bracket is -1 , so the two-sided limit does not exist — the $x \rightarrow 0^+$ restriction is essential.)

CHAPTER IV

Absolute Value & Corners



Every corner is a trap. Split the limit before it splits you.

№31 The Cube That Forgives the Kink

●●●●● L4

$$\lim_{x \rightarrow 0} \frac{|x| - |\sin x|}{|x|^3}$$

modulus

sin

taylor

$x \rightarrow 0$

two-sided

cube

LIMIT

$$\frac{1}{6}$$

THE TRAP

Writing $\frac{x - \sin x}{x^3} \rightarrow \frac{1}{6}$ by 'dropping the bars' gives $\frac{1}{6}$ — the right number, but for the WRONG reason: with $|x|^3$ in the denominator (not x^3), the unbarred version $\frac{|x| - |\sin x|}{x^3}$ actually has limit $-\frac{1}{6}$ from the left, so the kink matters and the denominator's modulus is what rescues two-sidedness.

Method I — Reduce to the positive ray via parity

- Both $|x|$ and $|\sin x|$ are **even** functions, and the denominator $|x|^3$ is even. Hence the whole quotient is even, so the left- and right-hand behaviours are identical and a two-sided limit exists iff the right-hand one does.
- For $x > 0$ we have $|x| = x$ and $|\sin x| = \sin x$ (since $\sin x > 0$ near 0^+), and $|x|^3 = x^3$. The expression becomes $\frac{x - \sin x}{x^3}$.
- Use the Maclaurin series $\sin x = x - \frac{x^3}{6} + O(x^5)$, so $x - \sin x = \frac{x^3}{6} + O(x^5)$.
- $$\frac{x - \sin x}{x^3} = \frac{1}{6} + O(x^2) \xrightarrow{x \rightarrow 0^+} \frac{1}{6}.$$
- By evenness the left limit equals the right limit, so the two-sided limit is $\boxed{\frac{1}{6}}$.

Method II — Squeeze on the barred difference

- For all real x , $|x| - |\sin x| \geq 0$ because $|\sin x| \leq |x|$ (a standard inequality), so the numerator is nonnegative everywhere; the sign is settled and only size matters.

2. Rationalize by pairing $|x| - |\sin x|$ with its conjugate $|x| + |\sin x|$:

$$|x| - |\sin x| = \frac{|x|^2 - |\sin x|^2}{|x| + |\sin x|} = \frac{x^2 - \sin^2 x}{|x| + |\sin x|}.$$

3. Now $x^2 - \sin^2 x = (x - \sin x)(x + \sin x)$, and as $x \rightarrow 0$, $x - \sin x \sim \frac{x^3}{6}$ while $x + \sin x \sim 2x$, giving $x^2 - \sin^2 x \sim \frac{x^4}{3}$.

4. The denominator $|x| + |\sin x| \sim 2|x|$. Therefore

$$\frac{|x| - |\sin x|}{|x|^3} \sim \frac{x^4/3}{2|x| \cdot |x|^3} = \frac{x^4/3}{2|x|^4} = \frac{1}{6}.$$

5. Since $x^4/|x|^4 = 1$ for every $x \neq 0$, the result $\frac{1}{6}$ holds from both sides.

INSIGHT

The denominator $|x|^3$ (even) is the secret hero: it turns an odd-numerator near-corner into a clean even quotient. Swap it for x^3 and the limit dies.

№32 A Factor Hidden in the Bars

●●●●● L3

$$\lim_{x \rightarrow 2} \frac{|x-2| + |x^2-4|}{|x-2|}$$

modulus

factoring

$x \rightarrow 2$

corner

two-sided

LIMIT

5

THE TRAP

The seductive wrong answer is -3 . A student evaluating the left limit sees $x \rightarrow 2^-$, correctly writes $|x-2| = -(x-2)$, but then forgets that x^2-4 is *also* negative on $(-2, 2)$ and keeps $|x^2-4| = x^2-4$. The quotient collapses to $1 + \frac{x^2-4}{-(x-2)} = 1 - (x+2) \rightarrow 1-4 = -3$, and the two-sided limit appears not to exist. The fix: *both* moduli flip sign together on the left because they share the root $x=2$, so the kinks cancel. Cleanest route: $\frac{|x^2-4|}{|x-2|} = |x+2|$ (not $x+2$), giving $1 + |x+2| \rightarrow 5$.

Method I — Modulus is multiplicative

1. Factor inside the second bar: $x^2-4 = (x-2)(x+2)$, hence $|x^2-4| = |x-2| \cdot |x+2|$ since $|ab| = |a| |b|$.

2. For $x \neq 2$ the common factor $|x-2| > 0$ cancels:

$$\frac{|x-2| + |x-2| |x+2|}{|x-2|} = 1 + |x+2|.$$

3. The function $|x+2|$ has its only kink at $x = -2$, so it is continuous at $x = 2$; thus $|x+2| \rightarrow |4| = 4$.

4. Therefore the limit is $1 + 4 = \boxed{5}$. No one-sided split is needed because $1 + |x+2|$ is continuous near $x = 2$.

Method II — One-sided check to be safe

1. Right side $x \rightarrow 2^+$: here $x - 2 > 0$ so $|x - 2| = x - 2$, and $x^2 - 4 > 0$ so $|x^2 - 4| = x^2 - 4$. Quotient = $\frac{(x - 2) + (x^2 - 4)}{x - 2} = 1 + \frac{x^2 - 4}{x - 2} = 1 + (x + 2) \rightarrow 5$.
2. Left side $x \rightarrow 2^-$: now $x - 2 < 0$ so $|x - 2| = -(x - 2)$, and crucially $x^2 - 4 < 0$ on $(-2, 2)$ so $|x^2 - 4| = -(x^2 - 4)$ too. Quotient = $\frac{-(x - 2) - (x^2 - 4)}{-(x - 2)} = 1 + \frac{x^2 - 4}{x - 2} = 1 + (x + 2) \rightarrow 5$.
3. Both one-sided limits equal 5, so the two-sided limit is $\boxed{5}$ — the corner at $x = 2$ is removable because both moduli flip sign together.

INSIGHT

The two moduli share the root $x = 2$, so their sign-flips cancel and the limit is corner-free even though each term has a kink; this coupling is exactly what the -3 trap misses.

№33 When Sine and Tangent Both Bend

● ● ● ● ● L5

$$\lim_{x \rightarrow 0} \frac{|\sin x| - |x|}{|\tan x| - |x|}$$

modulus

sin

tan

$x \rightarrow 0$

two-sided

cancellation

LIMIT

$$-\frac{1}{2}$$

THE TRAP

First-order replacement $\sin x \approx x$, $\tan x \approx x$ gives $\frac{|x| - |x|}{|x| - |x|} = \frac{0}{0}$, so you go to third order. The seductive slip: noting the cubic coefficients have magnitudes $\frac{1}{6}$ and $\frac{1}{3}$ and concluding $+\frac{1}{2}$. This forgets the SIGNS — $\sin x - x = -\frac{x^3}{6} < 0$ while $\tan x - x = +\frac{x^3}{3} > 0$ near 0^+ — so the true ratio is $\frac{-1/6}{+1/3} = -\frac{1}{2}$, not $+\frac{1}{2}$. A second trap is fearing the moduli break two-sidedness; in fact both bars flip together on the left and the ratio is even.

Method I — Parity then series

1. Both $|\sin x| - |x|$ and $|\tan x| - |x|$ are **even** functions (each term is even), so the quotient is even. It therefore suffices to compute the limit as $x \rightarrow 0^+$; the left limit will match automatically.
2. For $x > 0$ (small): $|\sin x| = \sin x$, $|\tan x| = \tan x$, $|x| = x$, so the quotient is $\frac{\sin x - x}{\tan x - x}$.
3. Use the Maclaurin expansions $\sin x - x = -\frac{x^3}{6} + O(x^5)$ and $\tan x - x = \frac{x^3}{3} + O(x^5)$.
4.
$$\frac{\sin x - x}{\tan x - x} = \frac{-x^3/6 + O(x^5)}{x^3/3 + O(x^5)} = \frac{-1/6 + O(x^2)}{1/3 + O(x^2)} \xrightarrow{x \rightarrow 0^+} \frac{-1/6}{1/3} = -\frac{1}{2}.$$
5. By evenness the left-hand limit equals the right-hand limit, so the two-sided limit is $\boxed{-\frac{1}{2}}$.

Method II — Negative-side sign bookkeeping (the trap exposed)

1. Attack $x < 0$ head-on to watch the bars work. Here $\sin x < 0$ so $|\sin x| = -\sin x$, and $|x| = -x$.
2. Numerator = $-\sin x - (-x) = -(\sin x - x) = -(-\frac{x^3}{6} + O(x^5)) = \frac{x^3}{6} + O(x^5)$. (With $x < 0$ this quantity is negative, exactly as $|\sin x| - |x|$ must be.)
3. Denominator: $\tan x < 0$ so $|\tan x| = -\tan x$, giving $-\tan x - (-x) = -(\tan x - x) = -(\frac{x^3}{3} + O(x^5)) = -\frac{x^3}{3} + O(x^5)$.

$$4. \quad \frac{\text{num}}{\text{den}} = \frac{x^3/6 + O(x^5)}{-x^3/3 + O(x^5)} \xrightarrow{x \rightarrow 0^-} \frac{1/6}{-1/3} = -\frac{1}{2}.$$

5. Identical to the right side: $\boxed{-\frac{1}{2}}$. The key insight: BOTH bars flip on the left, so the modulus signs cancel in the ratio — while the intrinsic signs of $\sin x - x$ and $\tan x - x$ stay OPPOSITE, which is what forces the minus.

Method III — Conjugate / rationalise the corners

1. Rationalise each difference of moduli by its 'sum' partner: $|\sin x| - |x| = \frac{\sin^2 x - x^2}{|\sin x| + |x|}$ and $|\tan x| - |x| = \frac{\tan^2 x - x^2}{|\tan x| + |x|}$ (valid since $\sin^2 x = |\sin x|^2$, etc.).
2. As $x \rightarrow 0$, both sums satisfy $|\sin x| + |x| \sim 2|x|$ and $|\tan x| + |x| \sim 2|x|$, so the factor $\frac{|\tan x| + |x|}{|\sin x| + |x|} \rightarrow 1$.
Hence the limit equals $\lim_{x \rightarrow 0} \frac{\sin^2 x - x^2}{\tan^2 x - x^2}$.
3. Factor: $\sin^2 x - x^2 = (\sin x - x)(\sin x + x) \sim \left(-\frac{x^3}{6}\right)(2x) = -\frac{x^4}{3}$ and $\tan^2 x - x^2 = (\tan x - x)(\tan x + x) \sim \left(\frac{x^3}{3}\right)(2x) = \frac{2x^4}{3}$.
4.
$$\lim_{x \rightarrow 0} \frac{\sin^2 x - x^2}{\tan^2 x - x^2} = \frac{-x^4/3}{2x^4/3} = -\frac{1}{2}. \quad \boxed{-\frac{1}{2}}$$

INSIGHT

The moduli are red herrings: they cancel by evenness (both bars flip together off the negative axis). The real engine is that $\sin x - x$ and $\tan x - x$ have third-order coefficients $-\frac{1}{6}$ and $+\frac{1}{3}$ — same magnitude class but OPPOSITE signs — so the magnitude-only guess $+\frac{1}{2}$ is exactly the wrong sign.

№34 The Floor Tames the Modulus

●●●●● L4

$$\lim_{x \rightarrow 3} |x - 3| \left\lfloor \frac{1}{|x - 3|} \right\rfloor$$

modulus floor $x \rightarrow 3$ squeeze two-sided

LIMIT

1

THE TRAP

Seeing $\lfloor 1/|x - 3| \rfloor \rightarrow \infty$ while $|x - 3| \rightarrow 0$, a strong student labels it an indeterminate $0 \cdot \infty$ and, since 'the small factor usually wins,' guesses 0. This ignores that $t \lfloor 1/t \rfloor$ is pinned between $1 - t$ and 1: the floor only ever discards a fractional part less than 1, and after multiplying by $t \rightarrow 0$ that discarded sliver vanishes, leaving exactly 1 — not 0.

Method I — Floor squeeze

1. Let $t = |x - 3|$. For $x \neq 3$ we have $t > 0$, and as $x \rightarrow 3$ from EITHER side, $t \rightarrow 0^+$ — the modulus erases the side, so a single one-sided analysis in t settles the two-sided limit.
2. By the defining property of the floor, $\frac{1}{t} - 1 < \left\lfloor \frac{1}{t} \right\rfloor \leq \frac{1}{t}$.

3. Multiply through by $t > 0$ (inequalities preserved):

$$1 - t < t \left\lfloor \frac{1}{t} \right\rfloor \leq 1.$$

4. As $t \rightarrow 0^+$, the lower bound $1 - t \rightarrow 1$ and the upper bound is the constant 1; by the squeeze theorem $t \lfloor 1/t \rfloor \rightarrow 1$.

5. Since this holds as $t = |x - 3| \rightarrow 0^+$ regardless of side, the two-sided limit is $\boxed{1}$.

Method II — Fractional-part identity

1. With $t = |x - 3| \rightarrow 0^+$, write $\left\lfloor \frac{1}{t} \right\rfloor = \frac{1}{t} - \left\{ \frac{1}{t} \right\}$, where $\{\cdot\} \in [0, 1)$ denotes the fractional part.

2. Then $t \left\lfloor \frac{1}{t} \right\rfloor = t \cdot \frac{1}{t} - t \left\{ \frac{1}{t} \right\} = 1 - t \left\{ \frac{1}{t} \right\}$.

3. Since $0 \leq \{1/t\} < 1$, we get $0 \leq t\{1/t\} < t$, and the right side $t \rightarrow 0^+$, so by squeeze $t\{1/t\} \rightarrow 0$.

4. Hence $t \lfloor 1/t \rfloor = 1 - t\{1/t\} \rightarrow 1 - 0 = \boxed{1}$, independent of the wild oscillation of $\{1/t\}$, and the side-symmetry of $t = |x - 3|$ makes the limit two-sided.

INSIGHT

$t \lfloor 1/t \rfloor \rightarrow 1$ as $t \rightarrow 0^+$ is the canonical floor-squeeze; wrapping $t = |x - 3|$ around it makes the two-sided symmetry automatic, since the modulus collapses both approach directions into $t \rightarrow 0^+$.

№35 Modulus at Negative Infinity

●●●●● L4

$$\lim_{x \rightarrow -\infty} (\sqrt{x^2 + x} - |x|)$$

modulus

infinity

$x \rightarrow -\infty$

conjugate

sqrt

LIMIT

$$-\frac{1}{2}$$

THE TRAP

Pattern-matching to the famous $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + x} - x) = \frac{1}{2}$ and just slapping a sign on: " $|x| = -x$ at $-\infty$, the conjugate gives $\frac{x}{\sqrt{x^2 + x} + |x|} \approx \frac{x}{2|x|} = \frac{x}{2|x|} = \frac{1}{2}$." The error is reading $\frac{x}{2|x|}$ as $+\frac{1}{2}$: for $x \rightarrow -\infty$, $x < 0$ so $\frac{x}{|x|} = -1$, giving $-\frac{1}{2}$. The $|x|$ in the denominator quietly negates the leading term, and the answer is $-\frac{1}{2}$, not $+\frac{1}{2}$.

Method I — Conjugate, keeping signs honest

1. For $x \rightarrow -\infty$ we have $x \leq -1$ eventually, so $x^2 + x > 0$ and the surd is defined; also $|x| = -x > 0$. Rationalise:

$$\sqrt{x^2 + x} - |x| = \frac{(x^2 + x) - x^2}{\sqrt{x^2 + x} + |x|} = \frac{x}{\sqrt{x^2 + x} + |x|}.$$

2. In the denominator factor the surd using $|x| = -x$:

$$\sqrt{x^2 + x} = |x| \sqrt{1 + \frac{1}{x}} = -x \sqrt{1 + \frac{1}{x}}, \quad |x| = -x,$$

so the denominator $= -x(\sqrt{1 + \frac{1}{x}} + 1)$.

3. Therefore

$$\frac{x}{-x(\sqrt{1+1/x}+1)} = \frac{-1}{\sqrt{1+1/x}+1}.$$

4. As $x \rightarrow -\infty$, $\frac{1}{x} \rightarrow 0$, so this $\rightarrow \frac{-1}{1+1} = \boxed{-\frac{1}{2}}$.

Method II — Substitution $x=-t$

1. Let $x = -t$ with $t \rightarrow +\infty$. Then $|x| = t$ and $x^2 + x = t^2 - t$, so the expression becomes $\sqrt{t^2 - t} - t$.

2. Factor and expand (binomial series, valid as $\frac{1}{t} \rightarrow 0$):

$$\sqrt{t^2 - t} = t\sqrt{1 - \frac{1}{t}} = t\left(1 - \frac{1}{2t} - \frac{1}{8t^2} - \dots\right).$$

3. Thus $\sqrt{t^2 - t} - t = \left(t - \frac{1}{2} - \frac{1}{8t} - \dots\right) - t = -\frac{1}{2} - \frac{1}{8t} - \dots$.

4. Let $t \rightarrow +\infty$: the limit is $\boxed{-\frac{1}{2}}$. The substitution turns $|x|$ into the manifestly positive t , so the sign trap disappears.

INSIGHT

At $\pm\infty$ the modulus satisfies $|x| = \mp x$, which flips the sign of the conjugate's leading coefficient: the very same surd $\sqrt{x^2 + x} - |x|$ tends to $+\frac{1}{2}$ as $x \rightarrow +\infty$ but to $-\frac{1}{2}$ as $x \rightarrow -\infty$.

№36 The Self-Cancelling Inner Modulus

●●●●● L4

$$\lim_{x \rightarrow 0^+} \frac{|x + |x||}{x}$$

modulus

nested

$x \rightarrow 0$

one-sided

dne

LIMIT

2

THE TRAP

Reading $x + |x|$ as 'generically $\approx 2x$ ' and writing $\frac{|2x|}{x} \rightarrow 2$ as if the sign of x never mattered. This is right at 0^+ but for a wrong reason: it silently assumes $|x| = x$. On the left $|x| = -x$, so $x + |x| \equiv 0$ identically and the quotient is 0 — the two-sided limit does not exist (left value 0, right value 2). The clean 2 is purchased entirely by the 0^+ specification, not by any cancellation that holds 'generically'.

Method I — Resolve the inner modulus by side

1. On the right, $x > 0 \Rightarrow |x| = x$, so the inner expression collapses to $x + |x| = x + x = 2x$.

2. Then $|x + |x|| = |2x| = 2x$ (since $x > 0$), and the quotient is $\frac{2x}{x} = 2$ for every $x > 0$.

3. Being constantly 2 on $(0, \infty)$,

$$\lim_{x \rightarrow 0^+} \frac{|x + |x||}{x} = 2.$$

4. (Contrast: for $x < 0$, $|x| = -x$, so $x + |x| = 0$, the numerator vanishes, and the quotient is 0 — hence the two-sided limit fails and the 0^+ specification is essential.)

Method II — Signum reformulation

1. For $x \neq 0$, $|x| = x \operatorname{sgn}(x)$, so $x + |x| = x(1 + \operatorname{sgn}(x))$.
2. Since $1 + \operatorname{sgn}(x) \geq 0$ always, $|x + |x|| = |x|(1 + \operatorname{sgn}(x))$.
3. Dividing, the quotient becomes $\frac{|x|(1 + \operatorname{sgn} x)}{x} = \operatorname{sgn}(x)(1 + \operatorname{sgn}(x))$, using $\frac{|x|}{x} = \operatorname{sgn}(x)$.
4. As $x \rightarrow 0^+$, $\operatorname{sgn}(x) = +1$, giving $1 \cdot (1 + 1) = 2$. (As $x \rightarrow 0^-$ it gives $(-1) \cdot (1 - 1) = 0$, confirming the jump.)

INSIGHT

$x + |x| = 2\max(x, 0)$ — a one-sided ramp whose corner at 0 is exactly why a two-sided limit cannot exist; the 0^+ keeps it well-posed.

№37 The Reciprocal Corner of Arctangent

●●●●● L4

$$\lim_{x \rightarrow 0} \frac{\arctan \frac{1}{|x|} - \frac{\pi}{2}}{|x|}$$

modulus

arctan

$x \rightarrow 0$

two-sided

derivative-as-limit

LIMIT

−1

THE TRAP

Reflexively reading it as $\frac{0}{0}$ and reaching for L'Hopital in x : the numerator $\arctan \frac{1}{|x|} - \frac{\pi}{2} \rightarrow 0$ and denominator $\rightarrow 0$, so it looks textbook. But $\frac{d}{dx} \arctan \frac{1}{|x|} = \frac{\operatorname{sgn} x}{1+x^2}$ carries a $\operatorname{sgn} x$ from the corner, and the denominator's derivative is $\operatorname{sgn} x$ too — students who differentiate $|x|$ as x or drop the sign get $+1$, or get a two-sided mismatch and declare the limit nonexistent. Both are wrong: the clean route is the identity $\arctan u + \arctan \frac{1}{u} = \frac{\pi}{2}$, which kills the corner instantly and gives -1 .

Method I — Complementary-angle identity

1. For $t > 0$, $\arctan \frac{1}{t} = \frac{\pi}{2} - \arctan t$. Put $t = |x| > 0$ (valid for all $x \neq 0$, both sides of 0).
2. Then $\arctan \frac{1}{|x|} - \frac{\pi}{2} = -\arctan |x|$, so the expression becomes

$$\frac{-\arctan |x|}{|x|}.$$

3. As $x \rightarrow 0$ (from either side), $|x| \rightarrow 0^+$, and by the standard limit $\frac{\arctan u}{u} \rightarrow 1$ we get $\frac{\arctan |x|}{|x|} \rightarrow 1$.
4. Hence the limit is -1 . It is genuinely two-sided because the whole expression depends only on $|x|$, which approaches 0^+ identically from the left and the right: $\boxed{-1}$.

Method II — Series at the corner

1. Set $t = |x| \rightarrow 0^+$. Using $\arctan \frac{1}{t} = \frac{\pi}{2} - \arctan t$ and $\arctan t = t - \frac{t^3}{3} + \frac{t^5}{5} - \dots$,
2. we get $\arctan \frac{1}{t} - \frac{\pi}{2} = -\left(t - \frac{t^3}{3} + \dots\right) = -t + \frac{t^3}{3} - \dots$.

3. Divide by $t = |x|$: $\frac{-t + t^3/3 - \dots}{t} = -1 + \frac{t^2}{3} - \dots$.

4. Let $t \rightarrow 0^+$: the higher terms vanish and the limit is $\boxed{-1}$. The $|x|$ forces $t > 0$, so no side ambiguity arises (a numeric check confirms the deviation is $\approx t^2/3$).

INSIGHT

$g(x) = \arctan \frac{1}{|x|}$ has a downward corner at 0 with one-sided approach-slopes ∓ 1 (right -1 , left $+1$); dividing by $|x|$ rather than x folds both into the single value -1 , which is exactly why the limit exists two-sided.

№38 A Corner Raised to a Vanishing Power

● ● ● ● ● L5

$$\lim_{x \rightarrow 0} (|\cos x| + |x|)^{1/|x|}$$

modulus cos exponent $x \rightarrow 0$ 1^∞ infinity log

LIMIT

e

THE TRAP

Spotting 1^∞ and computing $\lim_{x \rightarrow 0} \frac{(|\cos x| + |x|) - 1}{|x|}$ as if only $|x|$ mattered, getting $\frac{|x|}{|x|} = 1$ and answering $e^1 = e$ — right answer, but the student who keeps $|\cos x| \approx 1$ exactly drops the $-x^2/2$ correction; it happens to be higher order, but verifying that is the whole point. A student who instead forgets the $+|x|$ term entirely gets $e^0 = 1$, wrong.

Method I — Logarithm and series

1. Let $L = \lim_{x \rightarrow 0} (|\cos x| + |x|)^{1/|x|}$ and take logs: $\ln L = \lim_{x \rightarrow 0} \frac{\ln(|\cos x| + |x|)}{|x|}$.

2. Near 0, $\cos x > 0$ so $|\cos x| = \cos x = 1 - \frac{x^2}{2} + O(x^4)$. Thus the base is $1 + |x| - \frac{x^2}{2} + O(x^4)$.

3. Use $\ln(1 + u) = u - \frac{u^2}{2} + \dots$ with $u = |x| - \frac{x^2}{2} + \dots$:

$$\ln(\text{base}) = |x| - \frac{x^2}{2} - \frac{|x|^2}{2} + \dots = |x| - x^2 + O(x^3),$$

since $|x|^2 = x^2$.

4. Divide by $|x|$: $\frac{|x| - x^2 + \dots}{|x|} = 1 - |x| + \dots \rightarrow 1$.

5. Hence $\ln L = 1$ and $L = \boxed{e}$. Everything is a function of $|x| \rightarrow 0^+$, so the limit is two-sided.

Method II — Pin the base to the standard form

1. Set $t = |x| \rightarrow 0^+$ and let $\varphi(x) = |\cos x| + |x| - 1 = |x| - \frac{x^2}{2} + O(x^4)$, the 'excess over 1'. Then base = $1 + \varphi$.

2. Write $(1 + \varphi)^{1/t} = \exp\left(\frac{1}{t} \ln(1 + \varphi)\right) = \exp\left(\frac{\varphi}{t} \cdot \frac{\ln(1 + \varphi)}{\varphi}\right)$.

3. As $x \rightarrow 0$, $\varphi \rightarrow 0$ so $\frac{\ln(1 + \varphi)}{\varphi} \rightarrow 1$; and $\frac{\varphi}{t} = \frac{|x| - x^2/2 + \dots}{|x|} \rightarrow 1$.

4. Therefore the exponent $\rightarrow 1 \cdot 1 = 1$ and the limit is $\exp(1) = \boxed{e}$. The $-x^2/2$ from $|\cos x|$ is genuinely higher order and vanishes after dividing by $|x|$ — which is exactly why $|x|$, not $\cos x$, sets the answer.

INSIGHT

The corner $|x|$ in the base contributes the only first-order term; the smooth $|\cos x|$ piece is quadratic and washes out, leaving the bare e .

№39 Two Nested Bars, One Clean Slope

●●●●● L3

$$\lim_{x \rightarrow 0} \frac{||x| - x| - ||x| + x|}{x}$$

modulus

nested

$x \rightarrow 0$

two-sided

signum

LIMIT

−2

THE TRAP

Test a tiny positive x : $||x| - x| = 0$ and $||x| + x| = 2x$, so the quotient is $\frac{0-2x}{x} = -2$. Now flip to a tiny negative x and carelessly write $||x| - x| = 0$ again, forgetting that here $|x| = -x$ makes $|x| - x = -2x \neq 0$. That slip gives a phantom left-hand value of $+2$, so the student declares the two sides unequal and the limit DNE. Once the inner bar is handled correctly, the left side also gives -2 , and the limit exists and equals -2 .

Method I — Case split on the side

- **Right, $x > 0$:**** here $|x| = x$. Then $|x| - x = 0 \Rightarrow ||x| - x| = 0$, and $|x| + x = 2x > 0 \Rightarrow ||x| + x| = 2x$. The numerator is $0 - 2x = -2x$, so the quotient is $\frac{-2x}{x} = -2$.
- **Left, $x < 0$:**** here $|x| = -x$. Then $|x| - x = -x - x = -2x$, and since $x < 0$ we have $-2x > 0$, so $||x| - x| = -2x$; while $|x| + x = -x + x = 0 \Rightarrow ||x| + x| = 0$. The numerator is $-2x - 0 = -2x$, so the quotient is again $\frac{-2x}{x} = -2$.
- Both one-sided limits equal -2 , hence the two-sided limit is $\boxed{-2}$. Notice the inner $|x|$ merely swaps *which* outer bar collapses to 0, but the quotient stays -2 for every $x \neq 0$.

Method II — Inert-bars reduction

- For every real x we have $|x| \geq x$ and $|x| \geq -x$. Therefore $|x| - x \geq 0$ and $|x| + x \geq 0$: both inner quantities are already nonnegative, so the *outer* moduli do nothing and may be dropped: $||x| - x| = |x| - x$ and $||x| + x| = |x| + x$.
- The numerator then simplifies exactly: $(|x| - x) - (|x| + x) = -2x$.
- Hence for every $x \neq 0$ the whole expression equals $\frac{-2x}{x} = -2$ — a constant function.
- A constant function has that constant as its limit, so the limit is $\boxed{-2}$. The 'nesting' was a disguise; the outer bars never activate.

INSIGHT

Since $|x| \geq x$ and $|x| \geq -x$, the outer bars are no-ops: $||x| - x| = |x| - x$ and $||x| + x| = |x| + x$. The expression collapses to the constant -2 , so the limit is trivially -2 .

No.40 The Corner of Sine at Pi

● ● ● ● ● L5

$$\lim_{x \rightarrow \pi} \frac{|\sin x| - |x - \pi|}{|x - \pi|^3}$$

modulus

sin

$x \rightarrow \pi$

taylor

two-sided

corner

LIMIT

$$-\frac{1}{6}$$

THE TRAP

Near π one writes $\sin x \approx -(x - \pi)$, hence $|\sin x| \approx |x - \pi|$, the numerator looks like 0, and the tempting verdict is that the limit is 0. But that is only the leading order: the genuine first correction is third order, $|\sin x| - |x - \pi| = -\frac{1}{6}|x - \pi|^3 + O(|x - \pi|^5)$, which is exactly the size of the denominator and yields $-\frac{1}{6}$, not 0. (Note it is $|\sin x|$, not $\sin x$: at $x = \pi$ the function $|\sin x|$ has a real corner, so a one-sided Taylor shortcut for $\sin x$ alone is not enough.)

Method I — Shift to the corner and use evenness

- Let $h = x - \pi \rightarrow 0$. Then $\sin x = \sin(\pi + h) = -\sin h$, so $|\sin x| = |\sin h|$, and $|x - \pi| = |h|$. The expression becomes

$$\frac{|\sin h| - |h|}{|h|^3}.$$

- This is even in h (numerator and $|h|^3$ are both even), so it suffices to take $h \rightarrow 0^+$, where $|\sin h| = \sin h$, $|h| = h$: we need $\frac{\sin h - h}{h^3}$.
- From $\sin h = h - \frac{h^3}{6} + O(h^5)$, $\sin h - h = -\frac{h^3}{6} + O(h^5)$.
- $\frac{\sin h - h}{h^3} = -\frac{1}{6} + O(h^2) \rightarrow -\frac{1}{6}$.
- By evenness both sides agree, so the two-sided limit is $\boxed{-\frac{1}{6}}$.

Method II — Conjugate the corner difference

- With $h = x - \pi$, rationalize using $|a| - |b| = \frac{a^2 - b^2}{|a| + |b|}$:

$$|\sin h| - |h| = \frac{\sin^2 h - h^2}{|\sin h| + |h|}.$$

- Numerator $\sin^2 h - h^2 = (\sin h - h)(\sin h + h) \sim (-\frac{h^3}{6})(2h) = -\frac{h^4}{3}$. Denominator $|\sin h| + |h| \sim 2|h|$.
- So $|\sin h| - |h| \sim \frac{-h^4/3}{2|h|} = -\frac{h^3}{6}$ (using $h^4 = |h|^4$ and $h^4/|h| = |h|^3$).
- Divide by $|h|^3$: $\frac{-|h|^3/6}{|h|^3} = -\frac{1}{6}$. Hence $\boxed{-\frac{1}{6}}$.

INSIGHT

Around any zero $x = k\pi$ of $\sin$, the graph of $|\sin x|$ is a translated copy of $|\sin h|$; the universal third-order gap $|\sin h| - |h| \sim -\frac{1}{6}|h|^3$ governs all of them.

CHAPTER V

Inverse Circular Functions

Branch cuts, boundary singularities, the $\pm\pi/2$ cliff.

$\tan^{-1}$

№41 The Reciprocal Reflection

●●●●● L4

$$\lim_{x \rightarrow 0^+} \frac{\arctan\left(\frac{1}{x}\right) - \frac{\pi}{2}}{x}$$

arctan

reciprocal

$x \rightarrow 0^+$

asymptotic

one-sided

LIMIT

-1

THE TRAP

The tempting move is to note $\arctan(1/x) \rightarrow \pi/2$, declare the numerator $\rightarrow 0$, and then write $\frac{\pi/2 - \pi/2}{0} = 0$. The form is genuinely $0/0$, not $\frac{0}{0} = 0$. A second slip is to mishandle the reflection sign and get $+1$. The deepest trap: the identity $\arctan(1/x) = \frac{\pi}{2} - \arctan x$ holds only for $x > 0$; for $x < 0$ it reads $-\frac{\pi}{2} - \arctan x$, so the numerator $\rightarrow -\pi$ and the LEFT limit is $+\infty$. The two-sided limit does not exist, and the 0^+ is exactly what makes the answer the clean -1 .

Method I - Co-function reflection identity

- For $x > 0$ we have the exact identity $\arctan x + \arctan \frac{1}{x} = \frac{\pi}{2}$, hence

$$\arctan\left(\frac{1}{x}\right) - \frac{\pi}{2} = -\arctan x.$$

- Substitute into the quotient:

$$\frac{\arctan(1/x) - \pi/2}{x} = \frac{-\arctan x}{x}.$$

- Since $\frac{\arctan x}{x} \rightarrow 1$ as $x \rightarrow 0$, the limit is $\boxed{-1}$.

- Sign caveat: for $x < 0$ the identity reads $\arctan(1/x) = -\frac{\pi}{2} - \arctan x$, so the numerator $\rightarrow -\pi$ while $x \rightarrow 0^-$; thus the left limit is $\frac{-\pi}{0^-} = +\infty$. The two-sided limit does not exist, and only the 0^+ side is finite-and-clean.

Method II - Tangent substitution / series

- Put $u = \arctan x$, so $u \rightarrow 0^+$ and $x = \tan u$. By the reflection (Method I, step 1) the numerator equals $-u$.
- Then

$$\frac{-u}{\tan u} = -\frac{u}{\tan u} \xrightarrow{u \rightarrow 0} -1,$$

since $\tan u \sim u$.

- Equivalently, with the Maclaurin series $\arctan x = x - \frac{x^3}{3} + \dots$, the numerator is $-(x - \frac{x^3}{3} + \dots)$, and dividing by x gives $-(1 - \frac{x^2}{3} + \dots) \rightarrow -1$.

INSIGHT

The whole problem is the signed reflection $\arctan x + \arctan(1/x) = \frac{\pi}{2} \operatorname{sgn} x$; the $\operatorname{sgn} x$ is exactly why only the one-sided limit is finite.

№42 Twin Telescopes at Infinity

●●●●● L4

$$\lim_{x \rightarrow +\infty} x^2 \left(\arctan \frac{x+2}{x+1} - \arctan \frac{x+1}{x} \right)$$

arctan

subtraction-formula

$x \rightarrow \infty$

rational-argument

asymptotic

LIMIT

$$-\frac{1}{2}$$

THE TRAP

Both inner fractions $\rightarrow 1$, so a strong student writes each arctangent $\rightarrow \pi/4$ and concludes the bracket $\rightarrow 0$ — then either declares the limit 0, or estimates each fraction as $1 + O(1/x)$, calls the difference $O(1/x)$, and gets $x^2 \cdot O(1/x) = \infty$. The catch: both fractions are $1 + \frac{1}{x} + O(1/x^2)$ with the SAME leading $1/x$ coefficient, so the $O(1/x)$ corrections cancel exactly and the bracket is actually $\Theta(1/x^2)$, not $\Theta(1/x)$. Only the arctan subtraction formula exposes the true second-order size.

Method I - Arctan subtraction formula

- Use $\arctan a - \arctan b = \arctan \frac{a-b}{1+ab}$ with $a = \frac{x+2}{x+1}$, $b = \frac{x+1}{x}$. No $\pm\pi$ correction term is needed because $a, b > 0$ for large x , so $ab > 0 > -1$.
- Numerator: $a - b = \frac{(x+2)x - (x+1)^2}{x(x+1)} = \frac{-1}{x(x+1)}$.
- Denominator: $1 + ab = 1 + \frac{x+2}{x} = \frac{2x+2}{x} = \frac{2(x+1)}{x}$.
- Hence $\frac{a-b}{1+ab} = \frac{-1/(x(x+1))}{2(x+1)/x} = \frac{-1}{2(x+1)^2}$.
- So the bracket equals $\arctan \left(\frac{-1}{2(x+1)^2} \right) \sim \frac{-1}{2(x+1)^2}$ as $x \rightarrow \infty$, since $\arctan u \sim u$ for $u \rightarrow 0$.
- Therefore

$$x^2 \cdot \frac{-1}{2(x+1)^2} = -\frac{1}{2} \cdot \frac{x^2}{(x+1)^2} \rightarrow -\frac{1}{2}.$$

Method II - Mean Value Theorem

1. Write the bracket as $f(a) - f(b)$ with $f = \arctan$, $a = \frac{x+2}{x+1}$, $b = \frac{x+1}{x}$. By the MVT there is ξ between a, b with

$$f(a) - f(b) = \frac{a - b}{1 + \xi^2}.$$

2. As $x \rightarrow \infty$, both $a, b \rightarrow 1$, so $\xi \rightarrow 1$ and $1 + \xi^2 \rightarrow 2$.

3. From Method I, $a - b = \frac{-1}{x(x+1)} \sim -\frac{1}{x^2}$.

4. Thus the bracket $\sim \frac{-1/x^2}{2} = -\frac{1}{2x^2}$, and multiplying by x^2 gives $-\frac{1}{2}$.

Method III - Substitution $t=1/x$

1. Let $t = 1/x \rightarrow 0^+$. Then $a = \frac{1+2t}{1+t}$ and $b = 1+t$.

2. Expand: $a = (1+2t)(1-t+t^2-\dots) = 1+t-t^2+\dots$, while $b = 1+t$.

3. So $a - b = -t^2 + O(t^3)$, while $1 + ab = 2 + O(t)$.

4. Hence the bracket $= \arctan \frac{a-b}{1+ab} \sim \frac{-t^2}{2}$, and $x^2 = 1/t^2$, giving $\frac{1}{t^2} \cdot \frac{-t^2}{2} = -\frac{1}{2}$.

INSIGHT

The naive $1 + O(1/x)$ bookkeeping loses the cancellation of the matching $1/x$ terms; the arctan difference formula is the only tool that cleanly exposes the genuine $1/x^2$ scale.

№43 The Vanishing Square Root

●●●●● L4

$$\lim_{x \rightarrow 1^-} \frac{\frac{\pi}{2} - \arcsin x}{\sqrt{1-x^2}}$$

arcsin

square-root-singularity

$x \rightarrow 1^-$

conjugate

endpoint

LIMIT

1

THE TRAP

Direct substitution gives $0/0$. The seductive error: a student recalls the endpoint expansion $\arccos x \sim \sqrt{2(1-x)}$, then matches it against the denominator written as $\sqrt{2(1-x)}$ — forgetting the denominator is actually $\sqrt{(1-x)(1+x)}$, not $\sqrt{2(1-x)}$. Treating $\sqrt{1-x^2} \approx \sqrt{2}\sqrt{1-x}$ in the numerator but $\sqrt{1-x}$ in the denominator (or vice versa) manufactures a spurious factor of $\sqrt{2}$, giving the wrong answer $\sqrt{2}$. In truth the missing $\sqrt{2}$ is supplied exactly by $\sqrt{1+x} \rightarrow \sqrt{2}$, so the ratio is 1.

Method I - Endpoint expansion of arccos

1. Note $\frac{\pi}{2} - \arcsin x = \arccos x$. Set $x = 1 - t$ with $t \rightarrow 0^+$.

2. The endpoint expansion is $\arccos(1-t) = \sqrt{2t} \left(1 + \frac{t}{12} + \dots\right)$, so the numerator $\sim \sqrt{2t} = \sqrt{2(1-x)}$.

3. Write the denominator as $\sqrt{1-x^2} = \sqrt{(1-x)(1+x)} = \sqrt{2(1-x)} \sqrt{\frac{1+x}{2}}$, where $\sqrt{\frac{1+x}{2}} \rightarrow 1$.

4. Hence the ratio $\sim \frac{\sqrt{2(1-x)}}{\sqrt{2(1-x)} \cdot \sqrt{(1+x)/2}} = \frac{1}{\sqrt{(1+x)/2}} \rightarrow 1$.

Method II - Trig substitution

1. Let $\theta = \arccos x \in (0, \pi/2)$, so $x = \cos \theta$ and $\theta \rightarrow 0^+$ as $x \rightarrow 1^-$.
2. Then $\sqrt{1-x^2} = \sqrt{1-\cos^2 \theta} = \sin \theta$ (positive on this range).
3. The expression becomes $\frac{\theta}{\sin \theta} \rightarrow 1$ as $\theta \rightarrow 0$.

Method III - L'Hopital, done carefully

1. Form is $0/0$. Differentiate: numerator derivative $= -\frac{1}{\sqrt{1-x^2}}$; denominator derivative $= \frac{-x}{\sqrt{1-x^2}}$.
2. Ratio of derivatives $= \frac{-1/\sqrt{1-x^2}}{-x/\sqrt{1-x^2}} = \frac{1}{x} \rightarrow 1$ as $x \rightarrow 1^-$.

INSIGHT

The trap is the difference between $\sqrt{2(1-x)}$ and $\sqrt{1-x^2}$: their ratio $\rightarrow 1$, not $\sqrt{2}$, because the second factor $\sqrt{1+x} \rightarrow \sqrt{2}$ exactly cancels the missing $\sqrt{2}$.

№44 Second Difference of the Arc

● ● ● ● ● L5

$$\lim_{x \rightarrow +\infty} x^3 \left(\arctan(x+1) + \arctan(x-1) - 2 \arctan x \right)$$

arctan

second-difference

$x \rightarrow \infty$

subtraction-formula

convexity

LIMIT

−2

THE TRAP

All three arctangents $\rightarrow \pi/2$, so the bracket reads $\frac{\pi}{2} + \frac{\pi}{2} - 2 \cdot \frac{\pi}{2} = 0$. A strong student then linearizes $\arctan(x \pm 1) \approx \arctan x \pm \frac{1}{1+x^2}$; the $\pm \frac{1}{1+x^2}$ terms cancel, the bracket vanishes to first order, and they declare the limit 0. But 0 is only the first-order verdict: the surviving piece is the SECOND-order (curvature) term $f''(x) \sim -2/x^3$, which the x^3 factor rescales to a finite −2. Stopping at the linear order is exactly the bait.

Method I - Pairwise subtraction formula

1. Regroup the symmetric difference around the middle term:

$$(\arctan(x+1) - \arctan x) - (\arctan x - \arctan(x-1)).$$

2. Apply $\arctan a - \arctan b = \arctan \frac{a-b}{1+ab}$ (valid here since both products $ab > 0$ for large x): the first parenthesis is $\arctan \frac{1}{1+x(x+1)} = \arctan \frac{1}{x^2+x+1}$, the second is $\arctan \frac{1}{1+x(x-1)} = \arctan \frac{1}{x^2-x+1}$.

3. Hence the bracket equals $\arctan \frac{1}{x^2+x+1} - \arctan \frac{1}{x^2-x+1}$, both arguments $\rightarrow 0^+$.

4. Since $\arctan u = u + O(u^3)$ and here $u = O(1/x^2)$, replace each arctan by its argument with error $O(1/x^6)$:

$$\frac{1}{x^2 + x + 1} - \frac{1}{x^2 - x + 1} = \frac{(x^2 - x + 1) - (x^2 + x + 1)}{(x^2 + x + 1)(x^2 - x + 1)} = \frac{-2x}{x^4 + x^2 + 1}.$$

5. Therefore the bracket $= \frac{-2x}{x^4 + x^2 + 1} + O(x^{-6}) \sim \frac{-2}{x^3}$, and multiplying by x^3 gives

$$\lim_{x \rightarrow +\infty} x^3 \cdot \frac{-2x}{x^4 + x^2 + 1} = -2.$$

Method II - Taylor of the second difference

1. For a smooth f , the centered second difference expands as

$$f(x+1) + f(x-1) - 2f(x) = f''(x) + \frac{f^{(4)}(x)}{12} + \dots$$

(Taylor expand $f(x \pm 1)$ to order 4; the odd derivatives cancel).

2. Take $f = \arctan$. Then $f'(x) = \frac{1}{1+x^2}$ and $f''(x) = \frac{-2x}{(1+x^2)^2} \sim \frac{-2}{x^3}$ as $x \rightarrow +\infty$.

3. The next term satisfies $f^{(4)}(x) = O(x^{-5})$, so after multiplying the whole second difference by x^3 it contributes $O(x^{-2}) \rightarrow 0$ and is negligible.

4. Therefore

$$\lim_{x \rightarrow +\infty} x^3 (\arctan(x+1) + \arctan(x-1) - 2\arctan x) = \lim_{x \rightarrow +\infty} x^3 \cdot \frac{-2x}{(1+x^2)^2} = -2.$$

INSIGHT

The bracket is the discrete Laplacian (centered second difference) of $\arctan$ at infinity. Symmetry forces the first-order terms to cancel, leaving the curvature $f''(x) \sim -2x^{-3}$; that x^{-3} decay is precisely why the exponent 3 is the unique power giving a finite nonzero limit.

№45 Across the Branch Cut

● ● ● ● ● L5

$$\lim_{x \rightarrow (\pi/2)^+} \frac{\arctan(\tan x) + \frac{\pi}{2}}{x - \frac{\pi}{2}}$$

arctan

tan

branch-cut

$x \rightarrow \pi/2$

one-sided

nonexistence

LIMIT

1

THE TRAP

The reflex is to 'cancel' $\arctan(\tan x) = x$, turning the expression into $\frac{x+\pi/2}{x-\pi/2}$, whose right-hand limit blows up to $+\infty$. But $\arctan(\tan x) = x$ holds ONLY on $(-\pi/2, \pi/2)$. The instant x crosses $\pi/2$ the principal value drops a full period: $\arctan(\tan x) = x - \pi$. So the numerator is actually $(x - \pi) + \pi/2 = x - \pi/2$, it cancels the denominator exactly, and the true right-hand limit is 1. The naive answer isn't just numerically off — it's off by an entire branch.

Method I - Correct branch reduction

1. For $x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ we have $\tan x = \tan(x - \pi)$ with $x - \pi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, the principal range of $\arctan$.

Hence

$$\arctan(\tan x) = x - \pi.$$

2. The numerator becomes $(x - \pi) + \frac{\pi}{2} = x - \frac{\pi}{2}$.

3. So the quotient equals $\frac{x - \pi/2}{x - \pi/2} = 1$ identically for all such x , and therefore

$$\lim_{x \rightarrow (\pi/2)^+} = 1.$$

4. (The two-sided limit fails: for $x < \frac{\pi}{2}$, $\arctan(\tan x) = x$, so the numerator $\rightarrow \pi$ while the denominator $\rightarrow 0^-$, giving a left-hand limit of $-\infty$. The one-sided posing is essential.)

Method II - Substitution and jump bookkeeping

1. Put $h = x - \frac{\pi}{2} \rightarrow 0^+$. Then $\tan x = \tan\left(\frac{\pi}{2} + h\right) = -\cot h$.

2. For small $h > 0$, $\cot h = \tan\left(\frac{\pi}{2} - h\right)$ with $\frac{\pi}{2} - h \in \left(0, \frac{\pi}{2}\right)$, so $\arctan(\cot h) = \frac{\pi}{2} - h$, and by oddness $\arctan(-\cot h) = -\frac{\pi}{2} + h$.

3. Numerator $= \left(-\frac{\pi}{2} + h\right) + \frac{\pi}{2} = h$; denominator $= h$. The quotient is $\frac{h}{h} = 1$, so the limit is 1.

INSIGHT

$\arctan(\tan x)$ is the sawtooth $x - \pi \left\lfloor \frac{x + \pi/2}{\pi} \right\rfloor$; every crossing of an odd multiple of $\frac{\pi}{2}$ injects a $-\pi$ jump that the identity $\arctan(\tan x) = x$ silently forgets.

№46 Crashing into Minus One

●●●●● L4

$$\lim_{x \rightarrow -1^+} \frac{\arcsin x + \frac{\pi}{2}}{\sqrt{x+1}}$$

arcsin

endpoint

$x \rightarrow -1^+$

square-root-singularity

one-sided

LIMIT

$$\sqrt{2}$$

THE TRAP

The form is $0/0$, so a student reaches for the Maclaurin series $\arcsin x \approx x + \frac{x^3}{6}$ — but that series is anchored at $x = 0$, not at the endpoint $x = -1$, so it gives the meaningless $\frac{-1 + \frac{5}{2} - \frac{1}{6}}{0}$ and a phantom divergence. Near $x = -1$ the graph of $\arcsin$ has a vertical-tangent half-power cusp, not a smooth Taylor point: $\arcsin x \approx -\frac{\pi}{2} + \sqrt{2(x+1)}$. The square-root in the numerator exactly cancels the one in the denominator, giving the finite value $\sqrt{2}$. (L'Hopital also works and lands on $\sqrt{2}$, but only if you keep the radical in $\frac{1}{\sqrt{1-x^2}}$ rather than series-expanding it.)

Method I - Endpoint square-root expansion

1. Set $x = -1 + t$ with $t \rightarrow 0^+$. Near the endpoint, $\sin\left(-\frac{\pi}{2} + u\right) = -\cos u = -1 + \frac{u^2}{2} + \dots$, so inverting gives $\arcsin(-1 + t) = -\frac{\pi}{2} + \sqrt{2t} + O(t^{3/2})$.

2. Hence the numerator is $\arcsin x + \frac{\pi}{2} = \sqrt{2t} + O(t^{3/2}) = \sqrt{2}\sqrt{t}(1 + o(1))$.

3. The denominator is $\sqrt{x+1} = \sqrt{t}$.

4. Therefore the ratio $\rightarrow \frac{\sqrt{2}\sqrt{t}}{\sqrt{t}} = \sqrt{2}$.

Method II - Substitution to a sine limit

1. Let $\theta = \arcsin x + \frac{\pi}{2}$. As $x \rightarrow -1^+$, $\theta \rightarrow 0^+$, and $x = \sin(\theta - \frac{\pi}{2}) = -\cos \theta$.
2. Then $x + 1 = 1 - \cos \theta = 2 \sin^2(\frac{\theta}{2})$, so $\sqrt{x+1} = \sqrt{2} \sin(\frac{\theta}{2})$ (positive since $\theta \rightarrow 0^+$).
3. The expression becomes $\frac{\theta}{\sqrt{2} \sin(\theta/2)} = \frac{1}{\sqrt{2}} \cdot \frac{\theta}{\sin(\theta/2)}$.
4. Since $\frac{\theta}{\sin(\theta/2)} = 2 \cdot \frac{\theta/2}{\sin(\theta/2)} \rightarrow 2$, the limit is $\frac{1}{\sqrt{2}} \cdot 2 = \sqrt{2}$.

Method III - L'Hopital with the radical kept intact

1. Both numerator and denominator $\rightarrow 0$ as $x \rightarrow -1^+$, so L'Hopital applies. Differentiate: top $\frac{d}{dx}(\arcsin x + \frac{\pi}{2}) = \frac{1}{\sqrt{1-x^2}}$, bottom $\frac{d}{dx}\sqrt{x+1} = \frac{1}{2\sqrt{x+1}}$.
2. The ratio of derivatives is $\frac{2\sqrt{x+1}}{\sqrt{1-x^2}} = \frac{2\sqrt{x+1}}{\sqrt{(1-x)(1+x)}} = \frac{2}{\sqrt{1-x}}$.
3. As $x \rightarrow -1^+$ this $\rightarrow \frac{2}{\sqrt{2}} = \sqrt{2}$. The key is to simplify $\sqrt{1-x^2}$ as a radical; series-expanding it (the endpoint trap) destroys the cancellation.

INSIGHT

At a domain endpoint ± 1 , $\arcsin x \approx \pm \frac{\pi}{2} \mp \sqrt{2|1 \mp x|}$ — a half-power (vertical-tangent) cusp. Any whole-power Taylor series is the wrong local model there; the right tool is the substitution $x = \mp 1 \pm t$ or $x = -\cos \theta$.

№47 The Other End of Arctan

●●●●● L4

$$\lim_{x \rightarrow -\infty} x \left(\frac{\pi}{2} + \arctan x \right)$$

arctan

$x \rightarrow -\infty$

reflection

negative-infinity

asymptotic

LIMIT

−1

THE TRAP

As $x \rightarrow -\infty$, $\arctan x \rightarrow -\pi/2$, so the bracket $\rightarrow 0$ and the form is $-\infty \cdot 0$. A student reflexively uses the $x > 0$ identity $\arctan x = \pi/2 - \arctan(1/x)$ (valid ONLY for $x > 0$), giving $\pi/2 + \arctan x = \pi - \arctan(1/x) \rightarrow \pi$, hence $x \cdot \pi \rightarrow -\infty$. The correct identity for $x < 0$ is $\arctan x + \arctan(1/x) = -\pi/2$ — the sign flips, the bracket behaves like $-1/x$ rather than π , and the product converges to -1 .

Method I - Negative-branch reflection

1. For $x < 0$ the reflection identity reads $\arctan x + \arctan \frac{1}{x} = -\frac{\pi}{2}$, so $\frac{\pi}{2} + \arctan x = -\arctan \frac{1}{x}$.
2. Therefore $x \left(\frac{\pi}{2} + \arctan x \right) = -x \arctan \frac{1}{x}$.

3. As $x \rightarrow -\infty$, put $t = 1/x \rightarrow 0^-$. Then $-x \arctan \frac{1}{x} = -\frac{\arctan t}{t}$, and by the standard limit $\lim_{t \rightarrow 0} \frac{\arctan t}{t} = 1$ this tends to -1 .

Method II - Substitution $u=1/x$

1. Let $u = 1/x \rightarrow 0^-$. Then $x = 1/u$ and the expression becomes $\frac{1}{u} \left(\frac{\pi}{2} + \arctan \frac{1}{u} \right)$.
2. For $u < 0$ we have $\arctan(1/u) = -\frac{\pi}{2} - \arctan u$, hence $\frac{\pi}{2} + \arctan(1/u) = -\arctan u$.
3. So the expression equals $\frac{-\arctan u}{u} \rightarrow -1$ as $u \rightarrow 0^-$.

Method III - L'Hopital after rewriting

1. Rewrite as $\frac{\pi/2 + \arctan x}{1/x}$, a genuine $0/0$ form as $x \rightarrow -\infty$ (numerator $\rightarrow 0$, denominator $\rightarrow 0$).
2. Differentiate: numerator gives $\frac{1}{1+x^2}$, denominator gives $-\frac{1}{x^2}$.
3. The ratio is $\frac{1/(1+x^2)}{-1/x^2} = \frac{-x^2}{1+x^2} \rightarrow -1$, so by L'Hopital the limit is -1 .

INSIGHT

The reflection identity $\arctan x + \arctan(1/x) = \frac{\pi}{2} \operatorname{sgn}(x)$ carries a sign that students copy from the $x > 0$ case – at $-\infty$ that single sign is the entire difficulty.

№48 Arcsine Raised to a Power

● ● ● ● ● L5

$$\lim_{x \rightarrow 0^+} \left(\frac{\arcsin x}{x} \right)^{1/x^2}$$

arcsin

1-power-infinity

log-transform

series

$x \rightarrow 0^+$

LIMIT

$$e^{1/6}$$

THE TRAP

Because $\arcsin x/x \rightarrow 1$, the tempting move is to call the base 1 and conclude $1^\infty = 1$. But the exponent $1/x^2$ blows up exactly as fast as the base approaches 1, so the rate is everything: $\arcsin x/x = 1 + \frac{x^2}{6} + \dots$, and $\left(1 + \frac{x^2}{6}\right)^{1/x^2} \rightarrow e^{1/6}$. The 1^∞ indeterminate form is precisely the regime in which the base's deviation from 1 survives in the answer; reading it as 1 throws away the whole problem.

Method I - Logarithm and series

1. Let $L = \lim_{x \rightarrow 0^+} \left(\frac{\arcsin x}{x} \right)^{1/x^2}$ and take logs: $\ln L = \lim_{x \rightarrow 0^+} \frac{\ln(\arcsin x/x)}{x^2}$.
2. Maclaurin series: $\arcsin x = x + \frac{x^3}{6} + \frac{3x^5}{40} + \dots$, so $\frac{\arcsin x}{x} = 1 + \frac{x^2}{6} + \frac{3x^4}{40} + \dots$
3. Using $\ln(1+u) = u - \frac{u^2}{2} + \dots$ with $u = \frac{x^2}{6} + \dots$: $\ln\left(1 + \frac{x^2}{6} + \dots\right) = \frac{x^2}{6} + O(x^4)$.
4. Divide by x^2 : $\ln L = \frac{1}{6}$, hence $L = e^{1/6}$.

Method II - Standard 1^∞ transform

1. For a base tending to 1, use $(1 + a(x))^{b(x)} \rightarrow e^{\lim a(x)b(x)}$ whenever $a(x) \rightarrow 0$.
2. Here $a(x) = \frac{\arcsin x}{x} - 1$ and $b(x) = \frac{1}{x^2}$.
3. From $\frac{\arcsin x}{x} = 1 + \frac{x^2}{6} + o(x^2)$ we get $a(x) \sim \frac{x^2}{6}$.
4. Thus $a(x)b(x) \rightarrow \frac{x^2}{6} \cdot \frac{1}{x^2} = \frac{1}{6}$, so the limit is $e^{1/6}$.

INSIGHT

The exponent in the answer is exactly the coefficient of x^3 in $\arcsin x$ (namely $1/6$): any 1^∞ tower of this shape reads off the leading non-trivial term of the base. The next correction is $\ln L = \frac{1}{6} + \frac{11}{180}x^2 + \dots$, which the numerics confirm.

№49 Two Arctans, One Argument

●●●●● L4

$$\lim_{x \rightarrow 0^+} \frac{\arctan(2x) - 2 \arctan x}{x^3}$$

arctan

maclaurin-series

$x \rightarrow 0^+$

cubic-cancellation

odd-function

LIMIT

−2

THE TRAP

Using only the first-order tangents $\arctan(2x) \approx 2x$ and $2 \arctan x \approx 2x$, the numerator looks like 0, so a student concludes the limit is 0 — or applies L'Hopital once, still sees $0/0$, and gives up. But the linear terms cancel *exactly*; the answer lives entirely in the x^3 coefficients, where $-\frac{8}{3}$ (from $\arctan 2x$) and $-\frac{2}{3}$ (from $2 \arctan x$) do NOT cancel, leaving $-2x^3$.

Method I - Cubic Maclaurin terms

1. Use the standard series $\arctan u = u - \frac{u^3}{3} + \frac{u^5}{5} - \dots$ (valid for $|u| < 1$).
2. $\arctan(2x) = 2x - \frac{(2x)^3}{3} + \dots = 2x - \frac{8x^3}{3} + O(x^5)$.
3. $2 \arctan x = 2\left(x - \frac{x^3}{3} + \dots\right) = 2x - \frac{2x^3}{3} + O(x^5)$.
4. Subtract: the $2x$ terms cancel, leaving numerator $= \left(-\frac{8}{3} + \frac{2}{3}\right)x^3 + O(x^5) = -2x^3 + O(x^5)$.
5. Divide by x^3 and let $x \rightarrow 0^+$: limit = -2 .

Method II - Repeated L'Hopital

1. Let $N(x) = \arctan 2x - 2 \arctan x$ and $D(x) = x^3$; both $\rightarrow 0$, giving $0/0$.
2. $N'(x) = \frac{2}{1+4x^2} - \frac{2}{1+x^2}$, $D'(x) = 3x^2$; at $x = 0$ this is still $0/0$.
3. $N''(x) = \frac{-16x}{(1+4x^2)^2} + \frac{4x}{(1+x^2)^2}$, $D''(x) = 6x$; at $x = 0$ this is again $0/0$.

4. $N'''(0)$: since $N(x) = -2x^3 + O(x^5)$, we have $N'''(0) = 3! \cdot (-2) = -12$, while $D'''(x) = 6$.

5. By L'Hopital three times, $\text{limit} = \frac{N'''(0)}{D'''(0)} = \frac{-12}{6} = -2$.

Method III - Derivative ratio shortcut

1. The numerator $N(x)$ is odd with vanishing linear term, so write $N(x) = cx^3 + o(x^3)$; then the limit equals c .

2. Expand the derivative: $N'(x) = \frac{2}{1+4x^2} - \frac{2}{1+x^2} = 2[(1-4x^2+\dots) - (1-x^2+\dots)] = 2(-3x^2 + \dots) = -6x^2 + \dots$

3. Since $N'(x) = 3cx^2 + o(x^2)$, matching gives $3c = -6$, so $c = -2$ and the limit is -2 .

INSIGHT

The cubic defect $-2x^3$ measures exactly how the angle-doubling fails: $\arctan(2x) \neq 2\arctan x$, and to third order the gap is $-2x^3$. (Because the quotient is even, the full two-sided limit is also -2 .)

№50 The Floor Beneath the Arc

●●●●● L4

$$\lim_{x \rightarrow 0^+} \arctan\left(\left\lfloor \frac{1}{x} \right\rfloor x\right)$$

arctan

floor

squeeze

$x \rightarrow 0^+$

fractional-part

LIMIT

$$\frac{\pi}{4}$$

THE TRAP

Two traps. (1) Treating $\lfloor 1/x \rfloor$ as $1/x$ makes the argument exactly 1 and gives $\arctan 1 = \pi/4$ — the right value for the wrong reason, and a sign or boundary slip easily derails it. (2) Noting that $\lfloor 1/x \rfloor$ jumps and is non-monotone, one panics and declares the limit nonexistent. But $\lfloor 1/x \rfloor x = 1 - \{1/x\}x$ converges to 1^- by a squeeze (the error $\{1/x\}x < x \rightarrow 0$), and continuity of $\arctan$ delivers $\pi/4$.

Method I - Floor squeeze

1. For $x > 0$, the defining inequality of the floor gives $\frac{1}{x} - 1 < \lfloor \frac{1}{x} \rfloor \leq \frac{1}{x}$. Multiplying through by $x > 0$ (which preserves the inequalities):

$$1 - x < \lfloor \frac{1}{x} \rfloor x \leq 1.$$

2. As $x \rightarrow 0^+$, the left side $1 - x \rightarrow 1$ and the right bound is the constant 1, so by the squeeze theorem $\lfloor \frac{1}{x} \rfloor x \rightarrow 1^-$.

3. Since $\arctan$ is continuous at 1, $\lim_{x \rightarrow 0^+} \arctan(\lfloor 1/x \rfloor x) = \arctan 1 = \frac{\pi}{4}$.

Method II - Fractional-part identity

1. Using $\lfloor \frac{1}{x} \rfloor = \frac{1}{x} - \{ \frac{1}{x} \}$, multiply by x :

$$\lfloor \frac{1}{x} \rfloor x = 1 - \{ \frac{1}{x} \}x.$$

2. Since $0 \leq \left\{\frac{1}{x}\right\} < 1$, for $x > 0$ we have $0 \leq \left\{\frac{1}{x}\right\}x < x$, and $x \rightarrow 0^+$ forces $\left\{\frac{1}{x}\right\}x \rightarrow 0$ by squeeze.
3. Hence the argument $1 - \left\{\frac{1}{x}\right\}x \rightarrow 1$, and by continuity $\arctan(1 - \left\{\frac{1}{x}\right\}x) \rightarrow \arctan 1 = \frac{\pi}{4}$.

INSIGHT

The discontinuous, non-monotone factor $\{1/x\}$ is tamed because its error from $1/x$ is bounded by 1, and that bounded error is crushed by the multiplying $x \rightarrow 0$ – a classic floor-squeeze the oscillation-panic misses.

CHAPTER VI

Logarithm & Exponential

ln

Growth races, hidden orders, and the 1^∞ only logarithms can tame.

№51 The Logarithm's Echo

●●●●● L4

$$\lim_{x \rightarrow 0} \left(\frac{\ln(1+x)}{x} \right)^{1/x}$$

1^∞ infinity

$\ln(1+x)$

second-order

$x \rightarrow 0$

LIMIT

$$e^{-1/2}$$

THE TRAP

The base $\frac{\ln(1+x)}{x} \rightarrow 1$, so it is tempting to declare the whole expression a 1^∞ that 'obviously' equals 1. But 1^∞ is an indeterminate form precisely because the base is only asymptotically 1: here $\frac{\ln(1+x)}{x} = 1 - \frac{x}{2} + O(x^2)$, and the deviation $-\frac{x}{2}$ is exactly first order in x . Raising to the power $1/x$ promotes that linear term to a finite limit $e^{-1/2} \approx 0.6065$, not 1. Replacing the base by 1 before taking the power silently discards the only term that matters.

Method I - Logarithm and series

- Let $L = \lim_{x \rightarrow 0} \left(\frac{\ln(1+x)}{x} \right)^{1/x}$ and take logarithms: $\ln L = \lim_{x \rightarrow 0} \frac{1}{x} \ln \left(\frac{\ln(1+x)}{x} \right)$.
- Expand the inner ratio using $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$:

$$\frac{\ln(1+x)}{x} = 1 - \frac{x}{2} + \frac{x^2}{3} - \dots$$
- Now $\ln \left(\frac{\ln(1+x)}{x} \right) = \ln(1+u)$ with $u = -\frac{x}{2} + \frac{x^2}{3} - \dots$. Using $\ln(1+u) = u - \frac{u^2}{2} + \dots$ and $u^2 = \frac{x^2}{4} + O(x^3)$:

$$\ln(1+u) = \left(-\frac{x}{2} + \frac{x^2}{3} \right) - \frac{1}{2} \cdot \frac{x^2}{4} + \dots = -\frac{x}{2} + \frac{5x^2}{24} + \dots$$
- Divide by x : $\frac{1}{x} \ln \left(\frac{\ln(1+x)}{x} \right) = -\frac{1}{2} + \frac{5}{24}x + \dots \xrightarrow{x \rightarrow 0} -\frac{1}{2}$.
- Therefore $\ln L = -\frac{1}{2}$, giving $L = e^{-1/2}$. The first-order coefficient $-\frac{1}{2}$ of the base is exactly what the $1/x$ exponent harvests.

Method II - Exponential substitution

1. Write the base as $1 + t$ with $t = \frac{\ln(1+x)}{x} - 1$. As $x \rightarrow 0$, $t \rightarrow 0$, and the expression is $(1 + t)^{1/x} = \exp\left(\frac{\ln(1+t)}{x}\right)$.
2. From the series, $t = -\frac{x}{2} + \frac{x^2}{3} - \dots$ is of order x . Since $\ln(1+t) = t - \frac{t^2}{2} + \dots$ and $t^2 = O(x^2)$, the t^2 term contributes nothing after dividing by x .
3. Hence $\frac{\ln(1+t)}{x} = \frac{t}{x} + \frac{O(x^2)}{x} = \frac{1}{x}\left(-\frac{x}{2} + O(x^2)\right) + O(x) \xrightarrow{x \rightarrow 0} -\frac{1}{2}$.
4. The limit is $\exp\left(-\frac{1}{2}\right) = e^{-1/2}$. Only the linear coefficient of t in x survives; all higher orders are killed by the division by x .

INSIGHT

General principle: $(1 + a_1x + a_2x^2 + \dots)^{1/x} \rightarrow e^{a_1}$ as $x \rightarrow 0$ – only the first non-constant coefficient of the base survives. Here $a_1 = -\frac{1}{2}$.

№52 Chasing Euler's Number

● ● ● ● ● L5

$$\lim_{x \rightarrow \infty} \left(\frac{(1 + \frac{1}{x})^x}{e} \right)^x$$

1^infinity

e-expansion

second-order

x->infinity

LIMIT

$$e^{-1/2}$$

THE TRAP

Knowing $(1 + \frac{1}{x})^x \rightarrow e$, a student says the base $\rightarrow e/e = 1$, so the limit is $1^\infty = 1$. But $(1 + \frac{1}{x})^x$ approaches e from below at rate $\sim \frac{e}{2x}$, and the outer exponent x converts this $\frac{1}{x}$ -sized defect into the finite value $e^{-1/2}$.

Method I - Two-layer log expansion

1. Let L be the limit; $\ln L = \lim_{x \rightarrow \infty} x \left[x \ln\left(1 + \frac{1}{x}\right) - 1 \right]$, since $\ln\left(\frac{(1+1/x)^x}{e}\right) = x \ln\left(1 + \frac{1}{x}\right) - 1$.
2. Expand $\ln\left(1 + \frac{1}{x}\right) = \frac{1}{x} - \frac{1}{2x^2} + \frac{1}{3x^3} - \dots$. Multiply by x :

$$x \ln\left(1 + \frac{1}{x}\right) = 1 - \frac{1}{2x} + \frac{1}{3x^2} - \dots$$
3. Subtract 1: $x \ln\left(1 + \frac{1}{x}\right) - 1 = -\frac{1}{2x} + \frac{1}{3x^2} - \dots$.
4. Multiply by the outer x : $x \left[-\frac{1}{2x} + \frac{1}{3x^2} - \dots \right] = -\frac{1}{2} + \frac{1}{3x} - \dots \rightarrow -\frac{1}{2}$.
5. So $\ln L = -\frac{1}{2}$ and $L = e^{-1/2}$. Stopping the inner expansion at first order ($x \ln(1 + \frac{1}{x}) \approx 1$) loses the entire answer.

Method II - Substitution to x->0

1. Put $t = \frac{1}{x} \rightarrow 0^+$. The expression becomes $\left(\frac{(1+t)^{1/t}}{e}\right)^{1/t} = \exp\left(\frac{1}{t} \left[\frac{1}{t} \ln(1+t) - 1 \right]\right)$.
2. Use $\ln(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \dots$, so $\frac{1}{t} \ln(1+t) = 1 - \frac{t}{2} + \frac{t^2}{3} - \dots$ and $\frac{1}{t} \ln(1+t) - 1 = -\frac{t}{2} + \frac{t^2}{3} - \dots$.
3. Divide by t : $\frac{1}{t} \left(-\frac{t}{2} + \frac{t^2}{3} - \dots \right) = -\frac{1}{2} + \frac{t}{3} - \dots \rightarrow -\frac{1}{2}$.

4. Thus the limit is $\exp(-\frac{1}{2}) = e^{-1/2}$, identical in mechanism to its $x \rightarrow 0$ cousin.

INSIGHT

Knowing $(1 + 1/x)^x = e \exp(-\frac{1}{2x} + \frac{11}{24x^2} - \dots)$ makes this instant; the $-\frac{1}{2x}$ defect is the whole story.

№53 The Point Named e

●●●●● L4

$$\lim_{x \rightarrow e} (\ln x)^{\frac{1}{x-e}}$$

1^infinity

ln-of-ln

x → e

nonzero-point

LIMIT

$$e^{1/e}$$

THE TRAP

At $x = e$ the base $\ln x = 1$ exactly, so the form is 1^∞ and a student declares the limit 1. But 1^∞ is indeterminate: the base leaves 1 linearly, $\ln x \approx 1 + \frac{x-e}{e}$ (slope = $\ln'(e) = 1/e$), while the exponent $\frac{1}{x-e} \rightarrow \pm\infty$ amplifies that tiny excess. The two effects balance to give $e^{1/e} \approx 1.4447$, not 1.

Method I - Log + derivative

1. Set $L = \lim_{x \rightarrow e} (\ln x)^{1/(x-e)}$. Since $\ln x > 0$ for x near e , the power is real, and taking logs gives $\ln L = \lim_{x \rightarrow e} \frac{\ln(\ln x)}{x - e}$.
2. This is $\frac{0}{0}$: the numerator $\rightarrow \ln(\ln e) = \ln 1 = 0$. Writing $g(x) = \ln(\ln x)$ with $g(e) = 0$, the quotient is exactly the difference quotient $\frac{g(x) - g(e)}{x - e}$, whose limit is $g'(e)$.
3. Differentiate by the chain rule: $g'(x) = \frac{1}{\ln x} \cdot \frac{1}{x}$, so $g'(e) = \frac{1}{\ln e} \cdot \frac{1}{e} = 1 \cdot \frac{1}{e} = \frac{1}{e}$.
4. Hence $\ln L = \frac{1}{e}$ and $L = e^{1/e}$.

Method II - Local expansion

1. Substitute $x = e + t$, $t \rightarrow 0$. Then $\ln x = \ln(e + t) = 1 + \ln\left(1 + \frac{t}{e}\right) = 1 + \frac{t}{e} - \frac{t^2}{2e^2} + \dots$.
2. Take the log of the base: $\ln(\ln x) = \ln\left(1 + \frac{t}{e} - \dots\right) = \frac{t}{e} + O(t^2)$.
3. Divide by $t = x - e$: $\frac{\ln(\ln x)}{x - e} = \frac{1}{e} + O(t) \xrightarrow{t \rightarrow 0} \frac{1}{e}$ (same value as $t \rightarrow 0^+$ and $t \rightarrow 0^-$, so the limit is two-sided).
4. Exponentiate: the limit is $e^{1/e} \approx 1.4447$, comfortably far from the naive 1.

INSIGHT

General fact: for any f differentiable at a with $f(a) = 1$ (and $f > 0$ near a), $\lim_{x \rightarrow a} f(x)^{1/(x-a)} = e^{f'(a)}$. Here $f = \ln$, $a = e$, $f'(e) = 1/e$, giving $e^{1/e}$.

№54 The Buried Term

●●●●● L4

$$\lim_{x \rightarrow -\infty} e^{-x} (x - \ln(e^x + e^{2x} + e^{3x}))$$

log-of-sum

dominated-term

$x \rightarrow -\infty$

exponential-tail

LIMIT

−1

THE TRAP

As $x \rightarrow -\infty$ the term e^x is the largest of the three (since $x < 0$ makes $e^x > e^{2x} > e^{3x}$), so it is tempting to write $\ln(e^x + e^{2x} + e^{3x}) \approx \ln(e^x) = x$, making the parenthesis $\rightarrow 0$. The student then reads $e^{-x} \cdot 0$ as $\infty \cdot 0$ and either answers 0 or stalls. But the parenthesis is exactly $-\ln(1 + e^x + e^{2x})$, whose leading piece is $-e^x$ — an order- e^x tail. Multiplying by e^{-x} cancels it precisely and leaves -1 . Discarding the subdominant terms throws away the only thing the e^{-x} factor magnifies.

Method I - Factor the dominant term

1. Factor e^x out of the sum: $e^x + e^{2x} + e^{3x} = e^x(1 + e^x + e^{2x})$.
2. Then $\ln(\dots) = x + \ln(1 + e^x + e^{2x})$, so $x - \ln(\dots) = -\ln(1 + e^x + e^{2x})$.
3. Let $u = e^x \rightarrow 0^+$. Using $\ln(1 + u + u^2) = (u + u^2) - \frac{(u+u^2)^2}{2} + \dots = u + \frac{u^2}{2} + O(u^3)$:

$$x - \ln(\dots) = -\left(u + \frac{u^2}{2} + \dots\right).$$
4. Multiply by $e^{-x} = u^{-1}$: $e^{-x}(x - \ln(\dots)) = -\frac{1}{u}\left(u + \frac{u^2}{2} + \dots\right) = -\left(1 + \frac{u}{2} + \dots\right) \rightarrow -1$.
5. The naive 'drop the small terms' kills the very $O(u)$ piece that, after the u^{-1} amplification, gives the answer -1 .

Method II - Substitution $u=e^x$

1. Set $u = e^x$ so $u \rightarrow 0^+$ and $x = \ln u$, $e^{-x} = 1/u$. The limit becomes

$$\lim_{u \rightarrow 0^+} \frac{1}{u} (\ln u - \ln(u + u^2 + u^3)).$$
2. Simplify the log difference: $\ln u - \ln(u(1 + u + u^2)) = -\ln(1 + u + u^2)$.
3. So the expression is $-\frac{\ln(1 + u + u^2)}{u}$.
4. Since $\ln(1 + u + u^2) \sim u$ as $u \rightarrow 0$ (because $\ln(1 + t) \sim t$ with $t = u + u^2 \rightarrow 0$), the ratio

$$\frac{\ln(1 + u + u^2)}{u} \rightarrow 1$$
, giving the limit -1 .

INSIGHT

At $-\infty$ the exponential with the smallest growth rate dominates a sum of exponentials, so the leading log is just x ; the trap is that the discarded subdominant tail, of exact order e^x , is precisely what the e^{-x} factor magnifies into the finite answer.

№55 Three Floors Down

● ● ● ● ● L5

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e + \frac{ex}{2}}{x^2}$$

LIMIT

$$\frac{11e}{24}$$

THE TRAP

Reaching for the familiar first-order fact $(1+x)^{1/x} = e(1 - \frac{x}{2}) + o(x)$ makes the entire numerator collapse to $o(x)$, so dividing by x^2 seems to blow up or vanish — tempting the answers 0, ∞ , or 'indeterminate.' The trap is that $o(x)$ is far too crude: it hides a genuine $\frac{11e}{24}x^2$ term. You must expand $(1+x)^{1/x}$ to order x^2 , and crucially that coefficient is NOT $\frac{e}{3}$ (the exponent's quadratic term alone) — the $\frac{w^2}{2}$ contribution adds another $\frac{e}{8}$, giving $\frac{e}{3} + \frac{e}{8} = \frac{11e}{24}$.

Method I - Third-order expansion of $(1+x)^{1/x}$

1. Write $(1+x)^{1/x} = \exp\left(\frac{\ln(1+x)}{x}\right)$ and expand the exponent: $\frac{\ln(1+x)}{x} = 1 - \frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots$.
2. So $(1+x)^{1/x} = e \cdot \exp\left(-\frac{x}{2} + \frac{x^2}{3} - \dots\right)$. Let $w = -\frac{x}{2} + \frac{x^2}{3} - \dots$ and use $e^w = 1 + w + \frac{w^2}{2} + \dots$.
3. Collect to order x^2 : $w = -\frac{x}{2} + \frac{x^2}{3}$, $\frac{w^2}{2} = \frac{1}{2} \cdot \left(-\frac{x}{2}\right)^2 = \frac{x^2}{8}$, so

$$e^w = 1 - \frac{x}{2} + \left(\frac{1}{3} + \frac{1}{8}\right)x^2 + \dots = 1 - \frac{x}{2} + \frac{11}{24}x^2 + \dots$$
4. Therefore $(1+x)^{1/x} = e - \frac{ex}{2} + \frac{11e}{24}x^2 + \dots$.
5. The numerator $(1+x)^{1/x} - e + \frac{ex}{2} = \frac{11e}{24}x^2 + O(x^3)$, so dividing by x^2 gives $\boxed{\frac{11e}{24}}$.

Method II - Repeated L'Hopital with care

1. Let $f(x) = (1+x)^{1/x}$ for $x \neq 0$, $f(0) = e$. We need $\lim_{x \rightarrow 0} \frac{f(x) - e + \frac{ex}{2}}{x^2}$, a $\frac{0}{0}$ form. Since f is smooth at 0, Taylor's theorem gives this limit as $\frac{1}{2}f''(0)$ provided $f(0) = e$ and $f'(0) = -\frac{e}{2}$ (so the linear part of the numerator cancels).
2. Compute f' via $\ln f = \frac{\ln(1+x)}{x}$: $\frac{f'}{f} = \frac{d}{dx} \frac{\ln(1+x)}{x} = \frac{\frac{x}{1+x} - \ln(1+x)}{x^2}$.
3. Expanding the bracket, $\frac{x}{1+x} - \ln(1+x) = (x - x^2 + x^3 - \dots) - (x - \frac{x^2}{2} + \frac{x^3}{3} - \dots) = -\frac{x^2}{2} + \frac{2x^3}{3} - \dots$, so $\frac{f'}{f} = -\frac{1}{2} + \frac{2x}{3} - \dots$. At $x = 0$: $f'(0) = e \cdot \left(-\frac{1}{2}\right) = -\frac{e}{2}$, confirming the cancellation of the linear term.
4. For $f''(0)$, write $f' = f\left(-\frac{1}{2} + \frac{2x}{3} + \dots\right)$ and differentiate: $f'' = f'\left(-\frac{1}{2} + \dots\right) + f\left(\frac{2}{3} + \dots\right)$. At $x = 0$:

$$f''(0) = \left(-\frac{e}{2}\right)\left(-\frac{1}{2}\right) + e \cdot \frac{2}{3} = \frac{e}{4} + \frac{2e}{3} = \frac{11e}{12}.$$
5. The limit is $\frac{1}{2}f''(0) = \frac{1}{2} \cdot \frac{11e}{12} = \frac{11e}{24}$.

INSIGHT

$(1+x)^{1/x} = e\left(1 - \frac{x}{2} + \frac{11x^2}{24} - \frac{7x^3}{16} + \dots\right)$ is the engine behind a whole family of brutal limits.

№56 Change of Base, Change of Heart

●●●●● L3

$$\lim_{x \rightarrow 1} \left(\log_x e - \frac{1}{x-1} \right)$$

LIMIT

$$\frac{1}{2}$$

THE TRAP

A student writes $\log_x e = \frac{1}{\ln x}$, sees that both $\frac{1}{\ln x}$ and $\frac{1}{x-1}$ blow up, and uses $\ln x \approx x - 1$ to claim the two infinities cancel exactly, giving 0. The leading $\frac{1}{t}$ singularities do cancel — but stopping at the first-order approximation $\ln(1+t) \approx t$ discards the $-\frac{t^2}{2}$ correction, which is precisely what survives in the limit and contributes $+\frac{1}{2}$. The correct answer is $\frac{1}{2}$, not 0.

Method I - Common denominator + series

1. Change base: $\log_x e = \frac{\ln e}{\ln x} = \frac{1}{\ln x}$. So the expression is $\frac{1}{\ln x} - \frac{1}{x-1} = \frac{(x-1) - \ln x}{(x-1)\ln x}$.
2. Let $t = x - 1 \rightarrow 0$. Then $\ln x = \ln(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \dots$, so the numerator is $(x-1) - \ln x = t - (t - \frac{t^2}{2} + \frac{t^3}{3} - \dots) = \frac{t^2}{2} - \frac{t^3}{3} + \dots$.
3. The denominator is $(x-1)\ln x = t(t - \frac{t^2}{2} + \dots) = t^2 - \frac{t^3}{2} + \dots$.
4. Form the ratio and divide top and bottom by t^2 :

$$\frac{\frac{t^2}{2} - \frac{t^3}{3} + \dots}{t^2 - \frac{t^3}{2} + \dots} = \frac{\frac{1}{2} - \frac{t}{3} + \dots}{1 - \frac{t}{2} + \dots} \xrightarrow{t \rightarrow 0} \frac{1}{2}.$$

The same limit is reached from $t \rightarrow 0^+$ and $t \rightarrow 0^-$, so the two-sided limit is $\frac{1}{2}$.

Method II - Direct Laurent expansion of $1/\ln x$

1. With $t = x - 1$, write $\ln(1+t) = t(1 - \frac{t}{2} + \frac{t^2}{3} - \dots)$, so

$$\frac{1}{\ln(1+t)} = \frac{1}{t} \cdot \frac{1}{1 - \frac{t}{2} + \frac{t^2}{3} - \dots} = \frac{1}{t} \left(1 + \frac{t}{2} + \left(\frac{1}{4} - \frac{1}{3}\right)t^2 + \dots \right).$$
2. Hence $\frac{1}{\ln x} = \frac{1}{t} + \frac{1}{2} - \frac{t}{12} + \dots$, where the t -coefficient is $\frac{1}{4} - \frac{1}{3} = -\frac{1}{12}$.
3. Subtract $\frac{1}{x-1} = \frac{1}{t}$: $\frac{1}{\ln x} - \frac{1}{x-1} = \frac{1}{2} - \frac{t}{12} + \dots$.
4. As $x \rightarrow 1$ (i.e. $t \rightarrow 0$) the principal part cancels and the limit is $\frac{1}{2}$, valid from both sides.

INSIGHT

The expansion $\frac{1}{\ln(1+t)} = \frac{1}{t} + \frac{1}{2} - \frac{t}{12} + \dots$ is a workhorse; its constant term $\frac{1}{2}$ is the hidden residue that the naive cancellation argument misses.

№57 Sine Under the Log

●●●●● L4

$$\lim_{x \rightarrow 0^+} \frac{\ln(\sin x) - \ln x}{x^2}$$

log-of-ratio

one-sided

$x \rightarrow 0^+$

$\sin x$ -over- x

LIMIT

$$-\frac{1}{6}$$

THE TRAP

Two traps. (1) The numerator is $-\infty - (-\infty)$, so a student panics or, worse, declares the limit 0 by replacing $\sin x \approx x$ at first order, getting $\ln \frac{x}{x} = \ln 1 = 0$ over x^2 — the linear approximation is exactly what kills the answer, since the whole limit lives in the x^2 correction. (2) Brute-force L'Hopital on $\ln(\sin x) - \ln x$ produces $\frac{\cos x}{\sin x} - \frac{1}{x}$, an indeterminate $\infty - \infty$ that most students mishandle. The clean move is $\ln(\sin x) - \ln x = \ln \frac{\sin x}{x}$, turning a swamp into one line.

Method I - Combine logs, expand

1. Combine the logs: $\ln(\sin x) - \ln x = \ln\left(\frac{\sin x}{x}\right)$, valid for small $x > 0$ where $\sin x > 0$ (this is why the limit is taken one-sided).
2. Expand the ratio: $\frac{\sin x}{x} = \frac{x - \frac{x^3}{6} + \frac{x^5}{120} - \dots}{x} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \dots$.
3. Take its log with $\ln(1+u) = u - \frac{u^2}{2} + \dots$ and $u = -\frac{x^2}{6} + \frac{x^4}{120} - \dots$. The u^2 term is $O(x^4)$, so

$$\ln\left(\frac{\sin x}{x}\right) = -\frac{x^2}{6} + O(x^4).$$
4. Divide by x^2 : $\frac{1}{x^2}\left(-\frac{x^2}{6} + O(x^4)\right) = -\frac{1}{6} + O(x^2) \rightarrow -\frac{1}{6}$.

Method II - L'Hopital done carefully

1. Rewrite as $\frac{g(x)}{x^2}$ with $g(x) = \ln \frac{\sin x}{x}$. Since $\frac{\sin x}{x} \rightarrow 1$, we have $g(x) \rightarrow 0$, so this is a genuine $\frac{0}{0}$ form (unlike the raw $-\infty - (-\infty)$).
2. Differentiate once: $\frac{d}{dx} \ln \frac{\sin x}{x} = \frac{\cos x}{\sin x} - \frac{1}{x} = \frac{x \cos x - \sin x}{x \sin x}$, and $\frac{d}{dx} x^2 = 2x$. The new ratio is $\frac{x \cos x - \sin x}{2x^2 \sin x}$.
3. Expand the numerator: $x \cos x - \sin x = x\left(1 - \frac{x^2}{2} + \dots\right) - \left(x - \frac{x^3}{6} + \dots\right) = -\frac{x^3}{3} + O(x^5)$. The denominator $2x^2 \sin x = 2x^3 + O(x^5)$.
4. Hence the ratio $\rightarrow \frac{-1/3}{2} = -\frac{1}{6}$, consistent with Method I.

INSIGHT

Because $\ln \frac{\sin x}{x} = -\frac{x^2}{6} - \frac{x^4}{180} - \frac{x^6}{2835} - \dots$, this limit is just the leading coefficient of a classical even series; combining the logs first is the entire trick.

№58 The Slowly Widening Log

● ● ● ● ● L5

$$\lim_{x \rightarrow \infty} \left(\frac{\ln(x+1)}{\ln x} \right)^{x \ln x}$$

1^∞ infinity

base-change

$x \rightarrow$ infinity

$\ln(x+1)$ -over- $\ln-x$

LIMIT

e

THE TRAP

Since $\frac{\ln(x+1)}{\ln x} \rightarrow 1$, it is tempting to call the base 1 and declare the answer $1^\infty = 1$. But the base exceeds 1 by an amount $\sim \frac{1}{x \ln x}$ (because $\ln(x+1) - \ln x = \ln(1 + \frac{1}{x}) \sim \frac{1}{x}$, divided by $\ln x$), and the exponent is precisely $x \ln x$. Their product tends to $\lim x \ln(1 + \frac{1}{x}) = 1$, so $\ln L = 1$ and $L = e$, not 1.

Method I - Log and nested expansion

1. Let L be the limit; then $\ln L = \lim_{x \rightarrow \infty} x \ln x \cdot \ln\left(\frac{\ln(x+1)}{\ln x}\right)$.
2. Write $\ln(x+1) = \ln x + \ln(1 + \frac{1}{x})$, so $\frac{\ln(x+1)}{\ln x} = 1 + \frac{\ln(1+1/x)}{\ln x}$. Put $\delta = \frac{\ln(1+1/x)}{\ln x} \rightarrow 0$.
3. Since $\ln(1 + \delta) = \delta - \frac{\delta^2}{2} + \dots$ and $\delta = \frac{1}{x \ln x}(1 - \frac{1}{2x} + \dots)$ is of order $\frac{1}{x \ln x}$, the δ^2 term is negligible after multiplying by $x \ln x$:

$$\ln\left(\frac{\ln(x+1)}{\ln x}\right) = \frac{\ln(1 + 1/x)}{\ln x} + O\left(\frac{1}{(x \ln x)^2}\right).$$

4. Multiply by $x \ln x$; the $\ln x$ factors cancel: $x \ln x \cdot \frac{\ln(1+1/x)}{\ln x} = x \ln(1 + \frac{1}{x}) = 1 - \frac{1}{2x} + \dots \rightarrow 1$.
5. Therefore $\ln L = 1$, so $L = e$. The exponent $x \ln x$ is tuned exactly to cancel the $\frac{1}{\ln x}$ and leave $x \ln(1 + 1/x) \rightarrow 1$.

Method II - Substitution and asymptotics

1. Set $u = \frac{1}{x} \rightarrow 0^+$, so $\ln x = -\ln u$ and $\ln(x+1) = \ln x + \ln(1+u) = -\ln u + \ln(1+u)$. The base is $1 + \frac{\ln(1+u)}{-\ln u}$.
2. The exponent is $x \ln x = \frac{-\ln u}{u}$. Hence $\ln L = \lim_{u \rightarrow 0^+} \frac{-\ln u}{u} \cdot \ln\left(1 + \frac{\ln(1+u)}{-\ln u}\right)$.
3. Since $\frac{\ln(1+u)}{-\ln u} \rightarrow 0$, we have $\ln(1 + \dots) \sim \frac{\ln(1+u)}{-\ln u}$, so the $-\ln u$ factors cancel: $\ln L = \lim_{u \rightarrow 0^+} \frac{-\ln u}{u} \cdot \frac{\ln(1+u)}{-\ln u} = \lim_{u \rightarrow 0^+} \frac{\ln(1+u)}{u}$.
4. That standard limit equals 1, so $\ln L = 1$ and $L = e$.

INSIGHT

The exponent $x \ln x$ is the unique scaling that makes this 1^∞ form converge to a finite nonzero value: exponent $\sim x$ gives $L = 1$, exponent $\sim x(\ln x)^2$ gives $L = \infty$, and only $x \ln x$ lands on e .

№59 A Tale of Two Sides

● ● ● ● ● L4

$$\lim_{x \rightarrow 0} (1 + |x|)^{1/x}$$

nonexistence

one-sided

absolute-value

1^infinity

x→0

LIMIT

does not exist (right-hand limit e , left-hand limit e^{-1})

THE TRAP

Pattern-matching to the famous limit $(1+x)^{1/x} \rightarrow e$, a student mentally erases the bars and answers e . But the base carries $|x|$ (an even quantity) while the exponent carries $1/x$ (odd). Their interaction puts a $\text{sgn}(x)$ in the exponent of the limiting form, so the right side gives e^{+1} and the left side e^{-1} . Because $e \neq e^{-1}$, the two-sided limit cannot exist — yet both one-sided limits are perfectly finite, which is exactly what makes the wrong single answer feel safe.

Method I - Separate one-sided analysis

1. For $x > 0$ we have $|x| = x$, so the expression is the classic $(1 + x)^{1/x}$. Hence $\lim_{x \rightarrow 0^+} (1 + |x|)^{1/x} = e$.
2. For $x < 0$ we have $|x| = -x$, so the expression is $(1 - x)^{1/x} = \exp\left(\frac{\ln(1 - x)}{x}\right)$. Using $\ln(1 - x) = -x - \frac{x^2}{2} - \dots$ gives $\frac{\ln(1 - x)}{x} = -1 - \frac{x}{2} - \dots \rightarrow -1$, so $\lim_{x \rightarrow 0^-} (1 + |x|)^{1/x} = e^{-1}$.
3. The one-sided limits are e (right) and e^{-1} (left).
4. Since $e \neq e^{-1}$, the right- and left-hand limits disagree, so the two-sided limit does not exist.

Method II - Signum exponent via logarithm

1. Let $L(x) = \ln((1 + |x|)^{1/x}) = \frac{1}{x} \ln(1 + |x|)$. Expand $\ln(1 + |x|) = |x| - \frac{|x|^2}{2} + \dots$, so $L(x) = \frac{|x|}{x} \left(1 - \frac{|x|}{2} + \dots\right)$.
2. Here $\frac{|x|}{x} = \text{sgn}(x)$ equals $+1$ for $x > 0$ and -1 for $x < 0$, while the bracket $\rightarrow 1$ as $x \rightarrow 0$.
3. Therefore $L(x) \rightarrow \text{sgn}(x) \cdot 1 = \pm 1$, the sign depending solely on the side of approach.
4. Exponentiating, $(1 + |x|)^{1/x} \rightarrow e^{\pm 1} = e$ from the right and e^{-1} from the left. The two values differ, so the limit does not exist.

INSIGHT

Whenever a 1^∞ form pairs an even quantity in the base (here $|x|$) with an odd quantity in the exponent ($1/x$, or equivalently sgn), expect a sign-driven split: the limiting exponent picks up a factor $\text{sgn}(x)$, forcing reciprocal one-sided values $e^{\pm 1}$ and two-sided non-existence.

№60 The Root That Refuses to Settle

● ● ● ● ● L5

$$\lim_{x \rightarrow \infty} \frac{x(x^{1/x} - 1) - \ln x}{(\ln x)^2/x}$$

$x^{1/x}$

second-order

$x \rightarrow \infty$

exp-of-log-over-x

LIMIT

$$\frac{1}{2}$$

THE TRAP

Knowing $x^{1/x} \rightarrow 1$, a student writes $x^{1/x} - 1 \approx \frac{\ln x}{x}$, so $x(x^{1/x} - 1) \approx \ln x$ and the numerator looks like 0 — collapsing the whole thing to 0. That is only the first-order truth; the second-order term of $x^{1/x} - 1$, namely $\frac{(\ln x)^2}{2x^2}$, is exactly the surviving piece, and after multiplying by x and dividing by $(\ln x)^2/x$ it leaves $\frac{1}{2}$, not 0.

Method I - Second-order expansion

1. Write $x^{1/x} = \exp\left(\frac{\ln x}{x}\right)$ and set $u = \frac{\ln x}{x} \rightarrow 0$. Then $x^{1/x} = 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + \dots$.
2. So $x^{1/x} - 1 = u + \frac{u^2}{2} + \dots$ and $x(x^{1/x} - 1) = xu + \frac{xu^2}{2} + \dots$. Since $xu = \ln x$ and $xu^2 = x \cdot \frac{(\ln x)^2}{x^2} = \frac{(\ln x)^2}{x}$:
$$x(x^{1/x} - 1) = \ln x + \frac{(\ln x)^2}{2x} + O\left(\frac{(\ln x)^3}{x^2}\right).$$

3. Subtract $\ln x$: the numerator equals $\frac{(\ln x)^2}{2x} + O\left(\frac{(\ln x)^3}{x^2}\right)$.

4. Divide by $\frac{(\ln x)^2}{x}$: $\frac{\frac{(\ln x)^2}{2x} + \cdots}{(\ln x)^2/x} = \frac{1}{2} + O\left(\frac{\ln x}{x}\right) \rightarrow \frac{1}{2}$.

Method II - Substitution to a clean form

1. Let $t = \frac{\ln x}{x} \rightarrow 0^+$. Note $x^{1/x} = e^t$, and the numerator is $x(e^t - 1) - \ln x = x(e^t - 1) - xt = x(e^t - 1 - t)$.

2. The denominator is $\frac{(\ln x)^2}{x} = \frac{(xt)^2}{x} = xt^2$. Hence the whole expression simplifies exactly to

$$\frac{x(e^t - 1 - t)}{xt^2} = \frac{e^t - 1 - t}{t^2}.$$

3. This is a standard $\frac{0}{0}$ limit as $t \rightarrow 0$: since $e^t - 1 - t = \frac{t^2}{2} + \frac{t^3}{6} + \cdots$, the ratio $\rightarrow \frac{1}{2}$.

4. Therefore the limit is $\frac{1}{2}$, the universal second-order coefficient of the exponential, in disguise.

INSIGHT

After the substitution $t = \frac{\ln x}{x}$ the monster collapses to $\lim_{t \rightarrow 0} \frac{e^t - 1 - t}{t^2} = \frac{1}{2}$ – proof that the right change of variable can tame any $x^{1/x}$ expansion.

CHAPTER VII

Advanced Trigonometry

sin

Beyond $\sin x$ — nested angles, telescopes, Taylor cancellations.

№61 The Iterated Sine Crawls Home

●●●●● L4

Let $x_1 \in (0, \pi)$ and $x_{n+1} = \sin(x_n)$. $\lim_{n \rightarrow \infty} n x_n^2$

iterated-sine

recursion

cesaro-stolz

asymptotics

$x \rightarrow \infty$

LIMIT

3

THE TRAP

"Since $x_n \rightarrow 0$, $n x_n^2$ is $\infty \cdot 0$, indeterminate — and x_n surely shrinks geometrically, so $n x_n^2 \rightarrow 0$." Wrong: $\sin$ is tangent to the line $y = x$ at 0 ($\sin x = x - x^3/6 + \dots$, fixed-point multiplier exactly 1), so the iteration loses its contraction at the limit and the decay is only algebraic, $x_n \sim \sqrt{3/n}$, not geometric. The geometric instinct gives 0; the true value is 3.

Method I - Stolz–Cesàro on the reciprocal squares

1. Since $0 < x_{n+1} = \sin x_n < x_n$ for $x_n \in (0, \pi)$, the sequence is positive and strictly decreasing, hence converges to a limit $L \geq 0$ with $L = \sin L$, forcing $L = 0$. So $x_n \rightarrow 0^+$.
2. Study $y_n = 1/x_n^2$. Using $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$, so $\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \dots$
3. Squaring,

$$x_{n+1}^2 = \sin^2 x_n = x_n^2 \left(1 - \frac{x_n^2}{6} + \frac{x_n^4}{120} - \dots\right)^2 = x_n^2 \left(1 - \frac{x_n^2}{3} + \frac{2x_n^4}{45} - \dots\right).$$

4. Invert and expand $\frac{1}{1 - \frac{x_n^2}{3} + \frac{2x_n^4}{45} - \dots} = 1 + \frac{x_n^2}{3} + \left(\frac{1}{9} - \frac{2}{45}\right)x_n^4 + \dots = 1 + \frac{x_n^2}{3} + \frac{x_n^4}{15} + \dots$, giving

$$\frac{1}{x_{n+1}^2} = \frac{1}{x_n^2} + \frac{1}{3} + \frac{1}{15}x_n^2 + \dots$$

5. Hence $y_{n+1} - y_n \rightarrow \frac{1}{3}$ (the x_n^2 correction $\rightarrow 0$). By Stolz–Cesàro,

$$\frac{y_n}{n} \rightarrow \frac{1}{3} \implies \frac{1}{n x_n^2} \rightarrow \frac{1}{3}.$$

6. Therefore $n x_n^2 \rightarrow 3$.

Method II - Continuous comparison (heuristic made rigorous)

1. Model the recursion by the increment it produces on $1/x^2$: from Method I, one step changes $1/x_n^2$ by $\frac{1}{3} + O(x_n^2)$. Ignoring the small correction, treat $\frac{d}{dn}(\frac{1}{x^2}) \approx \frac{1}{3}$.
2. Integrating, $\frac{1}{x_n^2} \approx \frac{n}{3} + C$, i.e. $x_n^2 \approx \frac{3}{n + 3C}$.
3. The constant washes out in the limit: $nx_n^2 \approx \frac{3n}{n + 3C} \rightarrow 3$.
4. Rigor: the exact increment is $\Delta(1/x_n^2) = \frac{1}{3} + O(x_n^2)$ with $x_n^2 \rightarrow 0$, so the average of the increments $\frac{1}{n} \sum_{k=1}^{n-1} \Delta(1/x_k^2) \rightarrow \frac{1}{3}$ (Cesàro mean of a convergent sequence). Thus $\frac{1}{nx_n^2} = \frac{1}{n}(y_1 + \sum_{k=1}^{n-1} \Delta y_k) \rightarrow \frac{1}{3}$, confirming $nx_n^2 \rightarrow 3$.

INSIGHT

General principle: if $f(x) = x - cx^{k+1} + \dots$ with $c > 0$, then iterating $x_{n+1} = f(x_n)$ near the tangential fixed point 0 gives $x_n \sim (ckn)^{-1/k}$; here $\sin x = x - \frac{1}{6}x^3 + \dots$ so $k = 2$, $c = \frac{1}{6}$, yielding $x_n \sim \sqrt{3/n}$ and $nx_n^2 \rightarrow 3$.

№62 Nested Tangent Beats Nested Sine

●●●●● L4

$$\lim_{x \rightarrow 0} \frac{\tan(\tan x) - \sin(\sin x)}{\tan x - \sin x}$$

nested-trig

tan-tan

sin-sin

taylor

$x \rightarrow 0$

ratio

LIMIT

2

THE TRAP

"Both top and bottom vanish, so L'Hopital once and plug in $x = 0$." But one differentiation leaves a fresh 0/0 (numerator and denominator are both $O(x^3)$, not $O(x)$), so a single pass settles nothing. Worse, the tempting shortcut $\tan(\tan x) \approx x \approx \sin(\sin x)$ kills the numerator to first order and yields the bogus answer 1 (or even 0). The whole battle lives in the x^3 terms: keeping them gives numerator $\sim x^3$, denominator $\sim x^3/2$, hence 2.

Method I - Cubic Taylor of both nested functions

1. Expand to order x^3 . With $\tan x = x + \frac{x^3}{3} + \dots$ and $\tan u = u + \frac{u^3}{3} + \dots$,
2.
$$\tan(\tan x) = \left(x + \frac{x^3}{3}\right) + \frac{1}{3}(x)^3 + \dots = x + \frac{2x^3}{3} + \dots$$
3. With $\sin x = x - \frac{x^3}{6} + \dots$ and $\sin u = u - \frac{u^3}{6} + \dots$,
4.
$$\sin(\sin x) = \left(x - \frac{x^3}{6}\right) - \frac{1}{6}x^3 + \dots = x - \frac{x^3}{3} + \dots$$
5. Numerator $= \left(x + \frac{2x^3}{3}\right) - \left(x - \frac{x^3}{3}\right) + \dots = x^3 + \dots$
6. Denominator $= \tan x - \sin x = \left(x + \frac{x^3}{3}\right) - \left(x - \frac{x^3}{6}\right) + \dots = \frac{x^3}{2} + \dots$
7. Ratio $\rightarrow \frac{x^3}{x^3/2} = 2$.

Method II - Split and reuse one known limit

1. Write the numerator as $[\tan(\tan x) - x] - [\sin(\sin x) - x]$.
2. From the cubic expansions: $\tan(\tan x) - x \sim \frac{2}{3}x^3$ and $\sin(\sin x) - x \sim -\frac{1}{3}x^3$.

3. So numerator $\sim \frac{2}{3}x^3 - (-\frac{1}{3}x^3) = x^3$.

4. For the denominator use the standard $\tan x - \sin x = \tan x(1 - \cos x) \sim x \cdot \frac{x^2}{2} = \frac{x^3}{2}$.

5. Therefore the limit is $\frac{x^3}{x^3/2} = 2$.

INSIGHT

The cube coefficients are $+\frac{2}{3}, -\frac{1}{3}$ for $\tan \circ \tan, \sin \circ \sin$; the "2" is $\frac{2/3 - (-1/3)}{1/2}$ – a pure clash of curvatures.

№63 A Pisot Number Hugs the Integers

● ● ● ● ● L5

$$\lim_{n \rightarrow \infty} (2 + \sqrt{3})^n \sin(\pi(2 + \sqrt{3})^n)$$

$\sin(\pi \cdot f)$

near-integer

pisot

conjugate

$x \rightarrow \infty$

sequence

LIMIT

$$-\pi$$

THE TRAP

"sin is bounded by 1 while $(2 + \sqrt{3})^n \rightarrow \infty$, so the product oscillates without limit (or blows up)." Wrong: $(2 + \sqrt{3})^n$ is super-close to an integer, so $\sin(\pi \cdot \text{that}) \rightarrow 0$ at exactly the rate $(2 - \sqrt{3})^n = (2 + \sqrt{3})^{-n}$, and the two effects cancel to a finite, nonzero value. A subtler trap is to forget the parity of that integer and report $+\pi$ instead of $-\pi$.

Method I - Conjugate integer trick

1. Let $a = 2 + \sqrt{3}$, $b = 2 - \sqrt{3}$, so $a + b = 4$, $ab = 1$, and $b \in (0, 1)$. The sequence $N_n = a^n + b^n$ satisfies $N_n = 4N_{n-1} - N_{n-2}$ with $N_0 = 2$, $N_1 = 4$, so each $N_n \in \mathbb{Z}$. Since N_0, N_1 are even and even = 4·even – even, the recurrence preserves parity: every N_n is even.

2. Then $a^n = N_n - b^n$, so

$$\sin(\pi a^n) = \sin(\pi N_n - \pi b^n) = \sin(\pi N_n) \cos(\pi b^n) - \cos(\pi N_n) \sin(\pi b^n).$$

3. Since $N_n \in \mathbb{Z}$, $\sin(\pi N_n) = 0$ and $\cos(\pi N_n) = (-1)^{N_n} = +1$ (as N_n is even). Thus

$$\sin(\pi a^n) = -\sin(\pi b^n).$$

4. Multiply by $a^n = b^{-n}$:

$$a^n \sin(\pi a^n) = -b^{-n} \sin(\pi b^n) = -\pi \cdot \frac{\sin(\pi b^n)}{\pi b^n}.$$

5. As $n \rightarrow \infty$, $b^n \rightarrow 0$ so $\frac{\sin(\pi b^n)}{\pi b^n} \rightarrow 1$. Limit = $-\pi$.

Method II - Distance-to-nearest-integer estimate

1. Write $a^n = N_n - b^n$ where $N_n \in \mathbb{Z}$ and $0 < b^n < 1$, so the fractional distance of a^n below the integer N_n is exactly b^n .

2. Hence $\sin(\pi a^n) = \sin(-\pi b^n + \pi N_n)$; the parity of N_n (always even) makes this $-\sin(\pi b^n)$, with $\sin(\pi b^n) = \pi b^n + O(b^{3n})$.

3. So

$$a^n \sin(\pi a^n) = -a^n (\pi b^n + O(b^{3n})) = -\pi (ab)^n + O((ab)^n b^{2n}) = -\pi + O(b^{2n}),$$

using $ab = 1$.

4. The error $O(b^{2n}) \rightarrow 0$, so the limit is $-\pi$.

INSIGHT

$2 + \sqrt{3}$ is a Pisot number: its conjugate lies inside the unit disk, forcing a^n toward integers at a geometric rate – the engine behind every $\sin(\pi \text{Pisot}^n)$ miracle. The sign of the limit is dictated entirely by the parity of N_n .

№64 The Seventh-Order Duel

● ● ● ● ● L5

$$\lim_{x \rightarrow 0} \frac{\sin(\tan x) - \tan(\sin x)}{x^7}$$

sin-tan

tan-sin

high-order

taylor

$x \rightarrow 0$

cancellation

LIMIT

$$-\frac{1}{30}$$

THE TRAP

"To leading order $\tan x \approx \sin x \approx x$, so $\sin(\tan x) \approx \tan(\sin x)$ and the numerator is negligible compared to x^7 — the limit is 0." The two compositions agree through order x^5 ; the first surviving difference is at order x^7 , exactly the denominator. Truncating any earlier gives the wrong 0.

Method I - Brute Taylor to order seven

1. Use $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \dots$ and $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \dots$

2. Composing carefully (keeping all terms up to x^7):

$$3. \quad \sin(\tan x) = x + \frac{x^3}{6} - \frac{x^5}{40} - \frac{55x^7}{1008} + \dots$$

$$4. \quad \tan(\sin x) = x + \frac{x^3}{6} - \frac{x^5}{40} - \frac{107x^7}{5040} + \dots$$

5. Through x^5 the two are identical (this is the whole trap). Subtract the x^7 terms:

$$-\frac{55}{1008} - \left(-\frac{107}{5040}\right) = -\frac{275}{5040} + \frac{107}{5040} = -\frac{168}{5040} = -\frac{1}{30}.$$

$$6. \text{ Hence } \frac{\sin(\tan x) - \tan(\sin x)}{x^7} \rightarrow -\frac{1}{30}.$$

Method II - Odd symmetry + a single coefficient

1. Both $\sin(\tan x)$ and $\tan(\sin x)$ are odd, so their difference is odd: only odd powers appear, and the agreement through x^5 means the difference starts at x^7 .

2. Write $\sin(\tan x) - \tan(\sin x) = c_7 x^7 + O(x^9)$. The limit is exactly c_7 .

3. To pin c_7 , only the x^7 -coefficients matter. Evaluating the composition coefficients (the x^7 slots) gives $c_7 = -\frac{55}{1008} + \frac{107}{5040} = -\frac{1}{30}$.

4. Therefore the limit equals $-\frac{1}{30}$. (A numeric check at $x = 0.01$ gives $-0.0333372\dots$, matching $-1/30 = -0.0333333\dots$ as $x \rightarrow 0$.)

INSIGHT

$\sin \circ \tan$ and $\tan \circ \sin$ are "Taylor-twins" to fifth order; the first place they part ways is the famous x^7 coefficient $\mp 1/30$.

№65 Viète's Radical Telescope

●●●●● L4

$$\text{Let } a_n = \sqrt{2 + \underbrace{\sqrt{2 + \cdots + \sqrt{2}}}_{n \text{ twos}}}. \quad \lim_{n \rightarrow \infty} 2^n \sqrt{2 - a_n}$$

half-angle

nested-radical

telescope

viète

$x \rightarrow \infty$

sequence

LIMIT

$$\frac{\pi}{2}$$

THE TRAP

" $a_n \rightarrow 2$, so $2 - a_n \rightarrow 0$ and $\sqrt{2 - a_n} \rightarrow 0$, while $2^n \rightarrow \infty$; an $\infty \cdot 0$ form that 'usually' collapses to 0 (the root wins) or blows up to ∞ (the power wins)." Wrong on both counts: the gap obeys $2 - a_n = 4 \sin^2(\frac{\pi}{2^{n+2}}) \sim (\frac{\pi}{4})^2 4^{-n}$, so $\sqrt{2 - a_n} \sim \frac{\pi}{4} 2^{-n}$ decays at EXACTLY the rate 2^{-n} that cancels 2^n . Neither factor wins; they balance to the finite half-angle constant $\frac{\pi}{2}$. (Guessing 0 or ∞ from the indeterminate form, or guessing 1 by analogy with $\frac{\sin x}{x}$, all miss the surviving $\frac{\pi}{2}$.)

Method I - Closed form via half-angle

1. Claim $a_n = 2 \cos(\frac{\pi}{2^{n+1}})$. Base case: $a_1 = \sqrt{2} = 2 \cos \frac{\pi}{4}$, correct since $\frac{\pi}{2^{1+1}} = \frac{\pi}{4}$.
2. Inductive step. Assume $a_n = 2 \cos \frac{\pi}{2^{n+1}}$. Using $2 + 2 \cos \theta = 4 \cos^2 \frac{\theta}{2}$ with $\theta = \frac{\pi}{2^{n+1}}$ (so $\frac{\theta}{2} = \frac{\pi}{2^{n+2}}$ stays in $(0, \frac{\pi}{2})$, where cosine is positive):

$$3. \quad a_{n+1} = \sqrt{2 + a_n} = \sqrt{2 + 2 \cos \frac{\pi}{2^{n+1}}} = \sqrt{4 \cos^2 \frac{\pi}{2^{n+2}}} = 2 \cos \frac{\pi}{2^{n+2}},$$

which is the claim at index $n + 1$. The induction closes.

4. Now use $2 - 2 \cos \phi = 4 \sin^2 \frac{\phi}{2}$ with $\phi = \frac{\pi}{2^{n+1}}$:

$$2 - a_n = 2 - 2 \cos \frac{\pi}{2^{n+1}} = 4 \sin^2 \frac{\pi}{2^{n+2}} \implies \sqrt{2 - a_n} = 2 \sin \frac{\pi}{2^{n+2}}.$$

5. Hence

$$2^n \sqrt{2 - a_n} = 2^n \cdot 2 \sin \frac{\pi}{2^{n+2}} = 2^{n+1} \sin \frac{\pi}{2^{n+2}}.$$

6. Set $\theta_n = \frac{\pi}{2^{n+2}} \rightarrow 0^+$. Then

$$2^{n+1} \sin \theta_n = (2^{n+1} \theta_n) \frac{\sin \theta_n}{\theta_n} = \frac{\pi}{2} \cdot \frac{\sin \theta_n}{\theta_n} \xrightarrow{n \rightarrow \infty} \frac{\pi}{2},$$

since $2^{n+1} \theta_n = 2^{n+1} \cdot \frac{\pi}{2^{n+2}} = \frac{\pi}{2}$ for every n and $\frac{\sin \theta_n}{\theta_n} \rightarrow 1$. Therefore the limit is $\boxed{\frac{\pi}{2}}$.

Method II - Gap recursion + monotone convergence (no closed form assumed)

1. Let $\varepsilon_n = 2 - a_n > 0$ (positivity holds since $a_n < 2$, a standard induction on $a_n = \sqrt{2 + a_{n-1}} < \sqrt{2 + 2} = 2$). From $a_n^2 = 2 + a_{n-1}$, substitute $a_k = 2 - \varepsilon_k$:

$$(2 - \varepsilon_n)^2 = 2 + (2 - \varepsilon_{n-1}).$$

2. Expanding $4 - 4\varepsilon_n + \varepsilon_n^2 = 4 - \varepsilon_{n-1}$ gives the exact gap recursion

$$\varepsilon_{n-1} = 4\varepsilon_n - \varepsilon_n^2 = 4\varepsilon_n \left(1 - \frac{\varepsilon_n}{4}\right).$$

3. Define $c_n = 2^n \sqrt{\varepsilon_n}$ — this is precisely the quantity whose limit we want. Take the ratio of consecutive terms and use the recursion:

$$\frac{c_{n-1}}{c_n} = \frac{2^{n-1} \sqrt{\varepsilon_{n-1}}}{2^n \sqrt{\varepsilon_n}} = \frac{1}{2} \sqrt{\frac{\varepsilon_{n-1}}{\varepsilon_n}} = \frac{1}{2} \sqrt{4 - \varepsilon_n} = \sqrt{1 - \frac{\varepsilon_n}{4}}.$$

4. Since $0 < \varepsilon_n < 2$, we have $0 < \sqrt{1 - \varepsilon_n/4} < 1$, so $c_{n-1} < c_n$: the sequence c_n is strictly increasing. This already kills the trap — an $\infty \cdot 0$ form that is monotone increasing cannot tend to 0.
5. Boundedness: from the recursion $\varepsilon_n = 2 - a_n$ with $a_n = \sqrt{2 + a_{n-1}} \rightarrow 2$ one shows $\varepsilon_n \rightarrow 0$, and writing $\sqrt{\varepsilon_n} = 2 \sin \theta_n$ with $\theta_n = \frac{1}{2} \arccos(1 - \frac{\varepsilon_n}{2})$ gives $\theta_n = \frac{\pi}{2^{n+2}}$ (the unique angle reproducing the half-angle ratio $c_{n-1}/c_n = \cos \theta_n$). Because $\sin \theta \leq \theta$,

$$c_n = 2^{n+1} \sin \theta_n \leq 2^{n+1} \theta_n = 2^{n+1} \cdot \frac{\pi}{2^{n+2}} = \frac{\pi}{2}.$$

So c_n is increasing and bounded above by $\frac{\pi}{2}$, hence convergent to some $L \leq \frac{\pi}{2}$.

6. Identify L . From $c_n = 2^{n+1} \sin \theta_n$ with $2^{n+1} \theta_n = \frac{\pi}{2}$ fixed,

$$L = \lim_{n \rightarrow \infty} 2^{n+1} \sin \theta_n = \frac{\pi}{2} \lim_{n \rightarrow \infty} \frac{\sin \theta_n}{\theta_n} = \frac{\pi}{2}.$$

The monotone bound $L \leq \frac{\pi}{2}$ is attained, so

$$\lim_{n \rightarrow \infty} 2^n \sqrt{2 - a_n} = \frac{\pi}{2}.$$

INSIGHT

This is Viète's 1593 product in disguise: dividing the closed form telescopes into $\frac{2}{\pi} = \prod_{k \geq 1} \cos \frac{\pi}{2^{k+1}}$, and $a_n = 2 \cos \frac{\pi}{2^{n+1}}$ is its partial cosine tower. Equivalently $4^n(2 - a_n) \rightarrow \pi^2/4$.

№66 A Power Tower on the Sinc

●●●●● L4

$$\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{1 - \cos x}}$$

one-to-infinity

log-transform

sinc

second-order

$x \rightarrow 0$

LIMIT

$$e^{-1/3}$$

THE TRAP

"The base $\frac{\sin x}{x} \rightarrow 1$, and 1 raised to anything is 1, so the limit is 1." Wrong: this is a 1^∞ indeterminate form. The exponent blows up like $2/x^2$, exactly matching the x^2 -order deviation of the base below 1, so the two effects balance into a nontrivial power of e . A subtler trap is to use only the first-order base estimate $\sin x/x \approx 1$ (giving 0 in the exponent product) instead of the x^2 term $-x^2/6$.

Method I - Logarithm and second-order expansion

1. Let $L = \left(\frac{\sin x}{x} \right)^{1/(1 - \cos x)}$. Then

$$\ln L = \frac{\ln(\sin x/x)}{1 - \cos x}.$$

2. Numerator: $\frac{\sin x}{x} = 1 - \frac{x^2}{6} + O(x^4)$, so $\ln \frac{\sin x}{x} = -\frac{x^2}{6} + O(x^4)$.

3. Denominator: $1 - \cos x = \frac{x^2}{2} + O(x^4)$.

4. Therefore

$$\ln L = \frac{-x^2/6 + O(x^4)}{x^2/2 + O(x^4)} \rightarrow \frac{-1/6}{1/2} = -\frac{1}{3}.$$

5. Exponentiating, $L \rightarrow e^{-1/3}$. (The form is two-sided: base and exponent are both even functions of x .)

Method II - Reduce to the canonical 1^∞ rule

1. For a 1^∞ form $f(x)^{g(x)}$ with $f \rightarrow 1$, $g \rightarrow \infty$, the limit is $e^{\lim g(x)(f(x)-1)}$.

2. Here $f - 1 = \frac{\sin x}{x} - 1 \sim -\frac{x^2}{6}$ and $g = \frac{1}{1 - \cos x} \sim \frac{2}{x^2}$.

3. Product: $g(f - 1) \sim \frac{2}{x^2} \cdot \left(-\frac{x^2}{6}\right) = -\frac{1}{3}$.

4. Hence the limit is $e^{-1/3}$. (Numeric: at $x = 0.01$ the expression is $0.716529\dots$ versus $e^{-1/3} = 0.716531\dots$.)

INSIGHT

Swapping the exponent $1 - \cos x$ for x^2 would halve the constant to $-1/6$; the half-angle factor $1 - \cos x \sim x^2/2$ is precisely what doubles the exponent.

67 The Floor That Drops to Zero

●●●●● L3

$$\lim_{x \rightarrow 0^+} \lfloor x \cot x \rfloor$$

floor

one-sided

$x \rightarrow 0^+$

cot

squeeze

LIMIT

0

THE TRAP

" $x \cot x = \frac{x \cos x}{\sin x} \rightarrow 1$, so $\lfloor x \cot x \rfloor \rightarrow \lfloor 1 \rfloor = 1$." The floor is discontinuous exactly at 1: since $x \cot x < 1$ for all small $x > 0$, the floor sits at 0, not 1. Substituting the limit before flooring is illegal.

Method I - Strict inequality $\tan x > x$

1. For $x \in (0, \pi/2)$ the strict inequality $\tan x > x$ holds (the tangent line lies below the curve).

2. Hence $\frac{x}{\tan x} < 1$, i.e. $x \cot x < 1$ for all such x .

3. Also $x \cot x > 0$ on $(0, \pi/2)$ since $\sin x$ and $\cos x$ are both positive. So $0 < x \cot x < 1$, giving $\lfloor x \cot x \rfloor = 0$ for every small $x > 0$.

4. A constant-0 function has limit 0: $\lim_{x \rightarrow 0^+} \lfloor x \cot x \rfloor = 0$.

Method II - Series sandwich

1. Expand $x \cot x = 1 - \frac{x^2}{3} - \frac{x^4}{45} - \dots$, a sum of 1 minus strictly positive terms for $0 < x < \pi$.
2. Thus $x \cot x = 1 - \frac{x^2}{3} + o(x^2) < 1$, and clearly $x \cot x > 0$ near 0.
3. So $0 < x \cot x < 1$ on a punctured right-neighbourhood of 0, forcing $\lfloor x \cot x \rfloor \equiv 0$ there.
4. Therefore the one-sided limit is 0. (By evenness of $x \cot x$, the left-hand limit is also 0, but the value 1 is never attained.)

INSIGHT

The deep point: $\lim$ and $\lfloor \cdot \rfloor$ commute only when the limit is not an integer; approaching an integer from below collapses the floor by one.

№68 Three Cosines in Lockstep

●●●●● L3

$$\lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x \cos 3x}{x^2}$$

product

cosine

second-order

$x \rightarrow 0$

weighted-sum

LIMIT

7

THE TRAP

"Each factor $\cos(kx)$ contributes a $\frac{x^2}{2}$ deficit, so just add three halves: $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$." This silently treats all three cosines as the same frequency. In fact $1 - \cos(kx) \sim \frac{(kx)^2}{2}$, so the deficits are $\frac{1^2}{2}, \frac{2^2}{2}, \frac{3^2}{2}$ and the limit is $\frac{1+4+9}{2} = 7$, not $\frac{3}{2}$. The faster-oscillating factors dominate.

Method I - Logarithmic linearization of the product

1. Near 0, $\ln \cos(kx) = -\frac{(kx)^2}{2} + O(x^4)$, so

$$\ln(\cos x \cos 2x \cos 3x) = -\frac{x^2}{2}(1^2 + 2^2 + 3^2) + O(x^4) = -7x^2 + O(x^4).$$

2. Exponentiate using $e^u = 1 + u + O(u^2)$ with $u = -7x^2 + O(x^4)$:

$$\cos x \cos 2x \cos 3x = 1 - 7x^2 + O(x^4).$$

3. Therefore

$$\frac{1 - \cos x \cos 2x \cos 3x}{x^2} = \frac{7x^2 + O(x^4)}{x^2} \rightarrow 7.$$

Method II - Telescoping the difference \$1-P\$

1. Use the exact identity $1 - abc = (1 - a) + a(1 - b) + ab(1 - c)$ with $a = \cos x$, $b = \cos 2x$, $c = \cos 3x$.
2. As $x \rightarrow 0$ we have $a \rightarrow 1$ and $ab \rightarrow 1$, while $1 - \cos(kx) \sim \frac{(kx)^2}{2}$. Hence

$$1 - \cos x \cos 2x \cos 3x \sim \frac{x^2}{2} + 1 \cdot \frac{(2x)^2}{2} + 1 \cdot \frac{(3x)^2}{2}.$$

3. That is $\frac{x^2}{2}(1 + 4 + 9) = 7x^2$.

4. Dividing by x^2 gives the limit 7. (Check: at $x = 10^{-3}$ the quotient is 6.999983...)

INSIGHT

For $\prod_{k=1}^m \cos(kx)$ the second-order rate is $\frac{1}{2} \sum_{k=1}^m k^2 = \frac{m(m+1)(2m+1)}{12}$; here $m = 3$ gives 7.

№69 Sine at the Edge of Minus Infinity

●●●●● L4

$$\lim_{x \rightarrow -\infty} x^3 \left(\sin \frac{1}{x} - \frac{1}{x} \right)$$

$x \rightarrow -\infty$

substitution

$\sin(1/x)$

third-order

cubic-term

LIMIT

$$-\frac{1}{6}$$

THE TRAP

The seductive wrong answer is 0: "As $x \rightarrow -\infty$, $\frac{1}{x} \rightarrow 0$ and $\sin \frac{1}{x} \approx \frac{1}{x}$, so the bracket vanishes and $x^3 \cdot 0 = 0$." The flaw: the bracket is not exactly 0, it is the cubic remainder $\sin \frac{1}{x} - \frac{1}{x} = -\frac{1}{6x^3} + \frac{1}{120x^5} - \dots$, an infinitesimal of the SAME order as the x^3 that multiplies it. The product is a $\frac{0}{0}$ -style $\infty \cdot 0$ race, and the cubic term survives exactly: $x^3 \cdot (-\frac{1}{6x^3}) = -\frac{1}{6}$. First-order truncation throws away the only term that matters.

Method I - Substitution to a $\frac{0}{0}$ -limit

1. Let $t = \frac{1}{x}$. As $x \rightarrow -\infty$, $t \rightarrow 0^-$, and $x^3 = \frac{1}{t^3}$.

2. The expression becomes

$$\frac{\sin t - t}{t^3}.$$

3. Use $\sin t = t - \frac{t^3}{6} + \frac{t^5}{120} - \dots$, so $\sin t - t = -\frac{t^3}{6} + O(t^5)$.

4. Hence $\frac{\sin t - t}{t^3} = -\frac{1}{6} + O(t^2) \rightarrow -\frac{1}{6}$ as $t \rightarrow 0^-$. The limit is $-\frac{1}{6}$.

5. (Equivalently, apply L'Hopital three times to $\frac{\sin t - t}{t^3}$: derivatives give $\frac{\cos t - 1}{3t^2}$, then $\frac{-\sin t}{6t}$, then $\frac{-\cos t}{6} \rightarrow -\frac{1}{6}$.)

Method II - Direct asymptotic expansion in $\frac{1}{x}$

1. For large $|x|$, $\sin \frac{1}{x} = \frac{1}{x} - \frac{1}{6x^3} + \frac{1}{120x^5} - \dots$

2. Subtract $\frac{1}{x}$: $\sin \frac{1}{x} - \frac{1}{x} = -\frac{1}{6x^3} + \frac{1}{120x^5} - \dots$

3. Multiply by x^3 : $x^3 \left(\sin \frac{1}{x} - \frac{1}{x} \right) = -\frac{1}{6} + \frac{1}{120x^2} - \dots$

4. As $x \rightarrow -\infty$ the correction $\frac{1}{120x^2} \rightarrow 0^+$, so the limit is $-\frac{1}{6}$. The side ($-\infty$ vs $+\infty$) is irrelevant: the leading term is the constant $-\frac{1}{6}$ and every correction depends only on x^2 , so both ends give $-\frac{1}{6}$.

INSIGHT

The minus sign of x^3 for $x < 0$ is cancelled by the minus sign of the cubic remainder, so the answer matches the $x \rightarrow +\infty$ case. The genuine size of the surviving piece is set by the $t^3/6$ Taylor coefficient of $\sin t$, which is why $-1/6$ appears.

$$\lim_{x \rightarrow 0} \frac{\tan(\sin x) - \sin(\tan x)}{\arcsin(\arctan x) - \arctan(\arcsin x)}$$

high-order

inverse-trig

arcsin-arctan

$x \rightarrow 0$

ratio

symmetry

LIMIT

−1

THE TRAP

"Inverses just undo the direct functions, so the denominator mirrors the numerator term for term and the ratio is +1." Both numerator and denominator really are $\sim cx^7$ with the SAME magnitude $|c| = \frac{1}{30}$ — which makes +1 feel inevitable. But swapping each direct function for its inverse flips the sign of the surviving seventh-order coefficient (numerator $+\frac{1}{30}x^7$, denominator $-\frac{1}{30}x^7$), so the ratio is −1, not +1.

Method I - Match the two seventh-order coefficients

- Both numerator and denominator are odd functions whose first six Taylor coefficients vanish: $\tan(\sin x)$ and $\sin(\tan x)$ agree through order x^5 , and likewise $\arcsin(\arctan x)$ and $\arctan(\arcsin x)$ agree through order x^5 . So each difference starts at x^7 .

- Numerator: expanding $\tan(\sin x) - \sin(\tan x)$ to order x^7 gives

$$\tan(\sin x) - \sin(\tan x) = +\frac{1}{30}x^7 + O(x^9).$$

- Denominator: using $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$ and $\arcsin x = x + \frac{x^3}{6} + \frac{3x^5}{40} + \dots$, the same order- x^7 computation gives

$$\arcsin(\arctan x) - \arctan(\arcsin x) = -\frac{1}{30}x^7 + O(x^9).$$

$$4. \text{ Ratio} = \frac{+\frac{1}{30}x^7 + O(x^9)}{-\frac{1}{30}x^7 + O(x^9)} = \frac{\frac{1}{30} + O(x^2)}{-\frac{1}{30} + O(x^2)} \xrightarrow{x \rightarrow 0} -1.$$

Method II - Odd-function bookkeeping

- Each of the four nested expressions is odd, and within each pair the two compositions agree through order x^5 , so both numerator $N(x)$ and denominator $D(x)$ are odd with vanishing x, x^3, x^5 terms:

$$N(x) = n_7x^7 + O(x^9), \quad D(x) = d_7x^7 + O(x^9).$$

- Since $D(x)$ has leading order exactly x^7 , D is nonzero on a punctured neighborhood of 0, and the limit equals n_7/d_7 . Computing the seventh coefficients: $n_7 = +\frac{1}{30}$ for the direct (tan/sin) pair and $d_7 = -\frac{1}{30}$ for the inverse (arcsin/arctan) pair.

- Therefore

$$\lim_{x \rightarrow 0} \frac{N(x)}{D(x)} = \frac{n_7}{d_7} = \frac{+1/30}{-1/30} = -1.$$

- Numeric sanity at $x = 0.01$: numerator $\approx +3.334 \times 10^{-16}$, denominator $\approx -3.333 \times 10^{-16}$, ratio ≈ -1.0002 — already close to −1, with the gap $O(x^2)$.

INSIGHT

Direct and inverse trigonometric compositions are seventh-order mirror images: $\pm \frac{1}{30}x^7$ — the cleanest possible expression of 'inverting flips the curvature anomaly.' The matching magnitudes are exactly what makes the +1 trap

so tempting.

CHAPTER VIII

Exotic Indeterminate Forms

1^∞

$0^0, \infty - \infty, \infty^0$ – forms that hide their answer until forced.

№71 The Arctan Mirage

●●●●● L4

$$\lim_{x \rightarrow 0} \left(\frac{\arctan x}{x} \right)^{1/x^2}$$

1^∞ infinity

arctan

$x \rightarrow 0$

taylor

log-transform

LIMIT

$$e^{-1/3}$$

THE TRAP

A strong student notes $\frac{\arctan x}{x} \rightarrow 1$, declares the base equal to 1, and concludes the limit is 1. The flaw: this is a 1^∞ form, where the exponent $1/x^2 \rightarrow \infty$ amplifies the tiny gap between the base and 1. What survives is the *rate*: the base equals $1 - \frac{x^2}{3} + O(x^4)$, and the surviving $O(x^2)$ coefficient is $-\frac{1}{3}$, not 0. So the answer is $e^{-1/3}$, not 1.

Method I — Logarithm + Taylor series

1. Set $L = \lim_{x \rightarrow 0} \left(\frac{\arctan x}{x} \right)^{1/x^2}$ and take logs: $\ln L = \lim_{x \rightarrow 0} \frac{1}{x^2} \ln \frac{\arctan x}{x}$.
2. Use $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$, so $\frac{\arctan x}{x} = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \dots$.
3. With $u = -\frac{x^2}{3} + O(x^4) \rightarrow 0$, $\ln(1+u) = u + O(u^2) = -\frac{x^2}{3} + O(x^4)$.
4. Therefore $\ln L = \lim_{x \rightarrow 0} \frac{1}{x^2} \left(-\frac{x^2}{3} + O(x^4) \right) = -\frac{1}{3}$, giving

$$L = e^{-1/3}.$$

Method II — Standard limit reduction

1. Write the base as $1 + t(x)$ where $t(x) = \frac{\arctan x}{x} - 1$, and note $t(x) \rightarrow 0$.
2. Then $(1+t)^{1/x^2} = \exp\left(\frac{1}{x^2} \cdot \frac{\ln(1+t)}{t} \cdot t\right)$, and $\frac{\ln(1+t)}{t} \rightarrow 1$ since $t \rightarrow 0$.
3. Compute $\lim_{x \rightarrow 0} \frac{t(x)}{x^2} = \lim_{x \rightarrow 0} \frac{\arctan x - x}{x^3}$. By the series $\arctan x - x = -\frac{x^3}{3} + O(x^5)$, this limit is $-\frac{1}{3}$.
4. Hence the exponent $\frac{1}{x^2} \cdot \frac{\ln(1+t)}{t} \cdot t \rightarrow 1 \cdot \left(-\frac{1}{3}\right) = -\frac{1}{3}$, so $L = e^{-1/3}$.

INSIGHT

For any odd function f smooth at 0 with $f(x) = x + ax^3 + O(x^5)$, $(f(x)/x)^{1/x^2} \rightarrow e^a$; here $f = \arctan$ gives $a = -\frac{1}{3}$. (Check: $f = \sin$ gives $a = -\frac{1}{6}$ and $(\frac{\sin x}{x})^{1/x^2} \rightarrow e^{-1/6}$.)

№72 Vanishing Tail at Infinity

●●●●● L4

$$\lim_{x \rightarrow +\infty} \left(\frac{\pi}{2} - \arctan x \right)^{1/\ln x}$$

0^0

$\arctan$

$x \rightarrow +\infty$

log-transform

asymptotics

LIMIT

$$e^{-1}$$

THE TRAP

One sees the base $\rightarrow 0^+$ and the exponent $\rightarrow 0^+$ and reflexively writes $0^0 = 1$. But 0^0 is indeterminate: take logs and the answer is decided by the race $\ln(\text{base})/\ln x$. Here $\text{base} = \arctan(1/x) \sim 1/x$, so $\ln(\text{base}) \sim -\ln x$ and the ratio is -1 , giving e^{-1} , not 1. (Equally tempting: replacing the base by its $O(x^{-2})$ -corrected value and hoping the constant survives — it does not; only the polynomial decay rate matters.)

Method I — Identity $\frac{\pi}{2} - \arctan x = \arctan(1/x)$

- For $x > 0$, $\frac{\pi}{2} - \arctan x = \arctan \frac{1}{x}$. Let L be the limit and take logs: $\ln L = \lim_{x \rightarrow \infty} \frac{\ln \arctan(1/x)}{\ln x}$.
- As $x \rightarrow \infty$, $\arctan(1/x) = \frac{1}{x}(1 + O(x^{-2}))$, so $\ln \arctan(1/x) = \ln \frac{1}{x} + \ln(1 + O(x^{-2})) = -\ln x + o(1)$.
- Thus $\ln L = \lim_{x \rightarrow \infty} \frac{-\ln x + o(1)}{\ln x} = -1$, so

$$L = e^{-1}.$$

Method II — Substitution $t = 1/x$

- Put $t = 1/x \rightarrow 0^+$. Then $\frac{\pi}{2} - \arctan x = \arctan t$ and $\ln x = -\ln t$.
- The expression becomes $(\arctan t)^{-1/\ln t}$, so $\ln L = \lim_{t \rightarrow 0^+} \frac{\ln \arctan t}{-\ln t}$.
- Since $\arctan t = t(1 + O(t^2))$, we get $\ln \arctan t = \ln t + O(t^2)$, giving $\ln L = \lim_{t \rightarrow 0^+} \frac{\ln t + O(t^2)}{-\ln t} = -1$.
- Therefore $L = e^{-1}$.

Method III — Squeeze

- On $[0, 1]$ the function $\arctan$ is concave (its second derivative $-\frac{2t}{(1+t^2)^2} < 0$ for $t > 0$), so it lies above the chord through $(0, 0)$ and $(1, \frac{\pi}{4})$: $\frac{\pi}{4}t \leq \arctan t \leq t$ for $t \in [0, 1]$.
- Set $t = 1/x$ and raise to the positive power $-1/\ln t = 1/\ln x > 0$: $(\frac{\pi}{4}t)^{1/\ln x} \leq (\arctan t)^{1/\ln x} \leq t^{1/\ln x}$.
- The right side $= \exp(\frac{-\ln x}{\ln x}) = e^{-1}$. The left side $= \exp(\frac{\ln(\pi/4) - \ln x}{\ln x}) \rightarrow e^{-1}$, since the bounded constant $\ln(\pi/4)$ is divided by $\ln x \rightarrow \infty$ and washes out.
- Both bounds tend to e^{-1} , so by the squeeze theorem $L = e^{-1}$.

INSIGHT

Only the leading polynomial decay rate of the base matters: any base $\sim c x^{-p}$ raised to the power $1/\ln x$ tends to e^{-p} , independent of the constant c – because $\ln(c x^{-p})/\ln x = -p + \ln c/\ln x \rightarrow -p$.

№73 The Floor That Forgives

●●●●● L5

$$\lim_{x \rightarrow 0^+} (1+x)^{\lfloor 1/x \rfloor}$$

1^infinity

floor

$x \rightarrow 0^+$

fractional-part

squeeze

LIMIT

e

THE TRAP

Because $\lfloor 1/x \rfloor$ jumps by huge integer steps and the fractional part $\{1/x\} = \frac{1}{x} - \lfloor \frac{1}{x} \rfloor$ keeps oscillating densely in $[0, 1)$ as $x \rightarrow 0^+$, many students declare the exponent too wild and call the limit oscillatory / nonexistent. The trap is forgetting that the leftover $\{1/x\}$ is multiplied by $\ln(1+x) = O(x) \rightarrow 0$ in the logarithm, so the oscillation is annihilated and only the clean value e survives.

Method I – Split off the fractional part

1. Write $\lfloor 1/x \rfloor = \frac{1}{x} - \{1/x\}$, where $0 \leq \{1/x\} < 1$ is bounded for every $x > 0$.
2. Take logarithms: $\ln[(1+x)^{\lfloor 1/x \rfloor}] = \left(\frac{1}{x} - \{1/x\}\right) \ln(1+x) = \frac{\ln(1+x)}{x} - \{1/x\} \ln(1+x)$.
3. As $x \rightarrow 0^+$: $\frac{\ln(1+x)}{x} \rightarrow 1$, while $\{1/x\} \ln(1+x)$ is (bounded) $\times O(x) \rightarrow 0$ — here the oscillation is killed.
4. Hence the logarithm tends to $1 - 0 = 1$, and by continuity of $\exp$ the limit is $e^1 = e$.

Method II – Squeeze between consecutive integers

1. For $x > 0$ the floor satisfies $\frac{1}{x} - 1 < \lfloor 1/x \rfloor \leq \frac{1}{x}$. Since the base $1+x > 1$, raising it to a larger exponent gives a larger value, so order is preserved:
2. $(1+x)^{\frac{1}{x}-1} < (1+x)^{\lfloor 1/x \rfloor} \leq (1+x)^{1/x}$.
3. As $x \rightarrow 0^+$, the upper bound $(1+x)^{1/x} \rightarrow e$. The lower bound $= (1+x)^{1/x} \cdot (1+x)^{-1} \rightarrow e \cdot 1 = e$.
4. Both bounds tend to e , so by the squeeze theorem the limit equals

e .

INSIGHT

Replacing $\lfloor 1/x \rfloor$ by $1/x$ is legal precisely because the discarded $\{1/x\}$ couples to $\ln(1+x) = O(x)$, which vanishes. The same substitution would fail if base -1 were not $o(1)$ relative to the size of the floor's jump – e.g. $\lim_{x \rightarrow 0^+} 2^{\lfloor 1/x \rfloor}$ has no finite value at all.

№74 The Logarithm's Second Secret

●●●●● L4

$$\lim_{x \rightarrow +\infty} \left(x - x^2 \ln\left(1 + \frac{1}{x}\right) \right)$$

infinity-minus-infinity

ln

x → +infinity

taylor

collapse

LIMIT

$$\frac{1}{2}$$

THE TRAP

Replacing $\ln(1 + 1/x)$ by its first-order term $1/x$ gives $x - x^2 \cdot \frac{1}{x} = x - x = 0$. That linear approximation is fatally too crude here: x^2 is large, so the next term $-\frac{1}{2x^2}$ in $\ln(1 + 1/x)$ is amplified by x^2 into a finite residue $\frac{1}{2}$. The cancellation you see is only of the divergent parts; the curvature term survives.

Method I — Second-order expansion

1. Substitute $t = 1/x \rightarrow 0^+$. The Maclaurin series is $\ln(1 + t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \dots$

2. Since $x^2 = 1/t^2$,

$$x^2 \ln(1 + 1/x) = \frac{1}{t^2} \left(t - \frac{t^2}{2} + \frac{t^3}{3} - \dots \right) = \frac{1}{t} - \frac{1}{2} + \frac{t}{3} - \dots = x - \frac{1}{2} + \frac{1}{3x} - \dots$$

3. Therefore

$$x - x^2 \ln(1 + 1/x) = x - \left(x - \frac{1}{2} + \frac{1}{3x} - \dots \right) = \frac{1}{2} - \frac{1}{3x} + \dots$$

4. The divergent x terms cancel and letting $x \rightarrow \infty$ leaves

$$\frac{1}{2}.$$

Method II — Substitute and use L'Hôpital

1. Let $t = 1/x \rightarrow 0^+$. Then $x = \frac{1}{t}$, $x^2 = \frac{1}{t^2}$, and the expression becomes

$$\frac{1}{t} - \frac{1}{t^2} \ln(1 + t) = \frac{t - \ln(1 + t)}{t^2}.$$

2. As $t \rightarrow 0^+$ both numerator and denominator vanish, a $\frac{0}{0}$ form. Applying L'Hôpital's rule once:

$$\frac{\frac{d}{dt}(t - \ln(1 + t))}{\frac{d}{dt} t^2} = \frac{1 - \frac{1}{1+t}}{2t} = \frac{\frac{t}{1+t}}{2t} = \frac{1}{2(1+t)}.$$

3. This is no longer indeterminate; as $t \rightarrow 0^+$ it tends to $\frac{1}{2}$. Hence the original limit is

$$\frac{1}{2}.$$

INSIGHT

The $\infty - \infty$ form collapses not to 0 but to the curvature coefficient $\frac{1}{2}$ of $\ln(1 + t)$ at $t = 0$; the seductive linear cancellation hides the quadratic truth. More generally $\lim_{x \rightarrow \infty} (x - x^2 \ln(1 + 1/x))$ equals $-c_2$ where $c_2 = -\frac{1}{2}$ is the second Taylor coefficient of $\ln(1 + t)$.

№75 Fractional Part Meets Itself

● ● ● ● ● L3

$$\lim_{x \rightarrow 1^+} \{x\}^{\{x\}}$$

LIMIT

1

THE TRAP

Seeing $\{x\} \rightarrow 0^+$, a student reads the base-and-exponent as 0^0 and either declares the limit undefined or, treating the base as dominant, guesses 0. The form is genuinely indeterminate, but it resolves to 1: in $t^t = e^{t \ln t}$ the product $t \ln t \rightarrow 0$ because $|\ln t|$ grows far slower than $1/t$, so the exponent — not the base — controls the outcome.

Method I — Substitution $t = \{x\}$

1. For $x \in (1, 2)$ we have $\lfloor x \rfloor = 1$, so $\{x\} = x - 1$. As $x \rightarrow 1^+$ set $t = \{x\} = x - 1 \rightarrow 0^+$.
2. The expression becomes $t^t = e^{t \ln t}$, so it suffices to find $\lim_{t \rightarrow 0^+} t \ln t$.
3. Write $t \ln t = \frac{\ln t}{1/t}$, an $\frac{-\infty}{+\infty}$ form. L'Hôpital gives $\frac{d}{dt} \ln t = \frac{1}{t}$ and $\frac{d}{dt} \frac{1}{t} = -\frac{1}{t^2}$, so the ratio is $\frac{1/t}{-1/t^2} = -t \rightarrow 0$.
4. Hence $t \ln t \rightarrow 0$ and $t^t = e^{t \ln t} \rightarrow e^0 = 1$.
5. Therefore $\lim_{x \rightarrow 1^+} \{x\}^{\{x\}} = 1$.

Method II — Squeeze

1. Again put $t = \{x\} = x - 1 \rightarrow 0^+$ and study $t^t = e^{t \ln t}$ for $0 < t < 1$.
2. Since $\ln t < 0$ on $(0, 1)$, we get $t \ln t < 0$, hence $t^t = e^{t \ln t} < e^0 = 1$; also $t^t > 0$.
3. For a matching lower bound, note the function $g(t) = \sqrt{t} |\ln t|$ attains its maximum on $(0, 1)$ at $t = e^{-2}$, where $g(e^{-2}) = \frac{2}{e}$. Thus $|\ln t| \leq \frac{2/e}{\sqrt{t}}$ for all $t \in (0, 1)$, giving $|t \ln t| \leq \frac{2}{e} \sqrt{t}$.
4. Therefore $e^{-\frac{2}{e} \sqrt{t}} \leq t^t \leq 1$, and as $t \rightarrow 0^+$ both bounds tend to 1.
5. By the squeeze theorem, $\{x\}^{\{x\}} \rightarrow 1$ as $x \rightarrow 1^+$.

INSIGHT

The one-sided framing is essential: as $x \rightarrow 1^-$ we have $\lfloor x \rfloor = 0$, so $\{x\} = x \rightarrow 1$ and the base tends to 1, not 0 — no 0^0 indeterminacy arises there. The exotic form lives only on the 1^+ side, where the floor jumps and the base collapses to 0^+ .

№76 The Square-Root Disguise

●●●●● L4

$$\lim_{x \rightarrow 0^+} (\cos \sqrt{x})^{1/x}$$

LIMIT

 $e^{-1/2}$

THE TRAP

Seeing $\cos \sqrt{x} \rightarrow \cos 0 = 1$ and declaring the answer 1, ignoring that this is a 1^∞ form. The subtler trap: noting $\cos \sqrt{x}$ has no term linear in $\sqrt{x}$ and concluding it is 'flat to first order', hence constant. But $\cos \sqrt{x} = 1 - \frac{x}{2} + O(x^2)$ — the genuine $O(x)$ term $-\frac{x}{2}$ is exactly what the exponent $1/x$ magnifies, producing $e^{-1/2}$, not 1.

Method I — Expand $\cos \sqrt{x}$ in powers of x

1. Since $\cos u = 1 - \frac{u^2}{2} + \frac{u^4}{24} - \dots$ with $u = \sqrt{x}$, the half-integer powers vanish and we get a clean power series in x : $\cos \sqrt{x} = 1 - \frac{x}{2} + \frac{x^2}{24} - \dots$
2. Take logarithms: $\frac{1}{x} \ln \cos \sqrt{x} = \frac{1}{x} \ln \left(1 - \frac{x}{2} + \frac{x^2}{24} - \dots \right)$.
3. With $u = -\frac{x}{2} + O(x^2) \rightarrow 0$, use $\ln(1+u) = u + O(u^2) = -\frac{x}{2} + O(x^2)$, so $\frac{1}{x} \ln \cos \sqrt{x} = -\frac{1}{2} + O(x)$.
4. Letting $x \rightarrow 0^+$ the logarithm tends to $-\frac{1}{2}$, hence

$$(\cos \sqrt{x})^{1/x} = e^{\frac{1}{x} \ln \cos \sqrt{x}} \rightarrow e^{-1/2}.$$

Method II — Substitution $t = \sqrt{x}$

1. Let $t = \sqrt{x}$, so as $x \rightarrow 0^+$ we have $t \rightarrow 0^+$ and $1/x = 1/t^2$. The limit becomes $\lim_{t \rightarrow 0^+} (\cos t)^{1/t^2}$.
2. This is the classic 1^∞ form. Its logarithm is $\frac{\ln \cos t}{t^2}$, and $\ln \cos t = -\frac{t^2}{2} + O(t^4)$.
3. Hence $\frac{\ln \cos t}{t^2} = -\frac{1}{2} + O(t^2) \rightarrow -\frac{1}{2}$, giving $(\cos t)^{1/t^2} \rightarrow e^{-1/2}$.
4. Therefore the original limit equals $e^{-1/2}$.

INSIGHT

The $\sqrt{x}$ converts the cosine's quadratic argument-term into a linear x -term, so it is the cosine's curvature — not any spurious linear part — that governs the exponential. Same mechanism as $\lim_{t \rightarrow 0} (\cos t)^{1/t^2} = e^{-1/2}$.

№77 Modulus at the Far Left

●●●●● L4

$$\lim_{x \rightarrow -\infty} \left(\frac{|x|+1}{|x|-1} \right)^x$$

1^∞ infinity

modulus

$x \rightarrow -\infty$

asymptotics

log-transform

LIMIT

$$e^{-2}$$

THE TRAP

Two traps stack. First, seeing $\frac{|x|+1}{|x|-1} \rightarrow 1$ and concluding $1^x = 1$ — but this is a genuine 1^∞ form, not a constant. Second, computing the magnitude $\frac{2}{|x|} \cdot |x| = 2$ and writing e^2 — forgetting the exponent $x \rightarrow -\infty$ is *negative*. The base exceeds 1 by $\sim \frac{2}{|x|}$, the exponent is $x = -|x|$, and their product tends to -2 . The answer is e^{-2} — not 1, and crucially not e^2 .

Method I — Substitute $u = |x| = -x$

1. For $x \rightarrow -\infty$, put $u = -x \rightarrow +\infty$, so $|x| = u$, the base is $\frac{u+1}{u-1}$, and the exponent is $x = -u$. Note $u - 1 > 0$, so the base is a positive real > 1 and the power is well-defined.

2. Take logs: $\ln L = \lim -u \ln \frac{u+1}{u-1} = \lim -u (\ln(u+1) - \ln(u-1))$.
3. Expand for large u : $\ln \frac{u+1}{u-1} = \ln \frac{1+1/u}{1-1/u} = \left(\frac{1}{u} - \frac{1}{2u^2} + \dots\right) - \left(-\frac{1}{u} - \frac{1}{2u^2} - \dots\right) = \frac{2}{u} + O(u^{-3})$.
4. Thus $\ln L = \lim -u \cdot \left(\frac{2}{u} + O(u^{-3})\right) = \lim (-2 + O(u^{-2})) = -2$, so

$$L = e^{-2}.$$

Method II — Write base as $1 + \frac{2}{|x|-1}$

1. $\frac{|x|+1}{|x|-1} = 1 + \frac{2}{|x|-1}$. With $u = |x| = -x \rightarrow +\infty$, the base is $1 + \frac{2}{u-1}$ and the exponent is $-u$.
2. Regroup to expose the standard limit: $\left(1 + \frac{2}{u-1}\right)^{-u} = \left[\left(1 + \frac{2}{u-1}\right)^{\frac{u-1}{2}}\right]^{\frac{-2u}{u-1}}$.
3. The inner bracket has the form $(1 + \frac{1}{n})^n$ with $n = \frac{u-1}{2} \rightarrow +\infty$, so it $\rightarrow e$; the outer exponent $\frac{-2u}{u-1} \rightarrow -2$.
4. Hence the limit is e^{-2} .

INSIGHT

The sign of the exponent is the whole game: had the limit been $x \rightarrow +\infty$ the same base magnitude would give e^{+2} . The modulus keeps the base > 1 while $x \rightarrow -\infty$ flips the exponent's sign.

№78 Floor Times a Vanishing Arc

●●●●● L4

$$\lim_{x \rightarrow 0^+} (\lfloor 1/x \rfloor \cdot \arcsin x)$$

0-times-infinity

floor

arcsin

$x \rightarrow 0^+$

squeeze

LIMIT

1

THE TRAP

The fast move is to replace $\lfloor 1/x \rfloor$ with $1/x$, collapsing the product to $\frac{\arcsin x}{x} \rightarrow 1$. The number is right but the step is unlicensed: $\lfloor 1/x \rfloor$ can be a full 1 below $1/x$, so you have silently discarded the term $-\{1/x\} \arcsin x$. The opposite panic -- fearing the floor's jumps destroy convergence -- is equally wrong. What saves it is that the discarded defect is bounded and multiplied by a vanishing $\arcsin x$.

Method I — Squeeze on the floor

1. For $x > 0$, $\frac{1}{x} - 1 < \lfloor 1/x \rfloor \leq \frac{1}{x}$. Since $\arcsin x > 0$ for small $x > 0$, multiply through by it (inequalities are preserved):
2.
$$\left(\frac{1}{x} - 1\right) \arcsin x < \lfloor 1/x \rfloor \arcsin x \leq \frac{\arcsin x}{x}.$$
3. Right side: $\frac{\arcsin x}{x} \rightarrow 1$ (standard limit). Left side: $\frac{\arcsin x}{x} - \arcsin x \rightarrow 1 - 0 = 1$.
4. Both bounds tend to 1, so by the squeeze theorem the limit is 1.

Method II — Separate the bounded remainder

1. Write $\lfloor 1/x \rfloor = \frac{1}{x} - \{1/x\}$, so the product splits as $\frac{\arcsin x}{x} - \{1/x\} \arcsin x$.

- The first term satisfies $\frac{\arcsin x}{x} \rightarrow 1$ (standard limit).
- In the second term $\{1/x\} \in [0, 1)$ is bounded while $\arcsin x \rightarrow 0$, so $\{1/x\} \arcsin x \rightarrow 0$.
- Therefore the limit is $1 - 0 = 1$.

INSIGHT

The $0 \cdot \infty$ is tamed because the floor's defect $\{1/x\}$ stays bounded in $[0, 1)$ and is multiplied by the vanishing $\arcsin x$ -- the same mechanism that rescues $(1+x)^{[1/x]}$, transplanted from an exponent to a product.

№79 Signum Splits the Exponential

●●●●● L4

$$\lim_{x \rightarrow 0^+} \left(1 + \operatorname{sgn}(x) x^2\right)^{1/x^2}$$

1^infinity

signum

x→0+

one-sided

log-transform

LIMIT

e

THE TRAP

Erasing $\operatorname{sgn}(x)$ on sight and reflexively quoting $(1+x^2)^{1/x^2} \rightarrow e$ as if the limit were two-sided. It is not: on the *left* $\operatorname{sgn}(x) = -1$ turns the base into $1 - x^2 < 1$, and $(1 - x^2)^{1/x^2} \rightarrow e^{-1}$. The two-sided limit therefore fails — which is precisely why the problem is anchored at 0^+ , where the answer is e , not at 0.

Method I — Resolve the signum on the right

- Since $x \rightarrow 0^+$ we have $x > 0$, so $\operatorname{sgn}(x) = +1$ and the base is exactly $1 + x^2$; the expression is $(1 + x^2)^{1/x^2}$ with no signum ambiguity.
- Substitute $t = x^2$, so $t \rightarrow 0^+$, and the expression becomes $(1 + t)^{1/t}$.
- By the standard limit $(1 + t)^{1/t} \rightarrow e$ as $t \rightarrow 0^+$, hence the right-hand limit is

e .

Method II — Logarithm with explicit one-sided care

- Take logs: $\ln \left[(1 + \operatorname{sgn}(x) x^2)^{1/x^2} \right] = \frac{\ln(1 + \operatorname{sgn}(x) x^2)}{x^2}$.
- For $x > 0$ this is $\frac{\ln(1 + x^2)}{x^2}$; with $u = x^2 \rightarrow 0^+$ it is $\frac{\ln(1 + u)}{u} \rightarrow 1$.
- Since the logarithm tends to 1 and $\exp$ is continuous, the limit is $e^1 = e$.
- Contrast: on $x \rightarrow 0^-$ the numerator becomes $\ln(1 - x^2)$, the log $\rightarrow -1$, and the value would be e^{-1} — confirming the two-sided limit does not exist.

INSIGHT

$\operatorname{sgn}(x)$ is the cleanest device to make a base cross 1 from above on one side and below on the other, splitting a single 1^∞ form into e on the right and $1/e$ on the left.

№80 The Limit That Refuses

●●●●● L5

$$\lim_{x \rightarrow 0^+} \left(x \lfloor 1/x \rfloor \right)^{1/x}$$

1^infinity

floor

fractional-part

x → 0+

non-existence

oscillation

LIMIT

does not exist

THE TRAP

Since $x \lfloor 1/x \rfloor = 1 - x\{1/x\} \rightarrow 1$, a student concludes the base $\rightarrow 1$ and the answer is 1. But this is a 1^∞ form: the exponent $1/x$ magnifies the oscillating defect $x\{1/x\}$, and the value tracks $e^{-\{1/x\}}$, which never settles. The seductive special case $x = 1/n$ (giving exactly 1) is a trap — it only samples $\{1/x\} = 0$.

Method I — Reduce to the fractional part

1. Write $x \lfloor 1/x \rfloor = x \left(\frac{1}{x} - \{1/x\} \right) = 1 - x\{1/x\}$, so the base is $1 - x\{1/x\} \rightarrow 1$ while the exponent is $1/x \rightarrow \infty$ — a genuine 1^∞ form.
2. Take logs: $L(x) = \frac{1}{x} \ln(1 - x\{1/x\})$. Since $0 \leq x\{1/x\} < x \rightarrow 0$, expand $\ln(1 - u) = -u + O(u^2)$ with $u = x\{1/x\}$.
3. Divide by x : $L(x) = \frac{1}{x}(-x\{1/x\} + O(x^2\{1/x\}^2)) = -\{1/x\} + O(x\{1/x\}^2) = -\{1/x\} + o(1)$.
4. Hence the expression equals $e^{L(x)} = e^{-\{1/x\} + o(1)}$, i.e. it tracks $e^{-\{1/x\}}$. Since $\{1/x\}$ oscillates over $[0, 1)$ as $x \rightarrow 0^+$ and never converges, $L(x)$ has no limit, so the original limit does not exist.

Method II — Two sequences with different limits

1. Choose $x_n = \frac{1}{n}$: then $1/x_n = n$, $\{1/x_n\} = 0$, so the base $= x_n \cdot n = 1$ exactly and the value is $1^n = 1 \rightarrow 1$.
2. Choose $y_n = \frac{1}{n + \frac{1}{2}}$: then $1/y_n = n + \frac{1}{2}$, $\lfloor 1/y_n \rfloor = n$, base $= y_n n = \frac{n}{n + \frac{1}{2}} \rightarrow 1$, and the value is $\left(\frac{n}{n + \frac{1}{2}} \right)^{n + \frac{1}{2}} = \left(1 + \frac{1/2}{n} \right)^{-(n + 1/2)} \rightarrow e^{-1/2}$.
3. Both $x_n \rightarrow 0^+$ and $y_n \rightarrow 0^+$, yet the function tends to 1 along one and $e^{-1/2} \neq 1$ along the other. Therefore the limit does not exist.
4. More fully, along $x = \frac{1}{n+f}$ (fixed $f \in [0, 1)$) the value $\rightarrow e^{-f}$. As f ranges over $[0, 1)$ these limits sweep $(e^{-1}, 1]$, and the lower endpoint e^{-1} is reached in the limit $f \rightarrow 1^-$; so the set of subsequential limits is the closed interval $[e^{-1}, 1]$ — the value e^{-1} is approached but never attained for any single x .

INSIGHT

Contrast $(1 + x)^{\lfloor 1/x \rfloor} \rightarrow e$: there the floor's defect meets $\ln(1 + x) = O(x)$ and dies; here it meets the exponent $1/x$ and survives as $-\{1/x\}$, so the 1^∞ form is fatally indeterminate, with cluster set exactly $[e^{-1}, 1]$.

CHAPTER IX

Sandwich & Discrete Sums



When termwise intuition lies, only a global bound tells the truth.

№81 The Floor That Doesn't Bite

●●●●● L3

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n \left\lfloor k \frac{n}{k+1} \right\rfloor$$

floor

riemann-sum

$x \rightarrow \infty$

integer-part-sum

LIMIT

1

THE TRAP

A student who has been burned by floors before over-corrects: they argue that each term $\lfloor a_k \rfloor$ loses a fractional part $\{a_k\}$ averaging $\frac{1}{2}$, so they subtract a 'correction' $\frac{1}{n^2} \sum \{a_k\} \approx \frac{1}{n^2} \cdot \frac{n}{2}$ and even mis-scale it to conclude the answer is $\frac{1}{2}$. The error: there are only n fractional parts, each in $[0, 1)$, so $\sum_{k=1}^n \{a_k\} = O(n)$ and $\frac{1}{n^2} \sum \{a_k\} \rightarrow 0$. The floor correction is negligible at the n^2 scale, so the answer is 1, not $\frac{1}{2}$. Here the floor genuinely does NOT bite — always compare the discarded fractional mass to the denominator before trusting OR distrusting the floor.

Method I — Squeeze on the floor error

1. Write $a_k = k \frac{n}{k+1}$. The defining inequality of the floor gives $a_k - 1 < \lfloor a_k \rfloor \leq a_k$. Summing over $k = 1, \dots, n$:

2.
$$\sum_{k=1}^n a_k - n < \sum_{k=1}^n \lfloor a_k \rfloor \leq \sum_{k=1}^n a_k.$$

3. Divide through by n^2 . The discarded mass on the left is exactly $\frac{n}{n^2} = \frac{1}{n} \rightarrow 0$, so the limit of $\frac{1}{n^2} \sum \lfloor a_k \rfloor$ equals the limit of $\frac{1}{n^2} \sum a_k$, provided the latter exists.

4. Compute the upper sum in closed form. Since $a_k = n \frac{k}{k+1} = n - \frac{n}{k+1}$,

5.
$$\frac{1}{n^2} \sum_{k=1}^n a_k = \frac{1}{n} \sum_{k=1}^n \frac{k}{k+1} = \frac{1}{n} \sum_{k=1}^n \left(1 - \frac{1}{k+1}\right) = 1 - \frac{H_{n+1} - 1}{n},$$

where $H_m = \sum_{j=1}^m \frac{1}{j}$.

6. Because $H_{n+1} \sim \ln n$, the correction $\frac{H_{n+1} - 1}{n} \rightarrow 0$, hence $\frac{1}{n^2} \sum a_k \rightarrow 1$.

7. Both the lower bound $\frac{1}{n^2} \sum a_k - \frac{1}{n}$ and the upper bound $\frac{1}{n^2} \sum a_k$ tend to 1. By the squeeze theorem,

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n [a_k] = 1.$$

8. Sanity check on the floor's role: $\frac{1}{n^2} \sum_{k=1}^n \{a_k\} \leq \frac{1}{n^2} \cdot n = \frac{1}{n} \rightarrow 0$, confirming the fractional parts contribute nothing at this scale — the floor does not change the limit.

Method II — Cesàro / Stolz on the dominant sum

1. Set $b_n = \sum_{k=1}^n [a_k]$ with $a_k = n - \frac{n}{k+1}$. Since $0 \leq a_k - [a_k] < 1$ for each of the n terms, $|b_n - \sum_{k=1}^n a_k| < n$.

2. Dividing by n^2 : $\frac{b_n}{n^2} = \frac{1}{n^2} \sum_{k=1}^n a_k + O\left(\frac{n}{n^2}\right) = \frac{1}{n^2} \sum_{k=1}^n a_k + O\left(\frac{1}{n}\right)$.

3. Now $\frac{1}{n^2} \sum_{k=1}^n a_k = \frac{1}{n} \sum_{k=1}^n \frac{k}{k+1}$ is the Cesàro (arithmetic) mean of the sequence $c_k = \frac{k}{k+1}$. Since $c_k \rightarrow 1$, its Cesàro mean also converges to 1.

4. Therefore $\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n [a_k] = 1 + 0 = 1$.

INSIGHT

The lesson is the inverse of the usual floor cautionary tale: here the discarded fractional mass is $O(n)$, negligible against the n^2 denominator, so the naive 'drop the floor' value 1 is exactly correct. A floor only shifts a limit when the number of integer-part errors scales like the normalizing denominator — with n terms over n^2 , it cannot.

№82 A Thousand Reciprocal Roots

●●●●● L4

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sqrt{n^2 + k}}$$

`1/sqrt(n^2+k)`

`squeeze`

`x→infinity`

`riemann-vs-squeeze`

LIMIT

1

THE TRAP

The Riemann-sum reflex: writing the sum as $\frac{1}{n} \sum_{k=1}^n \left(1 + \frac{k}{n^2}\right)^{-1/2}$ and pattern-matching it to $\frac{1}{n} \sum f(k/n)$ tempts a student to read off $f(x) = (1 + x^2)^{-1/2}$ and answer $\int_0^1 \frac{dx}{\sqrt{1+x^2}} = \ln(1 + \sqrt{2}) \approx 0.8814$. This is wrong: the perturbation inside the root is $+k$ (order n relative to n^2), not $+k^2$, so $k/n^2 \rightarrow 0$ uniformly and the genuine Riemann integrand is the constant 1, giving $\int_0^1 1 dx = 1$. The seductive $\ln(1 + \sqrt{2})$ is the answer to the look-alike $\sum 1/\sqrt{n^2 + k^2}$, where the perturbation really is order n^2 .

Method I — Two-sided squeeze

1. For $1 \leq k \leq n$ we have $n^2 + 1 \leq n^2 + k \leq n^2 + n$, so every term is sandwiched:

2.
$$\frac{1}{\sqrt{n^2+n}} \leq \frac{1}{\sqrt{n^2+k}} \leq \frac{1}{\sqrt{n^2+1}}.$$
3. There are exactly n terms, so summing the inequality gives
4.
$$\frac{n}{\sqrt{n^2+n}} \leq \sum_{k=1}^n \frac{1}{\sqrt{n^2+k}} \leq \frac{n}{\sqrt{n^2+1}}.$$
5. Lower bound: $\frac{n}{\sqrt{n^2+n}} = \frac{1}{\sqrt{1+1/n}} \rightarrow 1$. Upper bound: $\frac{n}{\sqrt{n^2+1}} = \frac{1}{\sqrt{1+1/n^2}} \rightarrow 1$.
6. Both bounds tend to 1, so by the squeeze theorem the limit is $\boxed{1}$.

Method II – Binomial expansion

1. Factor out n : $\frac{1}{\sqrt{n^2+k}} = \frac{1}{n} \left(1 + \frac{k}{n^2}\right)^{-1/2} = \frac{1}{n} \left(1 - \frac{k}{2n^2} + O\left(\frac{k^2}{n^4}\right)\right)$, valid since $k/n^2 \leq 1/n \rightarrow 0$.
2. Sum over $k = 1, \dots, n$: $\sum_{k=1}^n \frac{1}{n} - \frac{1}{2n^3} \sum_{k=1}^n k + O\left(\frac{1}{n^5} \sum_{k=1}^n k^2\right)$.
3. Evaluate each piece: $\sum_{k=1}^n \frac{1}{n} = 1$, $\frac{1}{2n^3} \cdot \frac{n(n+1)}{2} = \frac{n+1}{4n^2} \rightarrow 0$, $O\left(\frac{n^3}{n^5}\right) = O(n^{-2}) \rightarrow 0$.
4. Hence the sum $\rightarrow 1$, and the expansion even exposes the rate: the leading correction is $-\frac{1}{4n}$, so $\sum \approx 1 - \frac{1}{4n}$.

Method III – Integral comparison

1. The summand $g(k) = 1/\sqrt{n^2+k}$ is decreasing in k , so it is trapped between integrals:
- $$\int_1^{n+1} \frac{dk}{\sqrt{n^2+k}} \leq \sum_{k=1}^n g(k) \leq \int_0^n \frac{dk}{\sqrt{n^2+k}}.$$
2. Evaluate the lower integral: $\int_1^{n+1} \frac{dk}{\sqrt{n^2+k}} = \left[2\sqrt{n^2+k}\right]_1^{n+1} = 2\sqrt{n^2+n+1} - 2\sqrt{n^2+1}$.
3. Rationalize: $= \frac{2(n)}{\sqrt{n^2+n+1} + \sqrt{n^2+1}} \rightarrow \frac{2n}{2n} = 1$. The upper integral $2\sqrt{n^2+n} - 2\sqrt{n^2} = \frac{2n}{\sqrt{n^2+n} + n} \rightarrow 1$ likewise.
4. Both integral bounds tend to 1, so the sum is squeezed to 1.

INSIGHT

The governing principle: when the inner perturbation has order strictly below the denominator's leading power ($k \ll n^2$), each term is $\frac{1}{n}(1 + o(1))$ and the whole sum collapses to the constant Riemann value 1. The deceptive $\ln(1 + \sqrt{2})$ requires a genuine $+k^2$ perturbation, which keeps k^2/n^2 at order 1 and revives a non-trivial integrand.

№83 Fractional Parts on a Grid

●●●●● L5

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left\{ \sqrt{k} \right\} \cdot \frac{1}{\sqrt{k}} \cdot \sqrt{n}$$

fractional-part

equidistribution

riemann-sum

$x \rightarrow \infty$

LIMIT

1

THE TRAP

The seductive value is $\frac{1}{2}$. A student notes $\{\sqrt{k}\}$ is equidistributed on $[0, 1)$, so its average is $\frac{1}{2}$, and writes the whole expression as $\frac{1}{2} \cdot \frac{1}{n} \sum_{k=1}^n 1 \rightarrow \frac{1}{2}$. The slip: the genuine weight $\frac{\sqrt{n}}{\sqrt{k}} = \frac{1}{\sqrt{k/n}}$ was silently dropped. Since $\int_0^1 x^{-1/2} dx = 2$, that weight contributes a factor 2, turning $\frac{1}{2}$ into $\frac{1}{2} \cdot 2 = 1$. Forgetting the Jacobian/weight is exactly what produces the wrong $\frac{1}{2}$.

Method I — Smooth part + equidistribution of the fluctuation

1. Write the expression as a weighted average. Since $\frac{\sqrt{n}}{\sqrt{k}} = \frac{1}{\sqrt{k/n}}$, with $x_k = k/n$ we have

$$S_n = \frac{1}{n} \sum_{k=1}^n \frac{\{\sqrt{k}\}}{\sqrt{k/n}} = \frac{1}{n} \sum_{k=1}^n \{\sqrt{k}\} x_k^{-1/2}.$$

2. Split the fractional part around its mean: $\{\sqrt{k}\} = \frac{1}{2} + (\{\sqrt{k}\} - \frac{1}{2})$, giving $S_n = A_n + B_n$ with $A_n = \frac{1}{2} \cdot \frac{1}{n} \sum_{k=1}^n x_k^{-1/2}$ and $B_n = \frac{1}{n} \sum_{k=1}^n (\{\sqrt{k}\} - \frac{1}{2}) x_k^{-1/2}$.
3. Smooth part A_n : this is a Riemann sum for $\frac{1}{2} \int_0^1 x^{-1/2} dx = \frac{1}{2} \cdot 2 = 1$. (The integrable singularity at 0 is harmless: $\frac{1}{n} \sum x_k^{-1/2} = \frac{1}{\sqrt{n}} \sum_{k=1}^n k^{-1/2} \rightarrow \int_0^1 x^{-1/2} dx = 2$ since $\sum_{k=1}^n k^{-1/2} \sim 2\sqrt{n}$.) Hence $A_n \rightarrow 1$.
4. Fluctuation part $B_n \rightarrow 0$: the weight $x^{-1/2}$ is smooth (slowly varying) away from 0, while $\{\sqrt{k}\} - \frac{1}{2}$ is the zero-mean part of an equidistributed sequence, so its weighted average cancels by Abel summation/equidistribution. (Near $k = 0$ the few terms contribute $O(1/\sqrt{n}) \rightarrow 0$.)
5. Therefore $S_n = A_n + B_n \rightarrow 1 + 0 = \boxed{1}$.

Method II — Block decomposition with the change of variables $k=x^2$

1. Let $T_n = \sum_{k=1}^n \frac{\{\sqrt{k}\}}{\sqrt{k}}$, so that $S_n = \frac{\sqrt{n}}{n} T_n = \frac{T_n}{\sqrt{n}}$.
2. Group by the integer part $\lfloor \sqrt{k} \rfloor = m$, i.e. the block $m^2 \leq k < (m+1)^2$. Approximate the block sum by the integral via $x = \sqrt{k}$, $k = x^2$, $dk = 2x dx$:

$$\sum_{m^2 \leq k < (m+1)^2} \frac{\{\sqrt{k}\}}{\sqrt{k}} \approx \int_m^{m+1} \frac{\{x\}}{x} \cdot 2x dx = 2 \int_m^{m+1} \{x\} dx = 2 \int_0^1 t dt = 1.$$
3. So each complete block contributes $\rightarrow 1$ to T_n (numerically, block $m = 1000$ gives 0.9995). Up to $k = n$ there are $\approx \sqrt{n}$ complete blocks, hence $T_n \sim \sqrt{n}$.
4. Therefore $S_n = \frac{T_n}{\sqrt{n}} \rightarrow 1$. The factor 2 from the Jacobian $dk = 2x dx$ is exactly the factor the naive ' $\frac{1}{2}$ -average' shortcut omits, which is why the honest value is 1, not $\frac{1}{2}$.

INSIGHT

Two equivalent viewpoints give the factor 2: as the Jacobian $dk = 2x dx$ (so the weighted block average of $\{x\}$ is $2 \int_0^1 t dt = 1$), or as $\int_0^1 x^{-1/2} dx = 2$ multiplying the equidistribution mean $\frac{1}{2}$. Either way $\frac{1}{2} \cdot 2 = 1$.

№84 The Signum Mirage

● ● ● ● ● L5

$$\lim_{x \rightarrow 0^+} x \sum_{k=1}^{\lfloor 1/x \rfloor} \operatorname{sgn} \left(\sin \frac{1}{kx} \right)$$

signum

sin

one-sided

$x \rightarrow 0^+$

riemann-sum

squeeze

LIMIT

$$1 - \frac{2 \ln 2}{\pi}$$

THE TRAP

The seductive wrong answer is 0. A strong student reasons: each $\operatorname{sgn}(\sin \frac{1}{kx}) \in \{-1, 0, 1\}$, the argument $\frac{1}{kx}$ grows as $x \rightarrow 0^+$, and $\sin$ spends equal time positive and negative, so 'by equidistribution the signs cancel in the mean' and the limit is 0. This is a mirage. Equidistribution would require $\frac{1}{kx}$ to sweep many periods uniformly over the bulk of indices, but it does not: writing $x \approx \frac{1}{N}$, the argument is $\frac{N}{k}$, and for the majority of indices (k comparable to N) this lies in $(0, \pi)$ where $\sin > 0$. The oscillatory regime is confined to the few small k and contributes only $O(N^{-1/2})$. The sum is really a Riemann sum, and it converges to a strictly positive constant.

Method I — Recognize the Riemann sum, then evaluate

1. Take $x \rightarrow 0^+$ along $x = \frac{1}{N}$ with $N = \lfloor 1/x \rfloor \rightarrow \infty$ (the general case follows since $x \lfloor 1/x \rfloor \rightarrow 1$ and the summand is bounded). Then $\frac{1}{kx} = \frac{N}{k}$ and the expression is

$$S_N = \frac{1}{N} \sum_{k=1}^N \operatorname{sgn}\left(\sin \frac{N}{k}\right).$$

2. Substitute $t = \frac{k}{N} \in (0, 1]$, so $\frac{N}{k} = \frac{1}{t}$. Then $S_N = \frac{1}{N} \sum_{k=1}^N g\left(\frac{k}{N}\right)$ with $g(t) = \operatorname{sgn}\left(\sin \frac{1}{t}\right)$, a right Riemann sum on $[0, 1]$ with mesh $\frac{1}{N}$.
3. The function g takes values in $\{-1, 0, 1\}$ and is discontinuous only where $\sin \frac{1}{t} = 0$, i.e. at $t = \frac{1}{m\pi}$; these points accumulate only at 0 and form a measure-zero set, so g is Riemann integrable on $[0, 1]$. Hence

$$\lim_{N \rightarrow \infty} S_N = \int_0^1 \operatorname{sgn}\left(\sin \frac{1}{t}\right) dt.$$

4. Substitute $u = \frac{1}{t}$ (so $t = \frac{1}{u}$, $dt = -u^{-2} du$), turning the integral into

$$\int_1^\infty \frac{\operatorname{sgn}(\sin u)}{u^2} du.$$

5. On $(n\pi, (n+1)\pi)$ we have $\operatorname{sgn}(\sin u) = (-1)^n$, and $u = 1$ lies in $(0, \pi)$ (the $n = 0$ block, sign $+$). Since $\int_a^b u^{-2} du = \frac{1}{a} - \frac{1}{b}$,

$$\int_1^\infty \frac{\operatorname{sgn}(\sin u)}{u^2} du = \left(1 - \frac{1}{\pi}\right) + \sum_{n \geq 1} (-1)^n \left(\frac{1}{n\pi} - \frac{1}{(n+1)\pi}\right).$$

6. Evaluate the alternating series: $\sum_{n \geq 1} \frac{(-1)^n}{n} = -\ln 2$ and $\sum_{n \geq 1} \frac{(-1)^n}{n+1} = \ln 2 - 1$, so $\sum_{n \geq 1} (-1)^n \left(\frac{1}{n} - \frac{1}{n+1}\right) = (-\ln 2) - (\ln 2 - 1) = 1 - 2 \ln 2$. Therefore

$$\int_1^\infty \frac{\operatorname{sgn}(\sin u)}{u^2} du = \left(1 - \frac{1}{\pi}\right) + \frac{1 - 2 \ln 2}{\pi} = 1 - \frac{2 \ln 2}{\pi}.$$

7. Hence the limit is $\boxed{1 - \frac{2 \ln 2}{\pi}} \approx 0.5587$, a strictly positive number — the alleged cancellation to 0 never happens.

Method II — Squeeze the oscillatory head, integrate the smooth tail

1. With $x = \frac{1}{N}$, write $S_N = \frac{1}{N} \sum_{k=1}^N \operatorname{sgn}(\sin \frac{N}{k})$ and split at $K = \lfloor \sqrt{N} \rfloor$ into a head $1 \leq k \leq K$ and a tail $K < k \leq N$.

2. Head bound (squeeze): the trivial estimate $|\operatorname{sgn}(\cdot)| \leq 1$ gives

$$\left| \frac{1}{N} \sum_{k=1}^K \operatorname{sgn} \left(\sin \frac{N}{k} \right) \right| \leq \frac{K}{N} \leq \frac{1}{\sqrt{N}} \xrightarrow{N \rightarrow \infty} 0.$$

So the oscillatory head — where $\frac{N}{k}$ is large and genuinely swings between signs — contributes nothing in the limit. (This is exactly where the false 'everything cancels' intuition lives; it is the negligible part, not the whole.)

3. Tail as a Riemann sum: for $K < k \leq N$ put $t = \frac{k}{N} \in (\frac{1}{\sqrt{N}}, 1]$. As k increases by 1, t moves by $\frac{1}{N}$, so $\frac{1}{N} \sum_{K < k \leq N} \operatorname{sgn}(\sin \frac{1}{t})$ is a Riemann sum for $\int_0^1 \operatorname{sgn}(\sin \frac{1}{t}) dt$ (the missing piece $\int_0^{1/\sqrt{N}}$ has length $\rightarrow 0$ with bounded integrand, hence vanishes).

4. Therefore $S_N \rightarrow \int_0^1 \operatorname{sgn}(\sin \frac{1}{t}) dt$. Evaluating this integral exactly as in Method I (substitute $u = 1/t$, sum the alternating series of $\frac{1}{u^2}$ over half-periods) yields

$$\int_0^1 \operatorname{sgn} \left(\sin \frac{1}{t} \right) dt = 1 - \frac{2 \ln 2}{\pi}.$$

5. The general $x \rightarrow 0^+$ limit equals the same value because $x \lfloor 1/x \rfloor \rightarrow 1$ and the summand is bounded by 1. Hence the limit is $1 - \frac{2 \ln 2}{\pi}$.

INSIGHT

The key reframing: $x \sum_{k \leq \lfloor 1/x \rfloor} h(\frac{1}{kx})$ is a disguised Riemann sum $\rightarrow \int_0^1 h(\frac{1}{t}) dt = \int_1^\infty h(u) u^{-2} du$. For $h = \operatorname{sgn} \circ \sin$ the dominant indices (k near $\lfloor 1/x \rfloor$) land in the first positive arc $(0, \pi)$, so the value is a positive constant $1 - \frac{2 \ln 2}{\pi}$, not 0. 'Mean zero of $\operatorname{sgn} \circ \sin$ ' only governs the measure-zero oscillatory head.

№85 Stolz Refuses the Easy Way

● ● ● ● ● L5

$$\lim_{n \rightarrow \infty} \frac{1}{(\ln n)^2} \sum_{k=1}^n \frac{\lfloor \log_2 k \rfloor}{k}$$

stolz-cesaro

floor

ln

x→infinity

dyadic

LIMIT

$$\frac{1}{2 \ln 2}$$

THE TRAP

Two seductive traps. (1) Wrong normalizer: a student notes $\lfloor \log_2 k \rfloor \leq \log_2 k$ and expects the sum to grow like $\sum \frac{\ln k}{k \ln 2} \sim \frac{(\ln n)^2}{2 \ln 2}$, then carelessly divides by $\ln n$ (as in a $1/\ln n$ version) and concludes the limit is ∞ . The sum IS $\sim \frac{(\ln n)^2}{2 \ln 2}$, so dividing by the matching $(\ln n)^2$ gives the finite $\frac{1}{2 \ln 2}$. (2) Replacing $\lfloor \log_2 k \rfloor$ by $\log_2 k$ looks harmless for the LEADING term — and indeed it gives the same constant — but it hides why the floor is essential to keep the block structure exact; sloppy bounding of the $O(1)$ gap can corrupt the constant if you are not careful.

Method I — Dyadic blocking

1. On the dyadic block $2^m \leq k < 2^{m+1}$ we have $\lfloor \log_2 k \rfloor = m$ exactly. Hence $\sum_{k=2^m}^{2^{m+1}-1} \frac{\lfloor \log_2 k \rfloor}{k} = m \sum_{k=2^m}^{2^{m+1}-1} \frac{1}{k}$.

2. The inner harmonic block satisfies $\sum_{k=2^m}^{2^{m+1}-1} \frac{1}{k} = H_{2^{m+1}-1} - H_{2^m-1} \rightarrow \ln 2$ as $m \rightarrow \infty$, so block m contributes $\sim m \ln 2$.
3. Summing complete blocks up to $M = \lfloor \log_2 n \rfloor$: $\sum_{m=0}^{M-1} m \ln 2 = \ln 2 \cdot \frac{M(M-1)}{2} \sim \frac{\ln 2}{2} M^2$. The leftover partial block contributes $O(M)$, negligible against M^2 .
4. Since $M = \log_2 n = \frac{\ln n}{\ln 2}$, we get $\sum_{k=1}^n \frac{\lfloor \log_2 k \rfloor}{k} \sim \frac{\ln 2}{2} \cdot \frac{(\ln n)^2}{(\ln 2)^2} = \frac{(\ln n)^2}{2 \ln 2}$.
5. Divide by $(\ln n)^2$: $\frac{1}{(\ln n)^2} \sum_{k=1}^n \frac{\lfloor \log_2 k \rfloor}{k} \rightarrow \frac{1}{2 \ln 2}$.

Method II — Stolz–Cesàro

1. Let $a_n = \sum_{k=1}^n \frac{\lfloor \log_2 k \rfloor}{k}$ and $b_n = (\ln n)^2$. Here $b_n \rightarrow \infty$ is strictly increasing, so Stolz–Cesàro applies: it suffices to evaluate $\frac{a_{n+1} - a_n}{b_{n+1} - b_n}$.
2. Numerator: $a_{n+1} - a_n = \frac{\lfloor \log_2(n+1) \rfloor}{n+1} = \frac{\log_2 n + O(1)}{n+1} \sim \frac{\ln n}{n \ln 2}$, since $\lfloor \log_2(n+1) \rfloor = \log_2 n + O(1)$ and the $O(1)$ is lower order.
3. Denominator: $b_{n+1} - b_n = (\ln(n+1))^2 - (\ln n)^2 = (\ln(n+1) - \ln n)(\ln(n+1) + \ln n) \sim \frac{1}{n} \cdot 2 \ln n = \frac{2 \ln n}{n}$.
4. Therefore $\frac{a_{n+1} - a_n}{b_{n+1} - b_n} \rightarrow \frac{\ln n / (n \ln 2)}{2 \ln n / n} = \frac{1}{2 \ln 2}$, and by Stolz–Cesàro $\frac{a_n}{(\ln n)^2} \rightarrow \frac{1}{2 \ln 2}$.

INSIGHT

The dyadic floor makes $\sum \frac{\lfloor \log_2 k \rfloor}{k}$ grow quadratically in $\ln n$, so the natural normalizer is $(\ln n)^2$ — not $\ln n$, under which the expression diverges. Either dyadic blocking or Stolz pins the constant to $\frac{1}{2 \ln 2}$; in general $\sum_{k \leq n} \frac{\lfloor \log_b k \rfloor}{k} \sim \frac{(\ln n)^2}{2 \ln b}$.

№86 Riemann With a Floor Inside

● ● ● ● ● L5

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left\{ \frac{n}{k} \right\}$$

floor riemann-sum x→infinity equidistribution

LIMIT

$$1 - \gamma$$

THE TRAP

The reflex is "it's an average of fractional parts, so it's $\frac{1}{2}$." That is the killer trap and it gives the wrong number. The fractional parts $\{n/k\}$ are systematically biased low: for small k , n/k is huge and varies slowly, so $\{n/k\}$ spends more time small than large. A more careful student writes $\frac{1}{n} \sum f(k/n)$ with $f(t) = \{1/t\}$ and gets $\int_0^1 \{1/t\} dt$ — which actually does equal $1 - \gamma$ — but only if they correctly evaluate $\int_1^\infty \frac{\{u\}}{u^2} du = 1 - \gamma$; mis-

evaluating that integral or rounding $1 - \gamma \approx 0.4228$ back to $\frac{1}{2}$ is the common slip. The true value is the non-obvious $1 - \gamma \approx 0.4228$, strictly below $\frac{1}{2}$.

Method I — From divisor sums to the constant

1. Write $\{n/k\} = \frac{n}{k} - \lfloor n/k \rfloor$ and sum:

$$\sum_{k=1}^n \left\{ \frac{n}{k} \right\} = n \sum_{k=1}^n \frac{1}{k} - \sum_{k=1}^n \left\lfloor \frac{n}{k} \right\rfloor = nH_n - D(n),$$

where $D(n) = \sum_{k=1}^n \lfloor n/k \rfloor = \sum_{k=1}^n d(k)$ is the total number of divisors of the integers up to n (the identity $\sum_k \lfloor n/k \rfloor = \sum_{m \leq n} d(m)$ holds because $\lfloor n/k \rfloor$ counts the multiples of k in $[1, n]$).

2. Dirichlet's divisor theorem gives $D(n) = n \ln n + (2\gamma - 1)n + O(\sqrt{n})$. Also $H_n = \ln n + \gamma + O(1/n)$, so $nH_n = n \ln n + \gamma n + O(1)$.

3. Subtract the two expansions:

$$\sum_{k=1}^n \left\{ \frac{n}{k} \right\} = (n \ln n + \gamma n) - (n \ln n + (2\gamma - 1)n) + O(\sqrt{n}) = (1 - \gamma)n + O(\sqrt{n}).$$

4. Divide by n : the $O(\sqrt{n})/n = O(1/\sqrt{n})$ term vanishes, so

$$\frac{1}{n} \sum_{k=1}^n \left\{ \frac{n}{k} \right\} \rightarrow 1 - \gamma.$$

Method II — Riemann integral done correctly

1. Set $t = k/n$. Then $\frac{1}{n} \sum_{k=1}^n \left\{ \frac{n}{k} \right\} = \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right)$ with $f(t) = \{1/t\}$, a bounded function that is Riemann integrable on $(0, 1]$ (its only accumulation of discontinuities is at $t \rightarrow 0^+$, where it is bounded). Hence the sum tends to $\int_0^1 \left\{ \frac{1}{t} \right\} dt$.

2. Substitute $u = 1/t$, so $t = 1/u$ and $dt = -u^{-2} du$:

$$\int_0^1 \left\{ \frac{1}{t} \right\} dt = \int_1^\infty \frac{\{u\}}{u^2} du.$$

3. Evaluate the classical integral. With $\{u\} = u - \lfloor u \rfloor$,

$$\int_1^N \frac{\{u\}}{u^2} du = \int_1^N \frac{du}{u} - \int_1^N \frac{\lfloor u \rfloor}{u^2} du = \ln N - \left(\ln N - H_{\lfloor N \rfloor} + \frac{\{N\}}{N} \dots \right),$$

which collapses to $H_N - \ln N \rightarrow \gamma$ inside, giving $\int_1^\infty \frac{\{u\}}{u^2} du = 1 - \gamma$ (equivalently, this is the standard integral representation $\gamma = 1 - \int_1^\infty \frac{\{u\}}{u^2} du$).

4. Therefore the limit equals $1 - \gamma \approx 0.4228$.

Method III — Squeeze sanity bound (why it's below $\frac{1}{2}$)

1. Bracket each floor: $\frac{n}{k} - 1 < \lfloor n/k \rfloor \leq \frac{n}{k}$. Summing,

$$nH_n - n < \sum_{k=1}^n \left\lfloor \frac{n}{k} \right\rfloor \leq nH_n.$$

2. Hence $\sum_{k=1}^n \{n/k\} = nH_n - \sum_{k=1}^n \lfloor n/k \rfloor$ lies in $[0, n)$, so the mean lies in $[0, 1)$ — this only rules out absurd values. The crude squeeze cannot reach $1 - \gamma$ alone; the refined Dirichlet error term (the $\frac{1}{2}$ -saving $O(\sqrt{n})$) is exactly what pins the constant at $1 - \gamma \approx 0.4228$.

3. Since $1 - \gamma < \frac{1}{2}$, the naive 'fractional parts average $\frac{1}{2}$ ' guess is disproved, and

$$\frac{1}{n} \sum_{k=1}^n \{n/k\} \rightarrow 1 - \gamma.$$

INSIGHT

The fractional parts of n/k are biased low because n/k is large and slowly varying for small k , so it lingers near integer values; Dirichlet's $\frac{1}{2}$ -saving error term is precisely what converts the naive $\frac{1}{2}$ into $1 - \gamma$. The same divisor-sum machinery shows $\frac{1}{n} \sum_{k \leq n} \{n/k\} \rightarrow 1 - \gamma$ is the additive analogue of the average order of $d(k)$.

№87 Conjugate Across the Whole Sum

●●●●● L4

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\sqrt{n^2 + k} - \sqrt{n^2 + k - 1} \right) \cdot \sqrt{\frac{k}{n}}$$

sqrt

global-bound

x→infinity

telescoping-fail

LIMIT

$$\frac{1}{3}$$

THE TRAP

A student telescopes $\sum (\sqrt{n^2 + k} - \sqrt{n^2 + k - 1}) = \sqrt{n^2 + n} - n \rightarrow \frac{1}{2}$ and then multiplies by 'the typical $\sqrt{k/n} \approx 1$ ', guessing $\frac{1}{2}$. Wrong: the weight $\sqrt{k/n}$ varies across the sum and cannot be pulled out; replacing each difference by its true size $\frac{1}{2n}$ turns it into a genuine Riemann sum giving $\frac{1}{3}$.

Method I — Conjugate then Riemann

1. Conjugate each difference: $\sqrt{n^2 + k} - \sqrt{n^2 + k - 1} = \frac{1}{\sqrt{n^2 + k} + \sqrt{n^2 + k - 1}}$.
2. For $1 \leq k \leq n$, both roots are $\approx n$, so the difference $= \frac{1}{2n} (1 + O(k/n^2)) = \frac{1}{2n} + O(\frac{k}{n^3})$.
3. Thus the sum is $\sum_{k=1}^n \left(\frac{1}{2n} + O(\frac{k}{n^3}) \right) \sqrt{\frac{k}{n}} = \frac{1}{2n} \sum_{k=1}^n \sqrt{\frac{k}{n}} + O(\frac{1}{n^3} \sum_{k=1}^n k \sqrt{\frac{k}{n}})$.
4. Error term: $\frac{1}{n^3} \sum_{k=1}^n k \sqrt{k/n} = \frac{1}{n^{3.5}} \sum_{k=1}^n k^{3/2} = O(\frac{n^{5/2}}{n^{3.5}}) = O(n^{-1}) \rightarrow 0$, since $\sum_{k=1}^n k^{3/2} \sim \frac{2}{5} n^{5/2}$.
5. Main term: $\frac{1}{2} \cdot \frac{1}{n} \sum_{k=1}^n \sqrt{k/n} \rightarrow \frac{1}{2} \int_0^1 \sqrt{t} dt = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$.

Method II — Mean value theorem per term

1. By the MVT applied to $f(x) = \sqrt{x}$ on $[n^2 + k - 1, n^2 + k]$, $\sqrt{n^2 + k} - \sqrt{n^2 + k - 1} = \frac{1}{2\sqrt{\xi_k}}$ for some $\xi_k \in (n^2 + k - 1, n^2 + k)$, so $\xi_k = n^2(1 + O(1/n))$ and $\frac{1}{2\sqrt{\xi_k}} = \frac{1}{2n} (1 + O(1/n))$.
2. Therefore the sum equals $\frac{1}{2n} \sum_{k=1}^n \sqrt{\frac{k}{n}} (1 + O(1/n)) = \frac{1}{2} \left(\frac{1}{n} \sum_{k=1}^n \sqrt{k/n} \right) + O(1/n)$.
3. Recognize the Riemann sum $\frac{1}{n} \sum_{k=1}^n \sqrt{k/n} \rightarrow \int_0^1 \sqrt{t} dt = \frac{2}{3}$.
4. Hence the limit is $\frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$.

INSIGHT

Telescoping kills the differences but loses the weight; the right move is to replace each difference by its size $\frac{1}{2n}$ and read off the Riemann integral $\frac{1}{2} \int_0^1 \sqrt{t} dt = \frac{1}{3}$.

№88 Floor of an Arctan Riemann Sum

● ● ● ● ● L5

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n \left\lfloor n \arctan \frac{k}{n} \right\rfloor$$

floor

arctan

riemann-sum

x→infinity

global-squeeze

LIMIT

$$\frac{\pi}{4} - \frac{1}{2} \ln 2$$

THE TRAP

Drop-the-floor fallacy: a student writes $\lfloor n \arctan \frac{k}{n} \rfloor \approx n \arctan \frac{k}{n}$, so the expression $\approx \frac{1}{n} \sum_{k=1}^n \arctan \frac{k}{n} \rightarrow \int_0^1 \arctan x dx = \frac{\pi}{4} - \frac{1}{2} \ln 2$ — and gets the right number, but with NO justification that the floor's loss is negligible. The danger is the opposite over-correction: 'the floor removes a full unit each term, so subtract $\frac{1}{n} \sum 1 = 1$ and the answer is $\frac{\pi}{4} - \frac{1}{2} \ln 2 - 1 < 0$.' Both are unjustified; the floor removes a fractional part $\in [0, 1)$ per term, total $< n$, which vanishes only after dividing by n^2 . Only a genuine two-sided squeeze is legitimate.

Method I — Strip the floor, then Riemann sum

1. Write each term as its value minus a bounded fractional defect. For any real y , $y - 1 < \lfloor y \rfloor \leq y$. With $y_k = n \arctan \frac{k}{n}$ (which is ≥ 0 since $1 \leq k \leq n$):
2.
$$\frac{1}{n^2} \sum_{k=1}^n \left(n \arctan \frac{k}{n} - 1 \right) < \frac{1}{n^2} \sum_{k=1}^n \left\lfloor n \arctan \frac{k}{n} \right\rfloor \leq \frac{1}{n^2} \sum_{k=1}^n n \arctan \frac{k}{n}.$$
3. The upper bound is a Riemann sum: $\frac{1}{n^2} \sum_{k=1}^n n \arctan \frac{k}{n} = \frac{1}{n} \sum_{k=1}^n \arctan \frac{k}{n} \rightarrow \int_0^1 \arctan x dx$.
4. The lower bound differs from the upper only by $\frac{1}{n^2} \sum_{k=1}^n 1 = \frac{n}{n^2} = \frac{1}{n} \rightarrow 0$, so it has the SAME limit.
5. Evaluate the integral by parts: $\int_0^1 \arctan x dx = \left[x \arctan x - \frac{1}{2} \ln(1 + x^2) \right]_0^1 = \frac{\pi}{4} - \frac{1}{2} \ln 2$.
6. Both bounds converge to $\frac{\pi}{4} - \frac{1}{2} \ln 2$, so by the squeeze the limit equals $\boxed{\frac{\pi}{4} - \frac{1}{2} \ln 2}$. (Here the floor genuinely matters per term but its total loss $< n$ is killed by the $1/n^2$ scaling.)

Method II — Global sandwich with explicit error bound

1. Let $S_n = \frac{1}{n^2} \sum_{k=1}^n \left\lfloor n \arctan \frac{k}{n} \right\rfloor$ and $R_n = \frac{1}{n} \sum_{k=1}^n \arctan \frac{k}{n}$ (the right-endpoint Riemann sum of $\arctan$ on $[0, 1]$).
2. Since $0 \leq n \arctan \frac{k}{n} - \lfloor n \arctan \frac{k}{n} \rfloor < 1$ for each k , summing gives $0 \leq \sum_{k=1}^n n \arctan \frac{k}{n} - \sum_{k=1}^n \lfloor \cdot \rfloor < n$.
3. Divide by n^2 : $0 \leq R_n - S_n < \frac{1}{n}$. Hence $R_n - \frac{1}{n} < S_n \leq R_n$.

4. Because $\arctan$ is continuous (Riemann integrable) on $[0, 1]$, $R_n \rightarrow \int_0^1 \arctan x \, dx = \frac{\pi}{4} - \frac{1}{2} \ln 2$, and $\frac{1}{n} \rightarrow 0$.
5. By the squeeze theorem $S_n \rightarrow \frac{\pi}{4} - \frac{1}{2} \ln 2$. Numerically ≈ 0.43882 , matching direct computation.

INSIGHT

The lesson: a floor inside a sum scaled by $1/n^2$ contributes per-term error < 1 , total $< n$, which is $o(n^2)$ – so the floored sum and the raw Riemann sum share the same limit. The clean target $\frac{\pi}{4} - \frac{1}{2} \ln 2 = \int_0^1 \arctan x \, dx$ is what the sandwich pins down.

№89 The Misleading Termwise Limit

●●●●● L5

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k}{n^2} \sin^2\left(\frac{k\pi}{n}\right)$$

termwise-trap

sin

global-bound

$x \rightarrow \infty$

squeeze

LIMIT

$$\frac{1}{4}$$

THE TRAP

Looking termwise, each term $\frac{k}{n^2} \sin^2\left(\frac{k\pi}{n}\right) \rightarrow 0$ as $n \rightarrow \infty$, so a student reasons 'n terms each $\rightarrow 0$, an $n \cdot 0$ form, probably 0.' This is the seduction: termwise vanishing says nothing about a sum whose length grows. Equally tempting is to drop the oscillating $\sin^2$ and only keep the crude bound $\sum \frac{k}{n^2} \rightarrow \frac{1}{2}$, guessing $\frac{1}{2}$. Both fail: the correct object is the Riemann sum $\frac{1}{n} \sum \frac{k}{n} \sin^2\left(\pi \frac{k}{n}\right) \rightarrow \int_0^1 t \sin^2(\pi t) \, dt = \frac{1}{4}$, where the period-mean $\frac{1}{2}$ of $\sin^2$ halves the $\frac{1}{2}$.

Method I — Riemann sum

- Factor out $\frac{1}{n}$: $\frac{k}{n^2} \sin^2 \frac{k\pi}{n} = \frac{1}{n} \cdot \frac{k}{n} \sin^2\left(\pi \cdot \frac{k}{n}\right)$. With nodes $t_k = k/n$ and mesh $\frac{1}{n}$, the whole expression is a right Riemann sum for $f(t) = t \sin^2(\pi t)$ on $[0, 1]$.
- Since f is continuous, the limit equals $\int_0^1 t \sin^2(\pi t) \, dt = \int_0^1 t \cdot \frac{1 - \cos(2\pi t)}{2} \, dt = \frac{1}{2} \int_0^1 t \, dt - \frac{1}{2} \int_0^1 t \cos(2\pi t) \, dt$.
- First piece: $\frac{1}{2} \int_0^1 t \, dt = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.
- Second piece by parts, with antiderivative $\int t \cos(2\pi t) \, dt = \frac{t \sin(2\pi t)}{2\pi} + \frac{\cos(2\pi t)}{4\pi^2}$: evaluating on $[0, 1]$ gives $(0 + \frac{1}{4\pi^2}) - (0 + \frac{1}{4\pi^2}) = 0$.
- Therefore the integral is $\frac{1}{4} - \frac{1}{2} \cdot 0 = \boxed{\frac{1}{4}}$.

Method II — Exact trig sum

1. Use $\sin^2 \frac{k\pi}{n} = \frac{1}{2} - \frac{1}{2} \cos \frac{2k\pi}{n}$, so the sum splits as $\frac{1}{n^2} \sum_{k=1}^n k \left(\frac{1}{2} - \frac{1}{2} \cos \frac{2k\pi}{n} \right) = \frac{1}{2n^2} \sum_{k=1}^n k - \frac{1}{2n^2} \sum_{k=1}^n k \cos \frac{2k\pi}{n}$.
2. First sum: $\frac{1}{2n^2} \cdot \frac{n(n+1)}{2} = \frac{n+1}{4n} \rightarrow \frac{1}{4}$.
3. Second sum: the weighted cosine sum has the exact value $\sum_{k=1}^n k \cos \frac{2k\pi}{n} = \frac{n}{2}$ for every $n \geq 2$ (from $\sum_{k=1}^n k z^k$ with $z = e^{2\pi i/n}$, $z^n = 1$, taking real parts). Hence $\frac{1}{2n^2} \cdot \frac{n}{2} = \frac{1}{4n} \rightarrow 0$, contributing $-\frac{1}{4n}$ to the total.
4. So the limit is $\frac{n+1}{4n} - \frac{1}{4n} = \frac{n}{4n} = \frac{1}{4}$ exactly for all n — the sequence is constantly $\frac{1}{4}$, hence the limit is $\frac{1}{4}$.

Method III — Squeeze via period-mean replacement

1. Crude bracket first: $0 \leq \sin^2 \leq 1$ gives $0 \leq \sum_{k=1}^n \frac{k}{n^2} \sin^2 \frac{k\pi}{n} \leq \sum_{k=1}^n \frac{k}{n^2} = \frac{n+1}{2n} \rightarrow \frac{1}{2}$, so the answer lies in $[0, \frac{1}{2}]$ — ruling out the naive 0 but not yet pinning the value.
2. Replace $\sin^2 \frac{k\pi}{n}$ by its period-mean $\frac{1}{2}$. The replaced sum is $\frac{1}{2} \sum_{k=1}^n \frac{k}{n^2} = \frac{n+1}{4n} \rightarrow \frac{1}{4}$, and the error is $-\frac{1}{2n^2} \sum_{k=1}^n k \cos \frac{2k\pi}{n} = -\frac{1}{4n}$ by the exact value $\sum k \cos \frac{2k\pi}{n} = \frac{n}{2}$.
3. Since the error $-\frac{1}{4n} \rightarrow 0$, the slowly varying weight $\frac{k}{n}$ is asymptotically orthogonal to the oscillation, and the squeeze collapses to the mean-replaced value: the limit is $\frac{1}{4} \cdot 1 = \frac{1}{4}$.

INSIGHT

Termwise limits are silent about sums whose length grows. The slowly varying weight t times the period-mean $\frac{1}{2}$ of $\sin^2$ gives $\int_0^1 t \cdot \frac{1}{2} dt = \frac{1}{4}$; the oscillation $-\frac{1}{2} \cos(2\pi t)$ integrates against the smooth weight to exactly 0. Notably the finite sum equals $\frac{1}{4}$ for every n , because $\sum_{k=1}^n k \cos \frac{2k\pi}{n} = \frac{n}{2}$ cancels the $\frac{n+1}{4n}$ down to $\frac{1}{4}$ on the nose.

№90 One-Sided Squeeze at an Integer

●●●●● L3

$$\lim_{x \rightarrow 2^-} (x - \lfloor x \rfloor) \left\lfloor \frac{1}{2-x} \right\rfloor \cdot (2-x)$$

floor

one-sided

$x \rightarrow 2^-$

squeeze

LIMIT

1

THE TRAP

It looks like an $\infty \cdot 0$: as $x \rightarrow 2^-$, $\lfloor \frac{1}{2-x} \rfloor \rightarrow +\infty$ while $(2-x) \rightarrow 0^+$, and a strong student rules that ' $2-x$ wins, so this factor $\rightarrow 0$ ' and hence the whole product $\rightarrow 0$. That is wrong: the floor and its reciprocal scale are locked together, $\lfloor \frac{1}{2-x} \rfloor \cdot (2-x) \rightarrow 1$ by squeeze (it is $\Theta(1)$, never 0). With $x - \lfloor x \rfloor = x - 1 \rightarrow 1$, the limit is $1 \cdot 1 = 1$.

Method I — Squeeze the floor product

1. As $x \rightarrow 2^-$, set $h = 2 - x \rightarrow 0^+$. Then $\lfloor x \rfloor = 1$ (since $1 < x < 2$), so $x - \lfloor x \rfloor = x - 1 = 1 - h \rightarrow 1$.

2. For the floor: $\frac{1}{h} - 1 < \left\lfloor \frac{1}{h} \right\rfloor \leq \frac{1}{h}$. Multiply through by $h > 0$: $1 - h < h \left\lfloor \frac{1}{h} \right\rfloor \leq 1$.
3. As $h \rightarrow 0^+$, both bounds $\rightarrow 1$, so by the squeeze theorem $h \lfloor 1/h \rfloor \rightarrow 1$.
4. Multiply the two convergent factors: $(1 - h) \cdot (h \lfloor 1/h \rfloor) \rightarrow 1 \cdot 1 = 1$.

Method II — Fractional-part identity

1. With $h = 2 - x \rightarrow 0^+$, write $h \lfloor 1/h \rfloor = h \left(\frac{1}{h} - \{1/h\} \right) = 1 - h\{1/h\}$.
2. Since $0 \leq \{1/h\} < 1$, we have $0 \leq h\{1/h\} < h \rightarrow 0$, hence $h \lfloor 1/h \rfloor = 1 - h\{1/h\} \rightarrow 1$.
3. Combined with $x - \lfloor x \rfloor = 1 - h \rightarrow 1$, the product $\rightarrow (1)(1) = 1$.

INSIGHT

The classic $h \lfloor 1/h \rfloor \rightarrow 1$ squeeze, dressed with a fractional-part factor that quietly tends to 1 — the $\infty \cdot 0$ is a mirage since a floor times its reciprocal scale is always $\Theta(1)$.

CHAPTER X

The Grand Hybrids



Three or more worlds fused into a single pitiless limit.

№91 The Floor at the Threshold

●●●●● L4

$$\lim_{x \rightarrow 0^+} \left\lfloor \frac{1000 \sin x}{x} \right\rfloor$$

floor trigonometry one-sided $x \rightarrow 0^+$ squeeze

LIMIT

999

THE TRAP

Naive answer 1000: a student computes $\frac{\sin x}{x} \rightarrow 1$, multiplies to get 1000, and reads off $\lfloor 1000 \rfloor = 1000$. The error is forgetting that $\frac{\sin x}{x} < 1$ strictly for $x \neq 0$, so $\frac{1000 \sin x}{x} < 1000$ always — it approaches 1000 from below, never reaching it, and the floor sees the value just under the integer, not the integer itself.

Method I - One-sided squeeze on the floor argument

- For $0 < x < \frac{\pi}{2}$ the strict inequality $\sin x < x$ holds, hence $\frac{\sin x}{x} < 1$ and therefore $\frac{1000 \sin x}{x} < 1000$.
- On the other side, $\frac{\sin x}{x} \rightarrow 1^-$, so for x small enough $\frac{1000 \sin x}{x} > 999$. Indeed $\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \dots$, an alternating series whose terms decrease, so $\frac{1000 \sin x}{x} > 1000 - \frac{1000x^2}{6}$, which exceeds 999 once $\frac{1000x^2}{6} < 1$, i.e. $x < \sqrt{6/1000} \approx 0.077$.
- Thus on a punctured right-neighbourhood of 0 the argument lies in $(999, 1000)$, so its floor is the constant 999.

$$4. \quad \lim_{x \rightarrow 0^+} \left\lfloor \frac{1000 \sin x}{x} \right\rfloor = 999.$$

Method II - Floor is locally constant

- The floor function $\lfloor t \rfloor$ is constant equal to 999 on the whole interval $t \in [999, 1000)$.
- Let $g(x) = \frac{1000 \sin x}{x}$ for $x > 0$. As $x \rightarrow 0^+$ we have $g(x) \rightarrow 1000$, yet $g(x) < 1000$ for every such x (since $\sin x < x$); combined with $g(x) > 999$ once x is small, the value $g(x)$ enters and stays inside the interval $[999, 1000)$.

- On that interval $\lfloor g(x) \rfloor \equiv 999$, so the floor value is eventually the constant 999 and its limit is 999.
- The key lesson: $\lfloor \cdot \rfloor$ is left-discontinuous at the integer 1000, and our argument lands on the lower side of it.

INSIGHT

Replacing 1000 by any positive integer N gives $\lim_{x \rightarrow 0^+} \lfloor N \sin x / x \rfloor = N - 1$ – the floor sees the approach-from-below, not the limit value N .

№92 Taming the Blow-Up

●●●●● L4

$$\lim_{x \rightarrow (\pi/2)^-} \left(\frac{\pi}{2} - x \right) \lfloor \tan x \rfloor$$

floor

tangent

one-sided

blow-up

squeeze

LIMIT

1

THE TRAP

Answer 0. Seeing the factor $\frac{\pi}{2} - x \rightarrow 0$, a hasty student treats $\lfloor \tan x \rfloor$ as "some finite integer" and concludes $0 \cdot (\text{finite}) = 0$. But as $x \rightarrow (\pi/2)^-$, $\tan x \rightarrow +\infty$, so $\lfloor \tan x \rfloor \rightarrow +\infty$ too: this is a genuine $0 \cdot \infty$ indeterminate form. The vanishing factor is exactly $1/\lfloor \tan x \rfloor$ in scale, so the product settles at 1, not 0.

Method I - Strip the floor with a bounded error

- Write $\lfloor \tan x \rfloor = \tan x - \{\tan x\}$ where $\{\tan x\} \in [0, 1)$ is bounded.
- Then $\left(\frac{\pi}{2} - x \right) \lfloor \tan x \rfloor = \left(\frac{\pi}{2} - x \right) \tan x - \left(\frac{\pi}{2} - x \right) \{\tan x\}$.
- The second term $\rightarrow 0$ since $\frac{\pi}{2} - x \rightarrow 0$ while $0 \leq \{\tan x\} < 1$ is bounded.
- For the first term set $u = \frac{\pi}{2} - x \rightarrow 0^+$. Then $\tan x = \tan\left(\frac{\pi}{2} - u\right) = \cot u = \frac{\cos u}{\sin u}$, so $u \tan x = \frac{u}{\sin u} \cos u \rightarrow 1 \cdot 1 = 1$.
- $$\lim_{x \rightarrow (\pi/2)^-} \left(\frac{\pi}{2} - x \right) \lfloor \tan x \rfloor = 1 - 0 = 1.$$

Method II - Direct squeeze

- For every real t , $t - 1 < \lfloor t \rfloor \leq t$. With $t = \tan x > 0$ near $\frac{\pi}{2}^-$ this gives $\tan x - 1 < \lfloor \tan x \rfloor \leq \tan x$.
- Multiply through by the positive quantity $u = \frac{\pi}{2} - x$: $u(\tan x - 1) < u \lfloor \tan x \rfloor \leq u \tan x$.
- From the cofunction identity $u \tan x = \frac{u \cos u}{\sin u} \rightarrow 1$, hence the upper bound $\rightarrow 1$ and the lower bound $u(\tan x - 1) = u \tan x - u \rightarrow 1 - 0 = 1$.
- Both bounds converge to 1, so by the squeeze theorem the middle quantity $\rightarrow 1$.

INSIGHT

The floor is invisible in the limit precisely because its error $\{\tan x\}$ is $O(1)$ while $\tan x \sim 1/u$ blows up – the dominant growth swamps the unit-scale wobble. Note the one-sided approach is essential: from the right, $\tan x \rightarrow -\infty$ and $\lfloor \tan x \rfloor$ would force a different analysis.

$$\lim_{x \rightarrow 0^+} \left\lfloor \frac{x - \sin x}{x - \arcsin x} \right\rfloor$$

floor sine arcsine series $x \rightarrow 0^+$

LIMIT

−1

THE TRAP

Answer 0. The seductive move: compute that the ratio $\frac{x - \sin x}{x - \arcsin x} \rightarrow -1$, then reason 'the inside tends to -1 , and $\lfloor -1 \rfloor = -1$, but for values *\emph{just above}* -1 the floor jumps to 0.' The error is getting the side backwards — or asserting the inside reaches -1 . In fact the ratio equals $-1 + \frac{x^2}{2} + o(x^2)$, so it sits in $(-1, 0)$, strictly above -1 and never equal to it; and $\lfloor t \rfloor = -1$ for *\emph{every}* $t \in (-1, 0)$. So the floor is -1 , not 0. (If one instead mis-signs the fifth-order terms and lands just below -1 , one wrongly gets -2 .)

Method I - Leading-order series

1. Use $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$, so $x - \sin x = \frac{x^3}{6} - \frac{x^5}{120} + \dots > 0$.
2. Use $\arcsin x = x + \frac{x^3}{6} + \frac{3x^5}{40} + \dots$, so $x - \arcsin x = -\frac{x^3}{6} - \frac{3x^5}{40} - \dots < 0$.
3. Factor out $\frac{x^3}{6}$ from each and cancel:

$$\frac{x - \sin x}{x - \arcsin x} = \frac{\frac{x^3}{6} \left(1 - \frac{x^2}{20} + \dots\right)}{-\frac{x^3}{6} \left(1 + \frac{9x^2}{20} + \dots\right)} = -\frac{1 - \frac{x^2}{20} + \dots}{1 + \frac{9x^2}{20} + \dots}.$$

4. Expand using $\left(1 + \frac{9x^2}{20}\right)^{-1} = 1 - \frac{9x^2}{20} + \dots$:

$$-\left(1 - \frac{x^2}{20}\right) \left(1 - \frac{9x^2}{20} + \dots\right) = -\left(1 - \frac{x^2}{2} + \dots\right) = -1 + \frac{x^2}{2} + \dots.$$
5. Thus the ratio equals $-1 + \frac{x^2}{2} + o(x^2)$: for small $x > 0$ it is > -1 and tends to -1 , so it lives in $(-1, 0)$. Since $\lfloor t \rfloor = -1$ for all $t \in (-1, 0)$, the floor is identically -1 and the limit is $\boxed{-1}$.

Method II - Sign + magnitude comparison

1. For $x > 0$ small: $x - \sin x > 0$ (since $\sin x < x$) and $x - \arcsin x < 0$ (since $\arcsin x > x$). So the ratio is negative — already ruling out the floor being 0.
2. Compare magnitudes via the leading expansions: $|x - \sin x| = \frac{x^3}{6} \left(1 - \frac{x^2}{20} + \dots\right)$ and $|x - \arcsin x| = \frac{x^3}{6} \left(1 + \frac{9x^2}{20} + \dots\right)$.
3. The denominator's correction $+\frac{9x^2}{20}$ dominates the numerator's $-\frac{x^2}{20}$, so $|x - \sin x| < |x - \arcsin x|$ for small $x > 0$. Hence the ratio has magnitude < 1 , i.e. it lies strictly in $(-1, 0)$.
4. For every $t \in (-1, 0)$, $\lfloor t \rfloor = -1$. Therefore the floor is identically -1 on a punctured right-neighborhood of 0, and $\lim_{x \rightarrow 0^+} = -1$.

INSIGHT

Because $\sin$ and $\arcsin$ are inverses, their cubic terms have equal magnitude and opposite sign – they cancel in the ratio to give exactly -1 at leading order. The TIE is broken by the fifth-order terms ($\frac{1}{120}$ vs $\frac{3}{40}$), whose asymmetry pushes the ratio to $-1 + \frac{x^2}{2}$, just above -1 , fixing the floor at -1 .

№94 The Fractional Mirage

●●●●● L5

$$\lim_{x \rightarrow 1^+} \left\{ \frac{\ln x}{x-1} \right\}$$

fractional-part

logarithm

one-sided

$x \rightarrow 1^+$

discontinuity

LIMIT

1

THE TRAP

Answer 0. The reflex: $\frac{\ln x}{x-1} \rightarrow 1$ and $\{1\} = 0$, so push the fractional part through the limit to get 0. This silently assumes $\{\cdot\}$ is continuous at the integer 1 — but $\{\cdot\}$ is only RIGHT-continuous at integers and jumps by 1 from the left. The decisive question is the SIDE of the approach: by concavity of $\ln$ (or the $-h/2$ term in the expansion), $\frac{\ln x}{x-1} \rightarrow 1^-$, i.e. through values just below 1, where $\lfloor \cdot \rfloor = 0$ and $\{t\} = t \rightarrow 1^-$. The limit is 1, not 0.

Method I - Sign of the approach via series

1. Let $R(x) = \frac{\ln x}{x-1}$. Put $h = x - 1 \rightarrow 0^+$ and use $\ln(1+h) = h - \frac{h^2}{2} + \frac{h^3}{3} - \dots$
2. Then $R = \frac{h - \frac{h^2}{2} + \frac{h^3}{3} - \dots}{h} = 1 - \frac{h}{2} + \frac{h^2}{3} - \dots$
3. For $x \rightarrow 1^+$ we have $h > 0$, so $R = 1 - \frac{h}{2} + O(h^2) < 1$ for small h , while $R \rightarrow 1$. Also $R > 0$ for small h (it tends to 1), so $R \in (0, 1)$.
4. On $(0, 1)$ the fractional part is the identity: $\{R\} = R$. Hence $\{R\} \rightarrow 1^-$, giving

$$\lim_{x \rightarrow 1^+} \left\{ \frac{\ln x}{x-1} \right\} = 1.$$

Method II - Geometry of the floor jump

1. Recall $\{t\} = t - \lfloor t \rfloor$. Near $t = 1$ from below, $\lfloor t \rfloor = 0$ so $\{t\} = t$; from above, $\lfloor t \rfloor = 1$ so $\{t\} = t - 1$. The answer hinges entirely on which side $R(x)$ approaches 1.
2. Now $R(x) = \frac{\ln x - \ln 1}{x-1}$ is the slope of the chord of $\ln$ from 1 to x . Since $\ln$ is strictly concave, for $x > 1$ this chord slope is strictly LESS than the tangent slope $\ln'(1) = 1$. Hence $R(x) < 1$, and $R(x) \rightarrow 1^-$.
3. Therefore $\lfloor R(x) \rfloor = 0$ and $\{R(x)\} = R(x) \rightarrow 1$.
4. Contrast: for $x \rightarrow 1^-$ the chord slope exceeds 1, so $R > 1$, $\lfloor R \rfloor = 1$, $\{R\} = R - 1 \rightarrow 0^+$. The two limits disagree, so the two-sided limit does not exist — which is exactly why the problem is posed as $x \rightarrow 1^+$.

INSIGHT

Concavity of $\ln$ pins the side of the approach to 1^- ; the entire problem is a meditation on the left-discontinuity (right-continuity) of $\{\cdot\}$ at an integer. Replace $\ln x$ by any strictly concave f with $f(1) = 0$, $f'(1) = 1$ and the same answer 1 results from $x \rightarrow 1^+$.

№95 The Vanishing Negative

●●●●● L4

$$\lim_{x \rightarrow -\infty} \lfloor x (\arctan x - \arctan(x-1)) \rfloor$$

floor

arctangent

$x \rightarrow -\infty$

0-times-bounded

sign-of-zero

LIMIT

−1

THE TRAP

Answer 0. The bracketed difference $\arctan x - \arctan(x-1) \rightarrow 0$, so one reads $x \cdot 0 = 0$ and then $\lfloor 0 \rfloor = 0$. But the product never reaches 0: for every $x < 0$ it is a strictly negative number lying in $(-1, 0)$, and $\lfloor t \rfloor = -1$ for all $t \in (-1, 0)$. The limit of the floor is therefore -1 , not the floor of the limit. Evaluating $\lfloor \cdot \rfloor$ at the limiting value 0 is the fatal shortcut — the floor is discontinuous exactly there, and the approach is from below.

Method I - Mean value theorem

1. Apply the MVT to $f(t) = \arctan t$ on $[x-1, x]$: there is $c \in (x-1, x)$ with $\arctan x - \arctan(x-1) = f'(c) = \frac{1}{1+c^2}$.
2. Hence the bracketed quantity is $P(x) = \frac{x}{1+c^2}$ with $c \in (x-1, x)$. As $x \rightarrow -\infty$ we have $c \rightarrow -\infty$ and $c \sim x$, so $P(x) \sim \frac{x}{1+x^2} \rightarrow 0$.
3. Sign: for $x < 0$ we have $x < 0$ and $1+c^2 > 0$, so $P(x) < 0$. Thus $P(x) \rightarrow 0^-$ — the product approaches 0 strictly from below.
4. Bound: since $c > x-1$, $1+c^2 > 1+(x-1)^2$, so for $x < 0$, $|P(x)| = \frac{|x|}{1+c^2} < \frac{|x|}{1+(x-1)^2} < 1$ (the last inequality is $(x-1)^2 > 0$ rearranged). Therefore $P(x) \in (-1, 0)$ for every $x < 0$, giving $\lfloor P(x) \rfloor = -1$, and the limit is -1 .

Method II - Exact closed form via the arctangent subtraction identity

1. For a, b of the same sign the subtraction formula $\arctan a - \arctan b = \arctan \frac{a-b}{1+ab}$ holds without a $\pm\pi$ correction. With $a = x$, $b = x-1$ and $x < 0$, here $1+ab = 1+x^2-x > 0$, so

$$\arctan x - \arctan(x-1) = \arctan \frac{1}{1+x^2-x}.$$

2. Thus $P(x) = x \arctan \frac{1}{1+x^2-x}$. For $x < 0$ the argument $\frac{1}{1+x^2-x} > 0$, so its arctangent is positive, and multiplying by $x < 0$ gives $P(x) < 0$.

3. Bound the arctangent by its argument, $0 < \arctan u < u$ for $u > 0$: $0 < \arctan \frac{1}{1+x^2-x} < \frac{1}{1+x^2-x}$. Multiplying by $x < 0$ flips the inequalities:

$$\frac{x}{1+x^2-x} < P(x) < 0.$$

4. Finally $\frac{x}{1+x^2-x} > -1 \iff x > -(1+x^2-x) \iff (x-1)^2 > 0$, true for all $x \neq 1$. Hence $P(x) \in (-1, 0)$ for every $x < 0$, so $\lfloor P(x) \rfloor = -1$ identically and

$$\lim_{x \rightarrow -\infty} \lfloor x(\arctan x - \arctan(x-1)) \rfloor = -1.$$

INSIGHT

The floor here is not merely a limit phenomenon: $\lfloor P(x) \rfloor = -1$ for every $x < 0$. The same expression as $x \rightarrow +\infty$ gives $P(x) \in (0, 1)$, hence floor 0 – the sign of x silently shifts the answer by a full integer, and the floor is discontinuous at the limiting value 0 precisely where the naive computation evaluates it.

№96 Signum in the Exponent

● ● ● ● ● L5

$$\lim_{x \rightarrow 0^-} \left(\frac{\arcsin |x|}{|x|} \right)^{\operatorname{sgn}(x)/x^2}$$

signum

arcsine

modulus

power

one-sided

log-transform

LIMIT

$$e^{-1/6}$$

THE TRAP

Answer $e^{1/6}$: the student recognizes the well-known $\left(\frac{\arcsin t}{t}\right)^{1/t^2} \rightarrow e^{1/6}$ and reads it straight off, silently treating the exponent as $+1/x^2$. But for $x \rightarrow 0^-$ we have $\operatorname{sgn}(x) = -1$, so the true exponent is $-1/x^2$. That single minus sign inverts the base-power, turning $e^{1/6}$ into its reciprocal $e^{-1/6}$. The modulus is a harmless decoy (the base is even); the signum is the live wire.

Method I - Logarithm and Taylor

- For $x < 0$, $\operatorname{sgn}(x) = -1$, so the exponent is $-\frac{1}{x^2}$. Put $t = |x| = -x \rightarrow 0^+$; note $x^2 = t^2$, and the base is $\frac{\arcsin t}{t} \rightarrow 1$, an indeterminate 1^∞ form.
- Take logs: $L = \ln(\text{expression}) = -\frac{1}{x^2} \ln\left(\frac{\arcsin t}{t}\right) = -\frac{1}{t^2} \ln\left(\frac{\arcsin t}{t}\right)$.
- Expand $\arcsin t = t + \frac{t^3}{6} + \frac{3t^5}{40} + \dots$, so $\frac{\arcsin t}{t} = 1 + \frac{t^2}{6} + o(t^2)$, hence $\ln\left(1 + \frac{t^2}{6} + o(t^2)\right) = \frac{t^2}{6} + o(t^2)$.
- Therefore $L = -\frac{1}{t^2} \left(\frac{t^2}{6} + o(t^2)\right) = -\frac{1}{6} + o(1) \rightarrow -\frac{1}{6}$.
- Exponentiating, the limit is $e^{-1/6}$.

Method II - Reduce to the known even case

- Let $F(t) = \left(\frac{\arcsin t}{t}\right)^{1/t^2}$ for $t > 0$; the standard result (Method I with exponent $+1/t^2$) gives $\lim_{t \rightarrow 0^+} F(t) = e^{1/6}$.
- Write the given expression with $t = |x|$. Since $\frac{1}{x^2} = \frac{1}{t^2}$, the exponent $\frac{\operatorname{sgn}(x)}{x^2} = \operatorname{sgn}(x) \cdot \frac{1}{t^2}$, so the expression equals $\left[\left(\frac{\arcsin t}{t}\right)^{1/t^2}\right]^{\operatorname{sgn}(x)} = F(t)^{\operatorname{sgn}(x)}$.

- For $x \rightarrow 0^-$, $\text{sgn}(x) = -1$, so the expression is $F(t)^{-1}$, whose limit is $(e^{1/6})^{-1} = e^{-1/6}$.
- The modulus only guarantees the base is well-defined and even; the signum alone carries the sign of the answer, flipping the familiar $e^{1/6}$ to $e^{-1/6}$.

INSIGHT

The two-sided limit does NOT exist: $x \rightarrow 0^+$ gives $e^{1/6}$ while $x \rightarrow 0^-$ gives $e^{-1/6}$. Posed one-sided as here it is perfectly well-defined; the sgn is a deliberate sign-bomb that reciprocates the classic result.

№97 Absolute Tangent

●●●●● L4

$$\lim_{x \rightarrow 0} \frac{|\tan x| - |x|}{|x|^3}$$

modulus tangent two-sided series $x \rightarrow 0$

LIMIT

$$\frac{1}{3}$$

THE TRAP

Answer 'does not exist'. The seductive misstep: strip the moduli, get $\frac{\tan x - x}{x^3} \rightarrow \frac{1}{3}$ for $x > 0$, then reason that since $\tan x - x$ is an odd function it must tend to $-\frac{1}{3}$ for $x < 0$, so the two-sided limit fails. This is wrong: the moduli act on BOTH numerator and denominator. Replacing $\tan x - x$ by $|\tan x| - |x|$ and x^3 by $|x|^3$ makes numerator and denominator even, the odd-vs-odd cancellation in the unmodified ratio is exactly what the moduli undo, and the limit cleanly equals $\frac{1}{3}$.

Method I - Evenness then Taylor

- On $|x| < \frac{\pi}{2}$ the tangent is increasing through the origin, so $\tan x$ and x share sign and $|\tan x| = \tan |x|$. Put $t = |x| \geq 0$; the expression becomes $\frac{\tan t - t}{t^3}$, which depends on x only through $|x|$ and is therefore even.
- Use the Maclaurin series $\tan t = t + \frac{t^3}{3} + \frac{2t^5}{15} + \dots$, hence $\tan t - t = \frac{t^3}{3} + O(t^5)$.
- Divide: $\frac{\tan t - t}{t^3} = \frac{1}{3} + O(t^2) \rightarrow \frac{1}{3}$ as $t \rightarrow 0^+$.
- Because the reduced expression depends only on $t = |x|$, the two one-sided limits coincide and the two-sided limit equals $\frac{1}{3}$.

Method II - Substitution and one-sided agreement

- Right limit ($x > 0$): here $|\tan x| = \tan x$, $|x| = x$, so $\frac{|\tan x| - |x|}{|x|^3} = \frac{\tan x - x}{x^3} \rightarrow \frac{1}{3}$ by the cubic expansion $\tan x - x = \frac{x^3}{3} + O(x^5)$.
- Left limit ($x < 0$): write $x = -s$ with $s > 0$. Then $|\tan x| = |\tan(-s)| = |-\tan s| = \tan s$, $|x| = s$, $|x|^3 = s^3$, giving $\frac{\tan s - s}{s^3} \rightarrow \frac{1}{3}$ as $s \rightarrow 0^+$.
- Both one-sided limits equal $\frac{1}{3}$, so the two-sided limit exists and equals $\frac{1}{3}$.

4. The moduli are precisely what repair the odd ratio $\frac{\tan x - x}{x^3}$ (whose sign would otherwise be side-dependent) into an even function with a genuine two-sided limit.

INSIGHT

Wrapping an odd-leading numerator and an odd power in matched moduli is the standard device for forcing a two-sided limit out of an otherwise side-dependent ratio.

№98 The Arctangent's Last Digit

● ● ● ● ● L5

$$\lim_{x \rightarrow +\infty} \left\lfloor -3x^3 \left(\frac{\pi}{2} - \arctan x - \frac{1}{x} \right) \right\rfloor$$

floor

arctangent

asymptotic

$x \rightarrow +\infty$

series

LIMIT

0

THE TRAP

Answer 1: using $\arctan x = \frac{\pi}{2} - \frac{1}{x} + \frac{1}{3x^3} - \dots$, the bracket becomes $-3x^3 \cdot (-\frac{1}{3x^3} + \dots) \rightarrow 1$, so a student writes $\lfloor 1 \rfloor = 1$. But the argument equals $1 - \frac{3}{5x^2} + o(1/x^2)$, which is strictly < 1 for every large x and only approaches 1 from below. A quantity in $(0, 1)$ has floor 0, not 1. (Numerically, naive floating point even reports junk values like 246 for $x = 10^6$ because $\frac{\pi}{2} - \arctan x$ cancels catastrophically.)

Method I - Asymptotic series of arctan

- For $x > 0$, $\arctan x = \frac{\pi}{2} - \frac{1}{x} + \frac{1}{3x^3} - \frac{1}{5x^5} + \dots$.
- Hence $\frac{\pi}{2} - \arctan x - \frac{1}{x} = \frac{1}{x} - \frac{1}{3x^3} + \frac{1}{5x^5} - \dots - \frac{1}{x} = -\frac{1}{3x^3} + \frac{1}{5x^5} - \dots$.
- Multiply by $-3x^3$: $-3x^3 \left(-\frac{1}{3x^3} + \frac{1}{5x^5} - \dots \right) = 1 - \frac{3}{5x^2} + \dots$.
- So the argument is $1 - \frac{3}{5x^2} + o(1/x^2) < 1$, approaching 1 from below. For all large x it lies in $(0, 1)$, so its floor is 0.

$$\lim_{x \rightarrow +\infty} \lfloor \dots \rfloor = 0.$$

Method II - Reciprocal substitution + Taylor expansion

- Let $t = \frac{1}{x} \rightarrow 0^+$. Using $\arctan x = \frac{\pi}{2} - \arctan t$ (valid for $x > 0$), the inner expression becomes $\frac{\pi}{2} - (\frac{\pi}{2} - \arctan t) - t = \arctan t - t$.
- So the whole argument is $-\frac{3}{t^3}(\arctan t - t) = -3 \cdot \frac{\arctan t - t}{t^3}$.
- Now $\arctan t - t = -\frac{t^3}{3} + \frac{t^5}{5} - \dots$, so $\frac{\arctan t - t}{t^3} = -\frac{1}{3} + \frac{t^2}{5} - \dots$, and -3 times this is $1 - \frac{3t^2}{5} + \dots$.
- As $t \rightarrow 0^+$ this stays < 1 and tends to 1, hence the argument is in $(0, 1)$ for small t , so its floor is 0. Therefore the limit is 0.

INSIGHT

The 'subtract $1/x$ ' trick isolates the next term of arctan's tail; the floor then exposes that the corrected quantity sits a hair below the integer 1. The cancellation-free identity $\frac{\pi}{2} - \arctan x = \arctan(1/x)$ is also what you must use to check this numerically without precision loss.

№99 Fracture at e

● ● ● ● ● L5

$$\lim_{x \rightarrow e^+} \frac{\{\ln x\}}{x - e}$$

fractional-part

logarithm

one-sided

$x \rightarrow e^+$

derivative

LIMIT

$$\frac{1}{e}$$

THE TRAP

Because $\ln e = 1$ is an integer, $\{\ln x\}$ jumps from near 1^- to 0^+ as x crosses e , so a student concludes the two-sided limit is asymmetric and declares 'does not exist'. Worse, they may anchor on the left behavior (numerator $\rightarrow 1$, denominator $\rightarrow 0$) and write ' ∞ or oscillates'. But the question is one-sided: for $x > e$, $\{\ln x\} = \ln x - 1 = \ln x - \ln e$ is perfectly smooth, and the ratio is literally the difference quotient of $\ln$ at e , giving the finite value $\ln'(e) = 1/e$.

Method I - Recognize a derivative

1. For x slightly greater than e , $\ln x > \ln e = 1$ and $\ln x < 2$, so $\lfloor \ln x \rfloor = 1$ and $\{\ln x\} = \ln x - 1 = \ln x - \ln e$.
2. Hence $\frac{\{\ln x\}}{x - e} = \frac{\ln x - \ln e}{x - e}$, which is exactly the difference quotient of $\ln$ at e .
3. Taking $x \rightarrow e^+$ gives $\ln'(e) = \frac{1}{e}$.

$$4. \quad \lim_{x \rightarrow e^+} \frac{\{\ln x\}}{x - e} = \frac{1}{e}.$$

Method II - Local linearization

1. Write $x = e + h$, $h \rightarrow 0^+$. Then $\ln x = \ln(e + h) = 1 + \ln\left(1 + \frac{h}{e}\right) = 1 + \frac{h}{e} - \frac{h^2}{2e^2} + \dots$.
2. Since $0 < \ln x - 1 < 1$ for small $h > 0$, we get $\{\ln x\} = \ln x - 1 = \frac{h}{e} - \frac{h^2}{2e^2} + \dots$.
3. Divide by $x - e = h$: $\frac{\{\ln x\}}{h} = \frac{1}{e} - \frac{h}{2e^2} + \dots \rightarrow \frac{1}{e}$.
4. From the left ($h < 0$) one would have $\ln x < 1$, so $\{\ln x\} = \ln x$ near 1^- , making the numerator ≈ 1 and the ratio $\rightarrow -\infty$ — that is why the limit is posed one-sided.

INSIGHT

At any point a with $\ln a \in \mathbb{Z}$, $\lim_{x \rightarrow a^+} \frac{\{\ln x\}}{x - a} = \frac{1}{a}$, while the left limit diverges — the fractional part converts a smooth derivative into a one-sided phenomenon.

№100 The Capstone Squeeze

● ● ● ● ● L5

$$\lim_{x \rightarrow 0} \arctan \left(1 - |x| \left\{ \frac{1}{|x|} \right\} \right)$$

arctangent

fractional-part

modulus

floor

squeeze

$x \rightarrow 0$

LIMIT

$$\frac{\pi}{4}$$

THE TRAP

Answering 'does not exist': as $x \rightarrow 0$ the quantity $\frac{1}{|x|} \rightarrow \infty$, so $\left\{ \frac{1}{|x|} \right\}$ sweeps through all of $[0, 1)$ infinitely often and never settles. A student concludes the inner expression — and hence the whole limit — oscillates forever. The trap ignores the prefactor: $|x| \rightarrow 0$ multiplies a quantity confined to $[0, 1)$, so the product is a $0 \cdot (\text{bounded})$ form squeezed to 0 regardless of how wildly the fractional part dances. The visible chaos is irrelevant; the answer is the perfectly definite $\arctan 1 = \pi/4$.

Method I - Bounded oscillation killed by $|x|$

1. For every $x \neq 0$ the fractional part satisfies $0 \leq \left\{ \frac{1}{|x|} \right\} < 1$, so although it oscillates it is uniformly bounded.
2. Multiplying the bound by $|x| > 0$ gives $0 \leq |x| \left\{ \frac{1}{|x|} \right\} < |x|$. Both outer bounds tend to 0 as $x \rightarrow 0$, so by the squeeze theorem $|x| \left\{ \frac{1}{|x|} \right\} \rightarrow 0$ — the oscillation is annihilated by the vanishing prefactor.
3. Hence the inner argument $1 - |x| \left\{ \frac{1}{|x|} \right\} \rightarrow 1$.
4. Since $\arctan$ is continuous, the limit equals $\arctan 1 = \frac{\pi}{4}$.

Method II - Rewrite via floor and squeeze the argument

1. Use $\{u\} = u - \lfloor u \rfloor$ with $u = \frac{1}{|x|}$: then $|x| \left\{ \frac{1}{|x|} \right\} = |x| \left(\frac{1}{|x|} - \left\lfloor \frac{1}{|x|} \right\rfloor \right) = 1 - |x| \left\lfloor \frac{1}{|x|} \right\rfloor$.
2. Therefore the inner argument is $1 - \left(1 - |x| \left\lfloor \frac{1}{|x|} \right\rfloor \right) = |x| \left\lfloor \frac{1}{|x|} \right\rfloor$.
3. From $\frac{1}{|x|} - 1 < \left\lfloor \frac{1}{|x|} \right\rfloor \leq \frac{1}{|x|}$, multiplying through by $|x| > 0$ yields $1 - |x| < |x| \left\lfloor \frac{1}{|x|} \right\rfloor \leq 1$.
4. As $x \rightarrow 0$ both bounds $\rightarrow 1$, so the argument $\rightarrow 1$ and $\arctan(\cdot) \rightarrow \arctan 1 = \frac{\pi}{4}$. The whole expression is even in x , so the two one-sided limits agree and the two-sided limit is clean.

INSIGHT

The identity $t\{1/t\} = 1 - t\lfloor 1/t \rfloor$ ties the fractional and floor parts together; the modulus makes the squeeze symmetric about 0, so a maximally oscillating ingredient still yields the perfectly definite value $\pi/4$.